

BLACK HOLE ENTROPY FROM CAUSAL SETS.

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A thesis presented for the degree of Doctor of Philosophy of the University of London
and for the Diploma of Imperial College.

August 2007

To Simone Libman.

ABSTRACT.

The thermodynamics of black holes is considered through the application of a causal set model of spacetime. This analysis is motivated by results in classical general relativity and quantum field theoretic approaches. Several modifications to ‘link-counting’ models of black hole entropy are investigated. One of these modifications is shown to eliminate known infinities arising in the extension of existing 1+1-dimensional models to higher dimensions. The results obtained are used to support the conjecture that the fundamental scale of causal set theory is of the order of the Planck scale.

ACKNOWLEDGEMENTS.

I should like to thank all the members, both staff and students, of the Theoretical Physics Group at Imperial College, for their support during the course of my PhD. In particular, thanks must go to my supervisor, Dr H. Fay Dowker, who provided the focus for my thesis and guided me through to completion with patience and understanding. I am also grateful to Professor Chris Isham for his support during the past few years. I am indebted to my office mates, particularly Dr Daniel Hook and Eduardo Pellegrino, whose advice and fecund jocularity never failed to raise spirits when required, and to Kelly McCallum and my mother, Evelyn Marr, who were always there when needed.

DECLARATION.

The work presented in this thesis was carried out between October 2003 and August 2007 in the Theoretical Physics Group at Imperial College London, under the supervision of Dr H. Fay Dowker. This thesis has not been submitted for a degree or diploma at any other university.

The initial impetus for the work arose from the PhD thesis of Dr Djamel Dou [Dou (1999)] and subsequent related papers. Any work derived from, or relating to, this thesis or any other works by any author is clearly marked as such in the text.

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1. INTRODUCTION TO CAUSAL SETS

1.1. Physical Background. It is well known, at least among the physics cognoscenti, that there exist irreconciled disparities between the large-scale, macroscopic descriptions provided by general relativity, and the formalism of quantum theory. It is, in large part, these disparities which causal set theory seeks to address. Additionally, it is hoped that causal sets may explain certain features of the post-Einsteinian description of spacetime, to ameliorate the current situation, in which these features are presumptively accepted.

1.1.1. Quantum Gravity. In their paper of 1987, Bombelli et al. [Bombelli (1987)] refer to quantum gravity's relationship to the 'schematic evolution' of Taketani and Sakata [Sakata (1971), Taketani (1971)]. Under this schema, a physical theory passes through three stages. In the *phenomenological* stage initial observations occur which underpin the existence and knowledge of the new phenomenon. The *substantialistic* stage provides rules which describe the phenomenon of the first stage. Finally, the *essentialistic* stage leads to a full theory of the new phenomenon, within appropriate limits of validity. A more detailed discussion, with examples, can be found in [Reid (1999)].

The impossibility of probing the length-scales at which general relativity and quantum effects combine, means that quantum gravity is not afforded the luxury of a phenomenological stage. However, a quantum theory is necessary to understand the behaviour of black holes and the early universe, to unify the known fundamental forces, etc. This has led, in particular, to the development of canonical quantization and string theoretic approaches to the problem. These approaches, and any others proposed, must combine the substantialistic and essentialistic stages to formulate a theory which, hopefully, can find *ex post facto* confirmation through its predictions of previously unrecognized phenomenology.

Of particular interest for our purposes are those questions which relate to the topological and differentiable structure of spacetime: why does a spacetime structure exist which permits the definition of a spacetime metric with Lorentzian signature? This type of 'extrinsic' question, facing back to the phenomenological stage of development, is posed to be addressed by both existing and future theories of quantum gravity. On the other hand, existing theories face 'intrinsic' problems which, to some extent, undermine the validity of their own formulation in their substantialistic and essentialistic stages.

The persistence of a continuum to arbitrarily small spacetime volume leads to infinities arising in calculations undertaken within a quantum framework [Dou (1999), Sorkin (1991)]:

- the infinities of *quantum field theory*, which renormalization techniques seek to address. Dou (*op. cit.*) points out that this renormalization is problematic in the non-perturbative study of ϕ^4 theory (and, probably for any theory which is not asymptotically free), placing limits on the predictive power of the Standard Model, above some energy scale. Renormalization in the presence of a (classical) gravitational background provides only ambiguous results, unless the spacetime metric is restricted to being static or stationary.
- the infinities of *quantized gravity*, which cannot be removed using existing renormalization techniques, and remove any predictive power from theories of quantum gravity.
- the divergent spacetime curvatures which occur at the *singularities* of classical general relativity.
- the infinite *black hole entropy* arising when one attempts to count the degrees of freedom of the horizon. We address this issue below, particularly a suggested resolution through the use of causal sets, in §4 et seq.

1.1.2. *Discrete Spacetime.* In part to overcome the problem of these infinities, it has been suggested (by [Bombelli (1987)] amongst others) that spacetime is not truly continuous, but, rather, that one finds only a large (but finite) number of elements within a finite volume of spacetime: what Riemann called a ‘discrete manifold’. Measuring the volume of a region is then a matter of counting the ‘point events’ which it contains. The continuous manifold of everyday experience is then an approximation to this underlying, discrete structure.

Furthermore, a discrete spacetime possesses a fundamental length scale, l_f , which represents the shortest physically allowable length, and similarly a fundamental time scale, t_f . (In fact, from the perspective of Lorentz-invariance, the meaningful measure is a fundamental spacetime *volume*, v_f .)

It seems likely that the scale at which spacetime is discrete is of order of the Planck scale, which is defined through use of the fundamental constants, and provides the Planck

length, l_P , and Plank time, t_P , as follows:

$$l_P = \left(\frac{G\hbar}{c^3} \right)^{\frac{1}{2}} \sim 10^{-35} \text{m} \quad (1)$$

$$t_P = \frac{l_P}{c} \sim 10^{-43} \text{s} . \quad (2)$$

(Here, and throughout, unless otherwise specified, terms are as usually defined in the literature, and may also be found in Appendix F.) This gives a Planck 4-volume of $\sim 10^{-148} \text{m}^3 \text{s}$.

The concept of spacetime as a discrete spacetime is realized through the structures imposed by a causal set, in which the relationship between spacetime points is governed by their causal order. It is worthwhile our mentioning that such an ordering requires its associated metric to be Lorentzian, since only a signature $\{-, +, +, \dots, +\}$ has the light-cone structure necessary to impose the continuum's causal order [Bombelli (1987)].

1.2. Introduction to Causal Sets.

1.2.1. History. Recent interest in causal sets was fueled by a 1987 paper by Bombelli et al., [Bombelli (1987)], which was itself preceded by early work of Finkelstein, Myrheim, 't Hooft and Sorkin. (See [Bombelli (1987), Reid (1999)] for references.) By 2002, Sorkin [Sorkin (2002)] could list theoretical advances in causal sets including,

- an order-of-magnitude prediction for the cosmological constant,
- a method of counting (kinematic) black-hole horizon 'states' and an initial framework for the consideration of 2-dimensional Hawking radiation,
- a classical causal set dynamics, consistent with discrete analogues of covariance and causality.

Sorkin provides a more detailed history of the causal set program, *op. cit.* at pp. 2-5.

1.2.2. Mathematical Background and Definitions. A causal set is a set of events for which there exists an order relation $\prec$ obeying the following three axioms:

- | | |
|----------------------|---|
| (1) transitivity | $x \prec y \prec z \Rightarrow x \prec z$, |
| (2) acyclicity | $x \prec y$ and $y \prec x \Rightarrow x = y$, |
| (3) local finiteness | $\text{cardinality}([x, y]) < \infty$. |

We adopt the reflexive convention, $x \prec x$.

In the third axiom, above, the interval $[x, y]$ defines the *Alexandrov set*, $\mathcal{A}$, for x and y , that is $\mathcal{A}(x, y) = \{z \mid x \prec z \prec y\}$. This axiom, local finiteness, ensures that the set is discrete, and is akin to the requirement that a Lorentzian manifold be globally hyperbolic.

These axioms define a locally finite, partially ordered set. Physical meaning is given to the order relation, by saying that for an element x , $x \prec y$ if and only if it is to the causal past of y . That is, $x \prec y \Leftrightarrow x \in J^-(y)$. If one considers an embedding of a causal set $\mathcal{C}$ into some manifold $\mathcal{M}$ (see §1.3.3), then the set $J^-(y)$ may be defined as those elements $\{q \mid q \in \mathcal{C} \subset \mathcal{M}\}$ such that q can be reached by a past-directed time- or light-like curve from y . “The macroscopic light-cone structure of spacetime directly reflects the order relation of the underlying causal set” [Sorkin (1991)]. Similarly, we have $x \succ y \Leftrightarrow x \in J^+(y)$ or, in the continuum limit, the set $J^+(y)$ consists of those elements $\{p \mid p \in \mathcal{C} \subset \mathcal{M}\}$ such that p can be reached by a future-directed time- or light-like curve from y .

We define several terms, which will be used during our discussions:

- If $x, y \in \mathcal{C}$ with $x \prec y$, and $J^+(x) \cap J^-(y) = \{x, y\}$ (i.e. $\mathcal{A}(x, y)$ is empty) then the relation between x and y is called a *link*, for which we introduce the notation, $x \tilde{\prec} y$. The existence of local finiteness, coupled with transitivity, implies that specifying all the links within a causal set is sufficient to encode the entire structure of that set.
- The *interval* of two elements, $x, y \in \mathcal{C}$, is the set of elements $\{p \mid x \prec p \prec y\}$, $p \in \mathcal{C}$.
- An element of a causal (sub)set is *maximal* if and only if it is to the past of no other element in the causal (sub)set. Similarly, a *minimal* element has no elements to its past.
- A *chain* is a subset of a causal set in which any two elements are causally related. A subset in which no two elements are so related is an *antichain*. A *maximal (anti)chain* is an (anti)chain to which no further element can be added.
- A *path* between two elements is a maximal chain between those elements. Different paths between the same two points may not have the same *length* (number of links): the path with the greatest number of links is the *maximal path*.
- The *width* of a causal set is the size of the largest antichain. Its *height* is the length of the longest path.

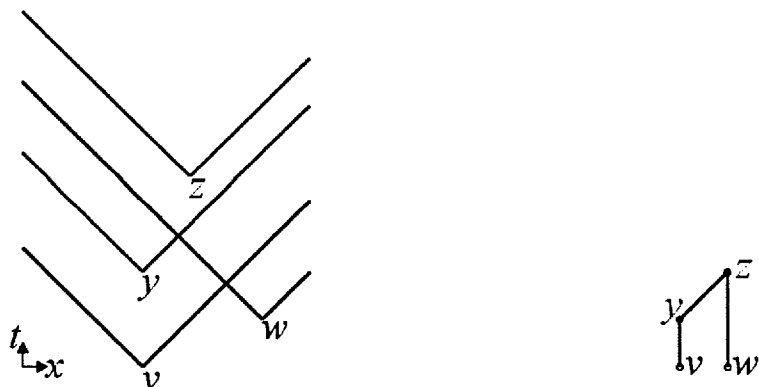


FIGURE 1. Hasse diagram (right) showing causal relationships between four spacetime points (v , w , y and z), shown, left, on a standard 1+1-dimensional spacetime diagram, with their ‘future light cones’. Note that the spatial arrangement on paper of points on the Hasse diagram is not of importance: causal connections are shown solely by connecting vertices. Also note that only links are shown (e.g. there is no vertex between v and z).

Causal sets are often represented in Hasse diagrams, in which elements of a causal set are represented by vertices, with relations appearing as edges (see Figure 1). Usually, only links are shown. Sorkin also suggests [Sorkin (2002)] that a causal set, $\mathcal{C}$, can be represented by some matrix M , with rows and columns labelled by elements of $\mathcal{C}$, with $M_{jk} = 1$ if $j \prec k$, else 0. Equally, one may think of a causal set as a ‘family tree’, indicating which elements of $\mathcal{C}$ are ‘ancestors’ of which others.

1.3. Causal Set Kinematics.

1.3.1. Overview. The kinematics of causal sets are concerned with the mathematics necessary to describe those sets, most of which belongs to the combinatorial theory of partial orders. It is to be used to extract, from causal sets, geometrical and topological information, and to facilitate the counting of elements enabling the evaluation of space-time volumes. A kinematical analysis must allow for some view of the causal set which replicates a 4-dimensional Lorentzian manifold structure in macroscopic limits: causal set theorists refer to the proposition that such a process can occur as the *Hauptvermutung* [Sorkin (1991)]. The change of scale, from microscopic discrete manifold to macroscopic

continuum, is brought about by a process called *coarse graining*. It is important to understand the nature of those causal sets for which *emergent* properties, such as length, topology and dimension, arise under coarse graining.

Much analysis has been done on the reverse problem: forming a causal set from a metric manifold $(\mathcal{M}, g_{\mu\nu})$, by randomly *sprinkling* points into $\mathcal{M}$. The causal, light-cone structure imposed by $g_{\mu\nu}$ then determines the causal relationships between the sprinkled points. Care must be taken as to the nature of the distribution of the points within $\mathcal{M}$, to ensure that the resulting sprinkling is Lorentz invariant.

1.3.2. Sprinkling. We may say that the manifold $(\mathcal{M}, g_{\mu\nu})$ ‘emerges from’ the causal set $\mathcal{C}$ if and only if an appropriately coarse-grained version of $\mathcal{C}$ could have come from sprinkling points into $\mathcal{M}$ at unit density (where unit density is taken with respect to the fundamental unit of volume, of order the Planck volume). Clearly the order relation between elements of $\mathcal{C}$ comes from the causal structure of $\mathcal{M}$, such that $x \prec y \Leftrightarrow y \in J^+(x)$ for $x, y \in \mathcal{C} \subset \mathcal{M}$.

The requirement of unit density sprinkling ensures that spacetime volumes are given by counting the number of elements of $\mathcal{C}$ which they contain. This type of sprinkling is achieved by generating the points of $\mathcal{C}$ using a Poisson process, under which the number of points, N , sprinkled into a region of volume V is directly proportional to V . This Poisson process may be expressed probabilistically: for each small volume $\varepsilon \subset \mathcal{M}$ a point is placed independently, with probability ε . The limit $\varepsilon \rightarrow 0$ then gives the required unit density sprinkling. The probability that there will be n points in a given volume, V , is given by the Poisson distribution:

$$\text{Prob}(n \text{ points in } V) = \frac{(\rho V)^n}{n!} e^{-\rho V} \quad (3)$$

where ρ is a fundamental density, of Planckian magnitude.

The *random* distribution of elements of $\mathcal{C}$ throughout $\mathcal{M}$ is also necessary to ensure Lorentz invariance of the spacetime. Sorkin illustrates the problem of an ordered or regular embedding by considering a ‘diamond lattice’ in 1+1-dimensional Minkowski spacetime, $\mathbb{M}^2$ [Sorkin (1991)]. The lattice consists of all points with integer values of null coordinates $u = t - x$ and $v = t + x$. Whilst this may appear to be uniform lattice in a certain frame, under some boost $u \rightarrow \lambda u$, $v \rightarrow \frac{v}{\lambda}$, the distribution ‘shifts’, with a very high density of

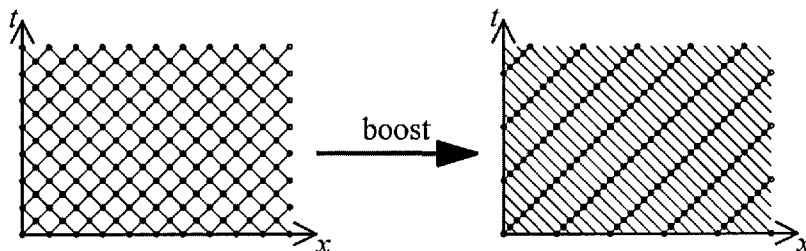


FIGURE 2. Standard 1+1-dimensional spacetime diagram (left) showing a ‘diamond lattice’ arrangement of causal elements in a Minkowski space-time. Boosting this frame results in the situation shown right, in which orthogonal directions show differing linear densities of elements, breaking Lorentz invariance.

points in certain directions, say $u = \text{constant}$, and large gaps between those lines. This is illustrated in Figure 2. It is clear that in this context, the *uniformity* of a distribution of $\mathcal{C}$ is ensured by its Lorentz invariance, rather than any order which privileges a specific frame.

A sprinkling based on a random Poisson process allows a proof of a limiting version of the Hauptvermutung [Bombelli (1989)]. Let $\mathcal{C}_1$ be the result of such a sprinkling into $\mathcal{M}$. Then $\mathcal{C}_2, \mathcal{C}_3, \dots, \mathcal{C}_j$ are a series of sprinklings of additional points at increasing densities, and we have $\mathcal{C}_m \subseteq \mathcal{C}_n \forall m < n, n > 1$. As further points are added, the fundamental length scale is being taken to zero, resulting in $(\mathcal{M}, g_{\mu\nu})$ in the limit as the number of sprinklings $\mathcal{C}_j$ is taken to infinity. Work is being done (by Bombelli, amongst others) to remove the requirement of this limiting process, since the fundamental volume is non-zero for a true discrete spacetime. The result sought is described in terms of the properties of faithful embeddings, discussed below.

1.3.3. Embedding. Whilst sprinkling involves the creation of a causal set through the random distribution of points in a continuous manifold, embedding is the inclusion of an *existing* causal set within a continuous manifold. The important concept is that of a *faithful embedding* [Dou (2003)]. If $\mathcal{M}$ is a Lorentzian manifold and $\mathcal{C}$ a causal set, then a faithful embedding of $\mathcal{C}$ into $\mathcal{M}$ is an injection $f : \mathcal{C} \rightarrow \mathcal{M}$ of the causal set into the manifold, for which:

- (1) the causal relations induced by the embedding agree with those of $\mathcal{C}$ itself,
i.e. $x \prec y \Leftrightarrow x \in J^-(y)$,
- (2) the embedded points are distributed with unit density,
- (3) the characteristic length over which the geometry varies is everywhere much greater than the mean spacing between the embedded points,

$(\mathcal{M}, g_{\mu\nu})$ is said to be *associated* to $\mathcal{C}$.

It may be the case that no $(\mathcal{M}, g_{\mu\nu})$ exists into which $\mathcal{C}$ can be embedded and, indeed, one would expect this to be the case for most causal sets. However, if such an $\mathcal{M}$ does exist, it is essentially unique. That is, for a pair of faithful embeddings, $f_1 : \mathcal{C} \rightarrow (\mathcal{M}_1, g_1)$ and $f_2 : \mathcal{C} \rightarrow (\mathcal{M}_2, g_2)$, there is a diffeomorphism $h : \mathcal{M}_1 \rightarrow \mathcal{M}_2$, where $f_2 = h \circ f_1$, which is an approximate isometry of g_1 to g_2 [Bombelli (1987)]. It is a proof of this statement which would remove the need for the limiting case analysis discussed at the end of the section on sprinklings, above.

1.3.4. Coarse Graining. It might be the case that small-scale fluctuations in the causal order of some causal set $\mathcal{C}$, will prevent its being embedded in any manifold [Bombelli (1987)]. To address this issue, we introduce the concept of *coarse graining*. Let $\mathcal{C}'$ be a coarse graining of $\mathcal{C}$. Then $\mathcal{C}'$ is a subset of $\mathcal{C}$ with an order relation inherited from $\mathcal{C}$. Furthermore, in order that $\mathcal{C}'$ be a ‘larger scale view’ of $\mathcal{C}$, $\mathcal{C}'$ must be ‘representative’ of $\mathcal{C}$, as it would be if, for example, each element of $\mathcal{C}$ was included in $\mathcal{C}'$ with probability p . In this case, the coarse graining would preserve only those features of $\mathcal{C}$ with a characteristic volume scale of order $\frac{1}{p}$ or larger.

This change of scale can lead to the ‘washing out’ of smaller scale structures. A causal set may contain dimensions compactified on, say S^1 , which are ‘lost’ when coarse graining is applied on a scale which is large in comparison to that characterizing the internal dimensions. This remains the case even in those instances where the original causal set, $\mathcal{C}$, *can* be faithfully embedded in some manifold $\mathcal{M}$: coarse graining then leaves $\mathcal{C}'$ with a simpler topology than $\mathcal{C}$, and, possibly, a different dimensionality. Similarly, topological features (holes, handles, etc.) may be lost in the coarse graining procedure.

1.3.5. *Geometry and Topology.* It is known that the causal structure of spacetime determines the great majority of information required to specify the metric upon that spacetime [Hawking (1976), Malament (1977)], and hence the gravitational field tensor (and see [Sorkin (2002)] at p. 2 for further references). Causal structure determines the metric up to a multiplicative, conformal factor, Ω , or equivalently, the ‘volume element’ $\sqrt{-g}d^4x$. However, in a discrete spacetime, the relationship between volume and the number of elements contained within that volume allows the conformal factor to be evaluated, and the full metric to be determined. As Sorkin puts it:

“In [a successful theory based on causal sets] the single relationship $\prec$ would unite within itself the topology, the differential structure, the metrical geometry, and of course the causal structure of spacetime.” [Sorkin (1991)]

We have already mentioned, in §1.1.2, that a discrete spacetime explains the Lorentzian signature of the metric.

Dou provides a more detailed description of the relationship between causal structure and geometry [Dou (1999)] which we summarize here. We take our spacetime to be a connected C^∞ Hausdorff manifold, $\mathcal{M}$, with a metric of Lorentzian signature, g , and time orientation. The causal relations on the manifold can be used to define the Alexandrov topology on $\mathcal{M}$, in which a set is open if and only if it is the union of one or more sets of the form $\mathcal{I}^+(x) \cap \mathcal{I}^-(y)$, $x, y \in \mathcal{M}$. (Here, $\mathcal{I}^\pm(x)$ are the chronological future/past of x , for which see §2.2) Any set which is open in the Alexandrov topology will be open in the manifold topology, but the reverse is not true. We may, however, impose the strong causality condition: $\forall p \in \mathcal{M}$ and every neighbourhood O of p , $\exists$ a neighbourhood N of p contained in O such that no causal curve intersects N more than once. The Alexandrov topology then corresponds with the manifold topology of $\mathcal{M}$, since $\mathcal{M}$ can be covered by local causal neighbourhoods. Thus the topology of spacetime is coded by its causal structure. This statement can be strengthened, and the strong causality condition relaxed: see [Dou (1999)] at pp. 7-8. Dou also outlines the extraction of topological information from a causal set, through the construction of simplicial complices, in [Dou (1999)] at pp. 25-26.

1.3.6. *Dimension of a Causal Set.* If it is indeed possible, as seems likely, to retrieve a macroscopic, continuous manifold from a causal set, then one may ask what the continuum dimension of such a manifold might be. Sorkin gives details of three estimators which assign a dimension to some interval I in a causal set $\mathcal{C}$, where $I \approx A$ for some interval A in Minkowski space $\mathbb{M}^d$ [Sorkin (2002)]:

- the *Myrheim-Meyer, or fractal, dimension* [Myrheim (1978), Meyer (1993)]. We define $N = |I|$, the number of elements in I , and let R be the number of relations in I , that is, the number of pairs x, y with $x \prec y$. Let $f(d) = \frac{3}{2} \binom{3d/2}{d}^{-1}$. Then $f^{-1} \left(R / \binom{N}{2} \right)$ is a good estimate of d for $N \gg (27/16)^d$.
- the *Midpoint scaling dimension*. We take $I = \text{interval}(a, b)$, and $m \in I$ to be the ‘mid-point’ defined to maximize $N' = \min \{ |\text{interval}(a, m)|, |\text{interval}(m, b)| \}$. Then $\log_2 \left(\frac{N}{N'} \right)$ estimates d .
- Let K be the total number of chains in I , then $\frac{\ln N}{\ln K}$ estimates d . This method is only accurate for exponentially large N .

In [Bombelli (1987)] a sketch argument is given to show that the space-time dimensionality must be the same for all faithful embeddings of a given causal set, $\mathcal{C}$. Let $f_1 : \mathcal{C} \rightarrow (\mathcal{M}_1, g_1)$ be a faithful embedding of dimension d_1 , and let $\mathcal{A}$ be an Alexandrov subset of $\mathcal{C}$, which is ‘small’ in that its image under f_1 is on a scale which is small with respect to any curvature of $\mathcal{M}_1$. Then the corresponding Alexandrov neighbourhood $\mathcal{A}_1 \subset \mathcal{M}_1$ will also be ‘small’, and approximately isometric to an Alexandrov neighbourhood in $\mathbb{M}^{d_1}$, otherwise its geometry would occur on length scales incompatible with a faithful embedding. It can be shown that such ‘small’ $\mathcal{A}$ cannot be embedded in fewer than d_1 flat dimensions. But then consider a second embedding of $\mathcal{C}$, f_2 , which will have some $\mathcal{A}_2$ with flat dimension d_2 . We require $d_2 \geq d_1$ (since the image, $\mathcal{A}_2$, of the Alexandrov subset $\mathcal{A}$ must still be approximately flat), but by symmetry $d_1 \geq d_2$ and hence $d_1 = d_2$.

As remarked in §1.3.4, the dimensionality of a causal set may vary with scale since compactified dimensions at small scales may be ‘washed out’ by the coarse graining process.

For a more detailed analysis of causal set dimensionality see [Dou (1999)] §§1.3.1-1.3.2. In particular, it may be noted that certain techniques used to determine the fractal dimension of a causal set can also be used to yield values for scalar curvatures.

1.3.7. Volumes and Distances. In Minkowski space the volume, V , of spacetime bounded by two causally connected events a and b is given by,

$$V = \frac{\pi \tau_{ab}^4}{24} \quad (4)$$

where τ_{ab} is the proper time between events a and b [’t Hooft (1979)]. In order to use this expression with causal sets, one needs to relate τ_{ab} to the number of links in the longest chain between a and b . V is then the number of elements in this region of spacetime. (We note that the dynamic nature of spacetime requires some consideration of the expectation value of τ , $\langle \tau \rangle$, rather than τ itself.)

The geodesic distance between two elements, a and b , of a causal set may be estimated by counting the number of elements, N_{ab} , in the longest chain between them. A maximal chain in this sense is analogous to a timelike geodesic, which, similarly, maximizes the proper-time between its endpoints. For Minkowski spacetime, work has been done to compare length defined in this way with the geodesic length of general relativity, and it is known that N_{ab} converges rapidly to a multiple of proper time, τ , as τ becomes large. The coefficient of proportionality is dependent upon the dimension, d , of the $\mathbb{M}^d$ spacetime, and is known exactly for $d = 1$ and $d = 2$. For higher values of d , the coefficient is known to fall within certain bounds. However, these results stand primarily in flat spacetimes, and their usefulness in curved spacetime has yet to be shown.

1.4. Causal Set Dynamics.

1.4.1. Overview. Just as general relativity is described by the dynamics of classical spacetime itself, so an understanding of the dynamics of the structure underlying that classical approximation will be necessary to construct a theory of quantum gravity. Thus, it will be necessary for the programme of causal sets to move beyond mere kinematics, to address this issue of dynamics. Some progress has been made in this direction, which, for our purposes here, we will only outline in the broadest terms, drawing mostly from [Reid (1999)].

However, Dou provides a detailed discussion of causal set dynamics in [Dou (1999)], §2, pp. 35-52.

The dynamics of causal sets seeks to provide a formal understanding of the formation and subsequent behaviour of such sets, and some insight into how such behaviour informs our perception of space and time. Time itself is discrete in this case, and it is not possible to write down some Hamiltonian generator of time-evolution; nor is it clear upon what Hilbert space any such Hamiltonian would act. Drawing on our present formulation of quantum dynamics, the best candidate to provide a dynamical framework for causal sets would seem to be the path integral approach, as (i) it deals directly with the propagation of one event configuration to another, and (ii) it can be applied as the discrete formalism of a sum over histories. So, to determine the propagator, U , between events a and b , we sum over paths, weighted by an amplitude function of the action, S , for each path:

$$U(a, b) \sim \sum_{\text{all paths}} \exp \left\{ \frac{iS}{\hbar} \right\} . \quad (5)$$

However, the formulation of a suitable amplitude function for causal sets is unclear. Sorkin suggests requirements which may help to determine such a function [Sorkin (1991)]: clearly, the dynamics must have a classical limit which appears to be a smooth manifold. Classical limits arise from the interference of many histories, and hence Lorentzian manifolds should be ‘stationary phase’ points of the causal set amplitude. The amplitude function of causal sets far from manifolds would then be ‘extremely rapidly varying’. However, as Sorkin points out, these amplitude functions cannot have a derivative, as a causal set cannot be varied continuously. In addition there is the question of locality: the effective action of the family of causal sets corresponding to some Lorentzian manifold should look like the integral of some locally defined quantity in that manifold.

Work by Rideout and Sorkin have led to the development of a general *classical* dynamics, under which causal sets grow sequentially, one element at a time [Rideout (2000)]. The element which takes the set from n to $n + 1$ elements has probability p_n of being unrelated to any of the existing n elements, such that:

$$\frac{1}{p_n} = \sum_{k=0}^n \binom{n}{k} t_k \quad (6)$$

with the restriction that $t_k \geq 0$. Freedom remains to explore many different growth models and their resulting causal sets, although the number of possibilities is limited by the requirements of discrete general covariance (that the labelling of elements be ‘pure gauge’ and play no part in the final probabilities) and Bell causality (to prevent the influence of an element on other elements space-like separated from it). See also [Sorkin (2002)] at pp. 12-14.

1.4.2. *The Cosmological Constant.* As early as 1991 Sorkin predicted that the cosmological constant, Λ , should be have some non-zero but very small value, as a consequence of the quantum nature of the stochastic growth process giving rise to the causal set underlying spacetime [Sorkin (1991)]. Subsequent astrophysical observations seem to have confirmed that the expansion of the universe is accelerating, as a result of a non-zero cosmological constant (or ‘dark energy’). Sorkin’s original argument is based on the supposition of a small Λ as a statistical effect of the large number of elements making up the universe. This being a statistical effect, whatever cancellations acted to keep Λ small should be subject to statistical variation, suggesting,

$$\Lambda \sim \frac{1}{\sqrt{N}} \tag{7}$$

where N is the number of elements in the universe, at the appropriate epoch. Assuming volume elements of order the Planck scale, the current observable universe has $N \approx 10^{240}$, giving $\Lambda \approx 10^{-120}$ in natural units, which is consistent with recent observations. A more refined version of this argument can be found in [Sorkin (2002)] at pp. 14-15.

2. BLACK HOLES: GENERAL RESULTS

2.1. Astrophysical Description. In 1783 the rector of St Michael and All Angels church at Thornhill, England, one John Michell, suggested that a body with sufficient gravitational attraction may prevent the escape of light from its surface to “infinity” [Coles (1998), Hart-Davis (1996)]. Modern interest in this possibility was kindled by Schwarzschild’s solution of Einstein’s equations of general relativity, which suggest that a body of mass M within some ‘Schwarzschild radius’ $R_S(M)$, will prevent the escape of light from its surface to spatial infinity. However, the existence of such a black hole requires an exceptionally high mass density. For example, the Earth, with mass $M_\oplus$ and radius $R_\oplus$, has $R_S \approx 1\text{cm} \ll R_\oplus$.

This density may occur, in the case of very massive black holes, through stellar collisions, particularly in the centres of large star clusters or galaxies [Shapiro (1983)]. For less massive bodies, black holes are an end state of stellar evolution: the end of energy production leads to gravitational collapse and the attainment of critical density. It is possible that the degeneracy pressure of electrons may halt the collapse to a black hole, resulting in a white dwarf, but the limiting mass for white dwarf formation (the Chandrasekhar mass) is approximately $1.4M_\odot$ (where $M_\odot$ is the mass of our sun). In a more massive star this electron degeneracy pressure is overcome by gravitational attraction, leading to the combination of electrons and protons, and the formation of a neutron star. Neutron degeneracy pressure may now prevent further collapse, but equation of state calculations based on general relativity and strong force interactions give a maximum mass for a neutron star of approximately $3M_\odot$. Beyond this mass, degeneracy pressure cannot prevent gravitational collapse to a theoretical singularity.

A succinct account of the astrophysical processes involved in black hole formation can be found in [Peacock (1999)].

2.2. Singularity Theorems. An examination of the causal structure of classical spacetime gives rise to series of theories concerning the existence and nature of spacetime singularities, as outlined in [Hawking (2000)]. These singularity theorems demonstrate the existence of singularities (which we take to be the occurrence of incomplete geodesics that cannot be extended to infinite values of the affine parameter), arising from the imposition

of certain restrictions upon that spacetime. These restrictions are believed to be physically reasonable. Furthermore, the singularity theorems of Penrose and Hawking show that geodesic incompleteness is a generic feature of gravitational collapse (as opposed to only the spherically symmetric case). We will not give further details here, but see, *inter alia*, [Townsend (1997)].

2.3. Uniqueness Theorems.

2.3.1. *Definitions.* Consider a spacetime $\mathcal{M}$:

- $\mathcal{M}$ is *asymptotically flat* if the regions $\mathcal{I}^\pm$ (future/past null infinities) and I_0 (space-like infinity) of $\mathcal{M}$ have the same structure as Minkowski spacetime. (See [Townsend (1997)], pp. 46-47, for a more rigorous definition.)
- $\mathcal{M}$ is *stationary* if and only if there exists a Killing vector field, $k = \frac{\partial}{\partial t}$, which is asymptotically timelike [Townsend (1997)]. In other words, there exists an isometry group with a timelike Killing vector at $\mathcal{I}^\pm$ [Hawking (1973)]. The metric is given by

$$ds^2 = g_{00}dt^2 + 2g_{0i}dtdx^i + g_{ij}dx^i dx^j \quad (8)$$

Here, and throughout, we take the Minkowski metric, $\eta_{\mu\nu}$, to have signature $\{-, +, +, +\}$.

- $\mathcal{M}$ is *static* if it is a stationary spacetime which also possesses a time reflection symmetry [Jacobson (1996)]. If $\mathcal{M}$ is static then the metric components $g_{0i} = 0$, giving a metric outside any horizon:

$$ds^2 = g_{00}dt^2 + g_{ij}dx^i dx^j \quad . \quad (9)$$

- $\mathcal{M}$ is *axisymmetric* if there exists an axial Killing vector field, $m = \frac{\partial}{\partial \phi}$, which is asymptotically spacelike, and for which all orbits are closed [Townsend (1997)]. This can be considered a rotational symmetry, and we have $\phi \in [0, 2\pi)$. [Hawking (1973)] shows that a stationary black hole must be axisymmetric. A stationary, axisymmetric black hole is *k - ϕ orthogonal* if the 2-planes spanned by the Killing vector fields $k = \frac{\partial}{\partial t}$ and $m = \frac{\partial}{\partial \phi}$ are orthogonal to a family of 2-dimensional surfaces: this is the case for stationary, axisymmetric solutions to the vacuum Einstein or Einstein-Maxwell equations [Wald (1999)].

2.3.2. *Theorems.* We state the following without proof, which may, however, be found in [Mazur (2000)].

- *Birkhoff's theorem:* [Birkhoff (1923)] If a solution of Einstein's field equations *in vacuo* gives a spherically symmetric spacetime, $\mathcal{M}$, then $\mathcal{M}$ is static and Schwarzschild, with metric

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 \quad (10)$$

in units where $G = c = 1$.

- *Israel's theorem:* [Israel (1967)] If $\mathcal{M}$ is an asymptotically flat, static, vacuum spacetime which is non-singular on and outside an event horizon (for which, see below), then $\mathcal{M}$ is Schwarzschild. Birkhoff's requirement of spherical symmetry is removed.
- *Carter-Robinson theorem:* [Carter (1971), Robinson (1975)] If $\mathcal{M}$ is an asymptotically flat, stationary and axisymmetric vacuum spacetime which is non-singular on and outside an event horizon, then $\mathcal{M}$ is parametrized by mass, M , and angular momentum, J , and only those two parameters.

Birkhoff's theorem may be extended to an Einstein-Maxwell system, in which a black hole may possess electric charge, Q . Then, the only spherically symmetric solution is the Reissner-Nordström solution [Reissner (1916), Nordström (1918)],

$$ds^2 = - \frac{\Delta}{r^2} dt^2 + \frac{r^2}{\Delta} dr^2 + r^2 d\Omega^2 \quad (11)$$

where $\Delta = r^2 - 2Mr + Q^2$.

Similarly, the Israel and Carter-Robinson theorems may be extended, so that any stationary black hole spacetime must have a Kerr-Newman solution [Kerr (1963), Newman (1965)] of the form:

$$\begin{aligned} ds^2 = & - \frac{(\Delta - a^2 \sin^2 \theta)}{\Sigma} dt^2 - 2a \sin^2 \theta \frac{(r^2 + a^2 - \Delta)}{\Sigma} dt d\theta \\ & + \left(\frac{(r^2 + a^2)^2 - \Delta a^2 \sin^2 \theta}{\Sigma} \right) \sin^2 \theta d\phi^2 + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 \end{aligned} \quad (12)$$

using notation as [Townsend (1997)] with Boyer-Lindquist [Boyer (1967)] coordinates in which $\Sigma = r^2 + a^2 \cos^2 \theta$ and $\Delta = r^2 - 2Mr + a^2 + Q^2$, where $a = \frac{J}{M}$.

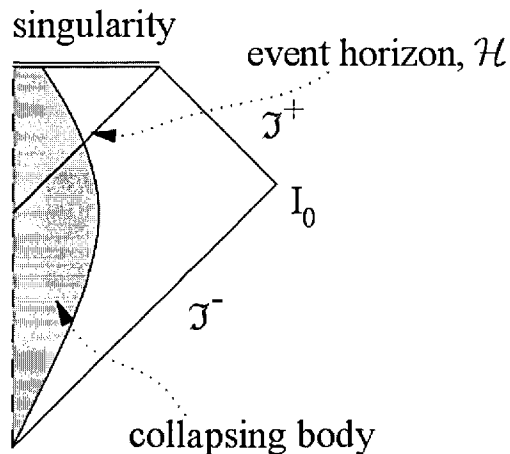


FIGURE 3. Carter-Penrose diagram of a spherically symmetric collapsing body producing a black hole with Schwarzschild metric. (After [Hawking (1975), Townsend (1997)].)

2.3.3. No Hair Theorem. The uniqueness theorems are summarized in the *No Hair Theorem*, which states that all black hole solutions are characterized by only three parameters: mass, M , angular momentum, J , and electric charge, Q (excluding considerations of magnetic monopoles and, possibly, Yang-Mills fields: [Hawking (2000)] at p. 39). Within the families of black holes discussed above, the initial nature of the matter forming the black hole is irrelevant for its physical description (although in the case of an initially spherically asymmetric system, the Kerr-Newman solution is only valid in the asymptotic future).

2.4. The Schwarzschild Black Hole. For a chargeless, static black hole $Q = J = 0$. Thus, the Kerr-Newman solution, eq.12, reduces to the Schwarzschild metric, eq.10, with only the mass parameter, M , remaining.

The apparent singularity at $r = r_H = 2M$ is a consequence of the coordinate system used. Defining the Regge-Wheeler coordinate $r^* = r + 2M \ln \left| \frac{r-2M}{2M} \right|$, we rewrite the Schwarzschild metric as

$$ds^2 = - \left(1 - \frac{2M}{r} \right) dv^2 + 2drdv + r^2 d\Omega^2 \quad (13)$$

where $v = t + r^*$ is the ingoing radial null coordinate and (v, r, Ω) are the ingoing Eddington-Finkelstein coordinates.

This metric is analytically extended, from $r > 2M$ to all $r > 0$ and is not singular at r_H . However, the solution remains non-analytic at $r = 0$. We have an irremovable singularity,

with an associated trapped surface, the boundary of which is formed by an *event horizon*, $\mathcal{H}$, at $r_{\mathcal{H}}$. This is shown in Figure 3.

2.4.1. Event Horizon. $\mathcal{H} = J^-(\mathcal{I}^+)$ describes the boundary of the set of points in $\mathcal{M}$ which can ‘send’ causal (null) curves to future infinity. This boundary, $\mathcal{H}$, is a null hypersurface: a spacelike slice through $\mathcal{M}$ will intersect $\mathcal{H}$ at a closed 2-surface, which is the event horizon of a black hole (see [Misner (1973)] at §34.4 for further details). This defines the black hole region of $\mathcal{M}$, $\mathcal{B} = \mathcal{M} - J^-(\mathcal{I}^+)$, and $\mathcal{H}$ is then $\partial\mathcal{B}$, the boundary of $\mathcal{B}$.

2.4.2. Cosmic Censorship Hypothesis. We quote from [Townsend (1997)]:

“Naked singularities cannot form from gravitational collapse in an asymptotically flat spacetime that is non-singular on some initial spacelike hypersurface (Cauchy surface).”

The Schwarzschild metric is clearly an asymptotically flat spacetime since, from eq.10, $\lim_{r \rightarrow \infty} [ds^2] = -dt^2 + dr^2 + r^2 d\Omega^2 = \eta_{\mu\nu} dx^\mu dx^\nu$.

2.5. Black Hole Parameters. We consider the definitions of the quantities M , J , and Q , required to specify a black hole. Whilst we will not treat charged black holes in later sections, we include some discussion of Q at this point, as it informs the formulation of the definition of M . This subsection is based closely on [Townsend (1997)], which contains more detailed proofs of the results given below.

2.5.1. Charge. Let Σ be a spacelike hypersurface with boundary $\partial\Sigma$. Then the volume element is a non-spacelike 1-form, dS_μ , and the charge, Q , bounded by $\partial\Sigma$ is given by

$$Q = \int_{\Sigma} dS_\mu j^\mu \quad (14)$$

$$= \int_{\Sigma} dS_\mu D_\nu F^{\mu\nu} \quad \text{by Maxwell's eq.} \quad (15)$$

$$= \frac{1}{2} \oint_{\partial\Sigma} dS_{\mu\nu} F^{\mu\nu} \quad \text{by Gauss' law} \quad (16)$$

where j^μ is the current density 4-vector and $dS_{\mu\nu}$ is the 2-form area element of $\partial\Sigma$.

2.5.2. Energy (Mass). The interaction between matter and gravitation, and its concomitant exchange of energy, means that there is no equivalent to local charge conservation

[Weinberg (1972)]. It is possible, however, to calculate the total energy of an asymptotically flat spacetime over a surface integral at infinity, since there it possesses a timelike Killing vector field.

We have, in the case of asymptotic flatness, $\lim_{r \rightarrow \infty} g_{\mu\nu} = \eta_{\mu\nu}$, and we take $h_{\mu\nu} = g_{\mu\nu} - \eta_{\mu\nu} = \mathcal{O}\left(\frac{1}{r}\right)$. Hence, we linearize Einstein's equations, in the asymptotic limit, to give the Pauli-Fierz equation,

$$\square h_{\mu\nu} + h_{,\mu\nu} - 2h_{(\mu,\nu)} = -16\pi G \left(T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}T \right) . \quad (17)$$

From this we derive the *ADM energy* formula for an asymptotically flat spacetime:

$$E = \frac{1}{16\pi G} \oint_{\infty} dS_i (\partial_j h_{ij} - \partial_i h_{jj}) \quad (18)$$

$$\text{or } E = -\frac{1}{8\pi G} \oint_{\infty} dS_i \partial_i h_{00} \quad (19)$$

$$\text{or } E = -\frac{1}{8\pi G} \oint_{\infty} dS_{\mu\nu} D^{\mu} k^{\nu} \quad (20)$$

where $k = \frac{\partial}{\partial t}$ is asymptotically Killing.

Alternatively, one may define a *Komar integral*, $Q_{\xi}(V)$, for any Killing vector field ξ , where V is a volume of spacetime on a spacelike hypersurface, with boundary ∂V :

$$Q_{\xi}(V) = \frac{c}{16\pi G} \oint_{\partial V} dS_{\mu\nu} D^{\mu} \xi^{\nu} \quad (21)$$

$$= \frac{c}{8\pi G} \int_V dS_{\mu} D_{\nu} D^{\mu} \xi^{\nu} \quad \text{by Gauss' law} \quad (22)$$

where c is some constant. Fixing $c = -2$ in eq.21 recovers eq.20. By analogy with eq.14 one may define a 'conserved current' $j^{\mu}(\xi)$, with $\partial_{\mu} j^{\mu}(\xi) = 0$. Then,

$$Q_{\xi}(V) = \int_V dS_{\mu} j^{\mu}(\xi) \quad (23)$$

with $j^{\mu}(\xi) = c \left(T^{\mu}_{\nu} \xi^{\nu} - \frac{1}{2} T \xi^{\mu} \right)$.

2.5.3. Angular Momentum. In an axisymmetric spacetime we may take $\xi = m = \frac{\partial}{\partial \phi}$ and put $c = 1$. Then the total angular momentum of an asymptotically flat spacetime is given by eq.21, with ∂V a 2-sphere at infinity (see [Townsend (1997)] at pp. 98-99):

$$J = \frac{1}{16\pi G} \oint_{\infty} dS_{\mu\nu} D^{\mu} m^{\nu} . \quad (24)$$

3. BLACK HOLES: CLASSICAL AND QUANTUM MECHANICS.

3.1. The Four Laws of Black Hole Mechanics. In their 1973 paper [Bardeen (1973)], Bardeen, Carter and Hawking proposed an analogy between certain quantities associated with black hole solutions and the four laws of thermodynamics. At the same time, Bekenstein published several observations relating to black hole entropy [Bekenstein (1973), Bekenstein (1974)]. The paper by Bardeen et al. deals with a stationary axisymmetric black hole orbited by rings of matter. Here we will give the results of that paper, but the derivations we give will assume an isolated black hole with no external matter.

3.2. The Zeroth Law.

“The surface gravity, κ , of a stationary black hole is constant over the event horizon.” [Bardeen (1973)]

3.2.1. The Analogy. The zeroth law of thermodynamics has: if two bodies Σ_1 and Σ_2 are in thermal equilibrium, and bodies Σ_2 and Σ_3 are also in thermal equilibrium, then Σ_1 and Σ_3 are in thermal equilibrium. The implication is that there is a function of state of the system which the three bodies have in common. This function of state is temperature, T . Thus, there is a suggestion of an analogy between surface gravity, κ , and temperature, T .

3.2.2. Definitions.

- Let $S(x)$ be a smooth function of spacetime coordinates x^μ , then a hypersurface $\mathcal{N}$ describe by $S(x) = \text{constant}$ is a *null hypersurface* if $l^2 = 0$, where l is a vector field normal to $\mathcal{N}$, given by

$$l = f'(x) (g^{\mu\nu} \partial_\nu S) \partial_\mu \quad (25)$$

where f' is some arbitrary non-zero function. [Townsend (1997)]

- A *Killing horizon*, $\mathcal{K}$, is a null hypersurface whose null generators coincide with the orbits of a one-parameter group of isometries, implying the existence of a Killing vector field, ξ , normal to $\mathcal{K}$ [Wald (1999)].

3.2.3. Derivation. It is clear that a vector, t , is defined as tangent to $\mathcal{N}$ if $t \cdot l = 0$. But since $\mathcal{N}$ is null, $l \cdot l = 0$, and hence l is a tangent vector: $l^\mu = \frac{dx^\mu}{d\lambda}$ for some null curve $x^\mu(\lambda)$ in $\mathcal{N}$. It can be shown that $x^\mu(\lambda)$ are geodesics (the proof is given in [Townsend (1997)])

at p. 28), and the function f' of eq.25 can be chosen such that $l \cdot Dl^\mu = 0$. In this case λ is an affine parameter, and the null geodesics $x^\mu(\lambda)$ are the *generators* of $\mathcal{N}$.

If $\mathcal{N}$ is a Killing horizon, $\mathcal{K}$, with normal l such that $l \cdot Dl^\mu = 0$, then on $\mathcal{K}$, $\xi = fl$ for some function f . Thus,

$$\begin{aligned} \xi \cdot D\xi^\mu &= fl \cdot D(ff'g^{\mu\nu}\partial_\nu S) \\ &= f(l^\rho\partial_\rho f)f'g^{\mu\nu}\partial_\nu S + f(l \cdot Dl^\mu) \end{aligned} \quad (26)$$

Now, $l \cdot Dl^\mu = 0$, and $l^\rho\partial_\rho f = f\left(l^\rho\frac{1}{f}\partial_\rho f\right) = fl^\rho\partial_\rho \ln|f| = f(l \cdot \partial \ln|f|)$. Hence,

$$\begin{aligned} \xi \cdot D\xi^\mu &= f^2(l \cdot \partial \ln|f|)f'g^{\mu\nu}\partial_\nu S \\ &= (\xi \cdot \partial \ln|f|)\xi^\mu \\ \Rightarrow \xi \cdot D\xi^\mu &= \kappa\xi^\mu \end{aligned} \quad (27)$$

where $\kappa = \xi \cdot \partial \ln|f|$ is *surface gravity*.

It can be shown (see [Schutz (1980)]) that $\xi_{[\mu}D_\nu\xi_{\rho]}|_{\mathcal{K}} = 0$ (where $|_{\mathcal{K}}$ indicates evaluation on the surface $\mathcal{K}$). Hence, for a Killing vector field, ξ , with $D_\mu\xi_\nu = D_{[\mu}\xi_{\nu]}$,

$$\xi_\rho D_\mu\xi_\nu|_{\mathcal{K}} + (\xi_\mu D_\nu\xi_\rho - \xi_\nu D_\mu\xi_\rho)|_{\mathcal{K}} = 0 \quad (28)$$

Then ([Townsend (1997)]) ,

$$\kappa^2 = -\frac{1}{2}(D^\mu\xi^\nu)(D_\mu\xi_\nu)\Big|_{\mathcal{K}} \quad (29)$$

We may now consider the situation described in [Bardeen (1973)], which concerns a stationary, axisymmetric black hole solution. In this case there exist time translation and rotational Killing vector fields, $k = \frac{\partial}{\partial t}$ and $m = \frac{\partial}{\partial \phi}$ respectively. The event horizon, $\mathcal{H}$, is a Killing horizon, by a result of Carter [Carter (1973)] showing that a stationary, axisymmetric black hole (which is $k - \phi$ orthogonal) possesses an associated Killing field $\xi = t + \Omega m$ (for some constant Ω), which is normal to the event horizon. This is also true for any stationary black hole, [Hawking (1973)]. This Killing horizon has tangent vectors $l^\mu = k^\mu + \Omega_{\mathcal{H}}m^\mu$, where $\Omega_{\mathcal{H}}$ is the angular velocity of the black hole, and is the same at all points of the horizon. l is itself Killing, as a linear superposition of Killing vectors.

A result obtained through the use of Raychaudhuri's equation (see §3.4.3), gives $R_{\mu\nu}l^\mu l^\nu|_{\mathcal{H}} = 0$, and we have $l^2|_{\mathcal{H}} = 0$, so by Einstein's equations,

$$\begin{aligned} R_{\mu\nu}l^\mu l^\nu|_{\mathcal{H}} &= \frac{1}{2}g_{\mu\nu}l^\mu l^\nu R|_{\mathcal{H}} - 8\pi GT_{\mu\nu}l^\mu l^\nu|_{\mathcal{H}} \\ &= -8\pi GT_{\mu\nu}l^\mu l^\nu|_{\mathcal{H}} \quad . \end{aligned} \quad (30)$$

Hence,

$$0 = -T_{\mu\nu}l^\mu l^\nu|_{\mathcal{H}} \equiv -L_\mu l^\mu|_{\mathcal{H}} \quad (31)$$

with $L = (-T_\nu^{\mu\nu})\partial_\mu$ which is clearly tangent to $\mathcal{H}$. Since this is the case, L can be expressed on a basis of tangent vectors to $\mathcal{H}$: $L = al + b\tau + b'\tau'$. But l is normal to $\mathcal{H}$ and hence $l \cdot \tau = l \cdot \tau' = 0$, so L must be spacelike or, when $b = b' = 0$, null. However, we impose the *dominant energy condition*¹ on $T_{\mu\nu}$ and thus we require L to be timelike or null. L must therefore be null, with $L \propto l$. If this is the case, L is Killing, from which it can be shown that

$$0 = l_{[\rho}\partial_{\sigma]}\kappa|_{\mathcal{H}} \quad (32)$$

using results given in [Townsend (1997)].

Then, eq.32 $\Rightarrow \partial_\sigma \kappa \propto l \Rightarrow t \cdot \partial \kappa = 0$ for any vector t tangent to $\mathcal{H}$.

Hence, κ is constant on $\mathcal{H}$ (providing the dominant energy condition is satisfied). We quote here without proof (see [Townsend (1997)] at pp. 32-33) the result

$$\kappa = \frac{1}{4M} \quad (33)$$

for a non-rotating, neutral black hole of mass M .

3.2.4. Discussion. On its own, the observation that κ is constant on $\mathcal{H}$ is not highly suggestive of the $\kappa - T$ analogy. However, it does supply supporting evidence for the equating of κ and T which arises from the first law.

¹The *dominant energy condition* is satisfied if the vector field $z(v)$ is future-directed timelike or null, for all future-directed timelike vector fields v , where $z(v) \equiv -v^\mu T_\mu^\nu \partial_\nu$. (Alternatively, per [Hawking (1973)], the dominant energy condition has $T^{00} \geq |T^{\mu\nu}| \forall \mu, \nu$.) This condition is satisfied by physically reasonable matter: see [Birrell (1982)] at p. 87, [Townsend (1997)] at p. 99.

3.3. The First Law.

“If a stationary black hole of mass M , charge Q , and angular momentum J , with future event horizon of surface gravity κ , electric surface potential $\Phi_{\mathcal{H}}$ and angular velocity $\Omega_{\mathcal{H}}$, is perturbed such that it settles down to another black hole with mass $M + \delta M$, charge $Q + \delta Q$, and angular momentum $J + \delta J$, then

$$dM = \frac{\kappa}{8\pi} dA + \Omega_{\mathcal{H}} dJ + \Phi_{\mathcal{H}} dQ \quad (34)$$

”[Townsend (1997)], where A is the area of a 2-dimensional spacelike cross-section of the horizon, $\mathcal{H}$, and we use units in which $G = 1$.

This is not the exact formulation given in [Bardeen (1973)], which considers solutions in which stationary matter, other than the electromagnetic field, exists outside the black hole, that is, outside $\mathcal{B} \cup \mathcal{H}$. In this case eq.34 contains extra terms associated with the chemical potential and temperature of the external matter. We will restrict ourselves here to the simpler situation described by Townsend.

3.3.1. The Analogy. The first law of thermodynamics can be expressed by the Gibbs equation as

$$dU = TdS - PdV \quad (35)$$

where dU is the change in internal energy of the system, T is temperature, dS is the change in entropy, P is pressure, and dV is the change in volume. The analogy is drawn between the first terms on the right-hand sides of eq.35 and eq.34, that is, TdS and $\frac{\kappa}{8\pi} dA$, respectively.

In agreement with the zeroth law, it is suggested that κ is analogous to T . Also, it is suggested that A is analogous to S : a proposal which is supported by the second law, given below.

3.3.2. Derivation. Smarr’s formula for the mass of a Kerr black hole has

$$M = \frac{\kappa A}{4\pi} + 2\Omega_{\mathcal{H}} J \quad (36)$$

where terms are as defined above. [Townsend (1997)] provides a derivation of Smarr’s formula at p. 110 ff. For a charged black hole the formula can be extended such that $M = \frac{\kappa A}{4\pi} + 2\Omega_{\mathcal{H}} J + \Phi_{\mathcal{H}} Q$. However, we will continue this derivation only in the case of

black holes without charge, i.e. $Q = 0$, as [Bardeen (1973)]. Then,

$$dM = \frac{\kappa}{8\pi} dA + \Omega_{\mathcal{H}} dJ \quad . \quad (37)$$

(A dimensional proof attributed to G. Gibbons is given in [Townsend (1997)].)

3.3.3. Discussion. Whilst parallels are drawn between temperature, T , and surface gravity, κ , Bardeen et al. consider this to be nothing more than an analogy, given that “the effective temperature of a black hole is absolute zero... [since]... a black hole cannot be in equilibrium with black body radiation at any non-zero temperature” [Bardeen (1973)]. Below, we show that a quantum mechanical treatment of the black hole demonstrates a direct relationship between T and κ which is more than a mere analogy.

3.4. The Second Law. If $T_{\mu\nu}$ satisfies the weak energy condition² and the cosmic censorship hypothesis is true then:

“The area A of the event horizon of each black hole does not decrease with time.” [Bardeen (1973)]

We may state this law, otherwise known as *Hawking’s Area Theorem*, using terms defined above, and following [Hawking (1973)]:

Let $\mathcal{M}$ be a space in which cosmic censorship and the weak energy condition hold, and which develops from a partial Cauchy surface $\mathcal{S}_C(\sigma)$. Let $\mathcal{B}_1(\tau)$ be a black hole on the surface $\mathcal{S}_C(\tau)$, and $\{\mathcal{B}_i(\tau_0)\}$, $i \in \{1, \dots, N\}$, be the set of black holes on an earlier surface $\mathcal{S}_C(\tau_0)$ such that $J^+(\mathcal{B}_i(\tau_0)) \cap \mathcal{B}_1(\tau) \neq \{\}$. Then if $A_i(\sigma)$ is the area of $\partial\mathcal{B}_i(\sigma)$ (the event horizon $\mathcal{H}(\sigma)$ of $\mathcal{B}_i(\sigma)$):

$$A_1(\tau) \geq \sum_{i=1}^N A_i(\tau_0) \quad (39)$$

with equality $\Leftrightarrow N = 1$. See Figure 4. Note that whilst this description refers to the case of merging black holes, the Area Theorem is also applicable to the evolution of an individual black hole, over time.

²The *weak energy condition* has

$$T_{\mu\nu} v^\mu v^\nu \geq 0 \quad (38)$$

where $T_{\mu\nu}$ is the stress-energy 2-form and v^μ is any non-spacelike vector. The energy density, T_{00} , is required to be positive semi-definite, in all frames, which is fulfilled by ‘ordinary’ matter.

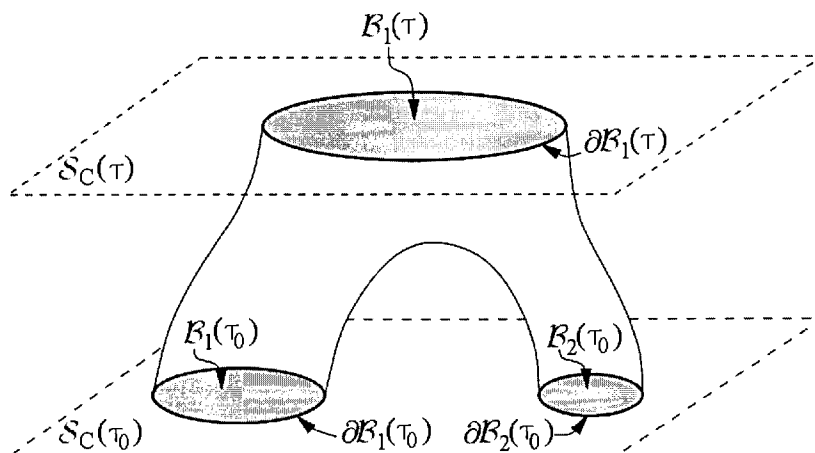


FIGURE 4. Two black holes, $\mathcal{B}_1$ and $\mathcal{B}_2$ combine. According to Hawking's Area Theorem, the surface area of the boundary of the combined holes is greater than the sum of the surface areas of the two separate holes.

3.4.1. *The analogy.* The Carathéodory form of the second law of thermodynamics is given as:

“In the neighbourhood of any state of a system in equilibrium there exist states which are inaccessible by adiabatic processes alone.” [Carrington (1994)]

This is a basis for the formulation of the principle of entropy increase, which has, for a closed system, $\Delta S \geq 0$, with equality if and only if the process causing the entropy change is reversible. The analogy is thus drawn between entropy, S , and the area, A , of $\mathcal{H}$.

3.4.2. *Definitions.*

- A *null geodesic congruence* is a family of null geodesic curves on some manifold $\mathcal{P}$, such that one and only one curve of the family passes through each point on $\mathcal{P}$.
- The *divergence*, ρ , of the congruence is the fractional rate of change of the infinitesimal cross-sectional area δA :

$$\rho := \frac{d}{d\lambda} \ln \delta A \quad (40)$$

where λ is an affine parameter for the null geodesics. We treat the congruence as itself a manifold, here, in our definition of δA : [Townsend (1997)] gives a more rigorous treatment.

3.4.3. *Derivation.* We give only a sketch of the proof.

The Raychaudhuri equation relates the focusing of a null geodesic congruence to the Ricci tensor, $R_{\mu\nu}$. Consider a null geodesic congruence originating from a spacelike 2-surface. Then it can be shown that

$$\frac{d\rho}{d\lambda} = -\frac{1}{2}\rho^2 - \sigma^2 - R_{\mu\nu}t^\mu t^\nu \quad (41)$$

where σ^2 is the (positive) square of the shear tensor of the congruence, and t^μ is the tangent vector to the geodesics. This is the *Raychaudhuri equation*, or *focusing equation* (with zero skew, ω). [Jacobson (1996)]

From eq.40 we have $\rho = \frac{d\delta A}{d\lambda} \frac{1}{\delta A} \Rightarrow \frac{d\delta A}{d\lambda} = \rho\delta A$. Hence, in order to prove the second law we must show that $\rho \geq 0$ on $\mathcal{H}$. If $\rho < 0$ the geodesics converge and this results in the existence of conjugate points beyond which points on the geodesic are timelike, not null, separated. Such a conjugate point to the future of a null geodesic generator of $\mathcal{H}$ would mean that this generator had a finite endpoint, which is not the case if the cosmic censorship conjecture is true. Hence, $\rho \geq 0$ on $\mathcal{H}$ and the second law follows.

3.4.4. *Discussion.* Consider two systems Σ_1 and Σ_2 , with entropies S_1 and S_2 respectively. Then, if the two systems interact, it is thermodynamically possible for the entropy of one of the systems, say Σ_1 , to decrease, that is $\Delta S_1 < 0$, providing that the overall entropy of the system obeys the second law, that is $\Delta S_1 + \Delta S_2 \geq 0$. This is not the case with the second law of black hole mechanics. For example, it is not possible for two black holes $\mathcal{B}_1$ and $\mathcal{B}_2$ with horizon areas A_1 and A_2 respectively, to coalesce and then bifurcate such that the area of $\mathcal{B}_1$ decreases whilst $\Delta A_1 + \Delta A_2 \geq 0$. That such a bifurcation cannot occur is easily seen in the case of the static Schwarzschild black hole, for which $A \propto M^2$. Let some black hole $\mathcal{B}_1$ bifurcate to produce $\mathcal{B}_2$ and $\mathcal{B}_3$, with areas A_i and masses M_i . Then by the area theorem, $A_2 + A_3 \geq A_1 \Rightarrow M_1 \leq \sqrt{M_2^2 + M_3^2} \leq M_2 + M_3$. But this violates energy conservation, which requires $M_1 \geq M_2 + M_3 + \text{radiated energy}$. Hence, the bifurcation is forbidden.

There remains the question of the source of black hole entropy. There are several suggestions as to the nature of this entropy, which are discussed in [Jacobson (1996)] at pp. 55-61,

and see [Ashtekar (1998)] for a quantum geometric approach. §4 initiates a discussion of causal set approaches to the resolution of this question.

Under classical general relativity, it is possible to drop a box containing matter into a black hole, thus reducing the entropy of the universe. Since the universe is regarded as a closed system, the second law of thermodynamics requires $\Delta S \geq 0$, a requirement which is broken in the situation described. This led Bekenstein to suggest a generalized entropy S' , given by

$$S' = S + S_{BH} \quad (42)$$

where S_{BH} is the entropy of the black hole and S is the entropy of matter outside the black hole [Bekenstein (1974)]. There is then a *Generalized Second Law* (GSL) which has $\Delta S' \geq 0$.

If one takes the area of the black hole as a measure of this entropy, then a gedankenexperiment may be proposed which seemingly contradicts the GSL. Suppose a box of matter is lowered arbitrarily close to the horizon before being ‘dropped in’ to the black hole beyond the horizon. Then one could recover all of the energy in the box as work at infinity. If this is the case, then the black hole gains no energy and hence the area of the horizon remains unchanged, by the first law. But then $\Delta S' = \Delta S + 0 < 0$, violating the GSL. As Wald points out, this is equivalent to a perfectly efficient Carnot cycle, reflecting the fact that, classically, the temperature of a black hole is absolute zero [Wald (1999)]. However, the first law requires the existence of a non-zero temperature if the (massive) black hole is to have a finite entropy. Thus, if the black hole can be shown to have a non-zero temperature, implying the existence of a radiation pressure against which a lowered box would have to do work, the GSL may not be violated. Again, such definite temperature arises in a quantum theoretic treatment of the black hole, which we discuss below.

3.5. The Third Law.

“It is impossible by any procedure, no matter how idealized, to reduce κ to zero by a finite sequence of operations.” [Bardeen (1973)]

3.5.1. *The Analogy.* Again, the analogy is between κ and T , since one form of the third law of thermodynamics states that it is impossible by any procedure, no matter how idealized, to reduce T to zero by a finite sequence of operations.

3.5.2. *Discussion.* Bardeen et al. point out that this law “has a rather different status from the others, in that it does not, so far at least, have a rigorous mathematical proof” [Bardeen (1973)]. The arguments in support of this law are therefore based on a failure to identify any process which could achieve the reduction of κ to zero in a finite number of steps, together with a concern that such process could presumably continue beyond this point, creating a naked singularity.

However, the status of the third law is not yet clear, particularly in the light of *extremal* black holes, for which the existence of a charge term in the definition of κ allows $\kappa = 0$ for finite A . This does not agree with the Planck-Nernst formulation of the third law, $\lim_{T \rightarrow 0} S = 0$. Alternative formulations of the third law may hold [Israel (1986)], and it may be the case that there is some form of discontinuity between the treatment of extremal and non-extremal states [Belgiorno (2002)].

3.6. The Quantum Approach: Particle Creation by Black Holes. Hawking’s 1975 paper, *Particle Creation by Black Holes*, shows that a black hole has a physical, non-zero temperature [Hawking (1975)]. Quantum effects cause a black hole to radiate particles to infinity, with a black-body spectrum at a temperature T_H , given as

$$T_H = \frac{\hbar \kappa}{2\pi k} \quad (43)$$

where κ is the surface gravity, and k is Boltzmann’s constant. This radiation is known as *Hawking radiation*. We will follow here the approach taken by Hawking and Townsend [Townsend (1997)], but note that other derivations are possible. In particular, Hawking radiation may be considered as an instance of the Unruh effect, by which the vacuum in Minkowski space appears to be a thermal state of non-zero temperature, when observed from a non-inertial frame. A derivation using this approach can be found in [Jacobson (1996)].

3.6.1. *Production of Hawking Radiation.* Consider a real, massless, scalar field, $\Phi(x)$, satisfying the Klein-Gordon equation:

$$\square \Phi \equiv \eta^{\mu\nu} \Phi_{,\mu\nu} = 0 \quad (44)$$

in Minkowski spacetime. The solution is given as an integral over k -space:

$$\Phi(x) = \int \frac{d^3k}{(2\pi)^3 2k^0} \left[\mathbf{a}(\vec{k}) e^{-ik \cdot x} + \mathbf{a}^\dagger(\vec{k}) e^{ik \cdot x} \right] \quad (45)$$

where $k = \begin{pmatrix} k^0 \\ \vec{k} \end{pmatrix}$. $\mathbf{a}$ and $\mathbf{a}^\dagger$ are annihilation and creation operators, respectively, obeying the usual commutation relations:

$$\left[\mathbf{a}(\vec{k}), \mathbf{a}(\vec{k}') \right] = \left[\mathbf{a}^\dagger(\vec{k}), \mathbf{a}^\dagger(\vec{k}') \right] = 0 \quad , \quad (46)$$

$$\left[\mathbf{a}(\vec{k}), \mathbf{a}^\dagger(\vec{k}') \right] = (2\pi)^3 2k^0 \delta^{(3)}(\vec{k} - \vec{k}') \quad , \quad (47)$$

$$\left[\mathbf{H}, \mathbf{a}^\dagger(\vec{k}) \right] = k^0 \mathbf{a}^\dagger(\vec{k}) \quad , \quad (48)$$

$$\left[\mathbf{H}, \mathbf{a}(\vec{k}) \right] = -k^0 \mathbf{a}(\vec{k}) \quad , \quad (49)$$

where $\mathbf{H}$ is the Hamiltonian, and $\delta^{(3)}$ the Dirac delta function in three dimensions. The operators exist in an Hilbert space which is a Fock space: we define a vacuum state $|0\rangle$ such that $\mathbf{a}|0\rangle = 0$, normalized such that $\langle 0 | 0 \rangle = 1$.

We can take a solution in discrete modes i :

$$\Phi = f_i \mathbf{a}_i + f_i^* \mathbf{a}_i^\dagger \quad (50)$$

where $\{f_i\}$ form a set of (complex) orthonormal solutions to $\square f_i = 0$. Then $[\mathbf{a}_i, \mathbf{a}_j^\dagger] = \delta_{ij}$.

Thus far, we have only considered fields in Minkowski spacetime, but a black hole clearly exists in a curved spacetime for which the metric is not $\eta_{\mu\nu}$. It may be suggested that the Klein-Gordon equation, eq.44, be modified to give $g^{\mu\nu} \Phi_{;\mu\nu} = 0$. However, in a curved spacetime there does not exist a global inertial frame enabling the unambiguous identification of f_i and f_i^* with positive and negative frequency solutions respectively (that is, solutions of the form $e^{-k \cdot x}$ and $e^{k \cdot x}$, cf. eq.45). We therefore consider a globally hyperbolic, asymptotically flat spacetime consisting of a sandwich of three regions, which we take here to be $\mathcal{M}_{BH}$, $\mathcal{I}^+$ and $\mathcal{I}^-$: see Figure 5.

On $\mathcal{I}^+$ we may define a field:

$$\Phi = f_i \mathbf{a}_i + f_i^* \mathbf{a}_i^\dagger \quad (51)$$

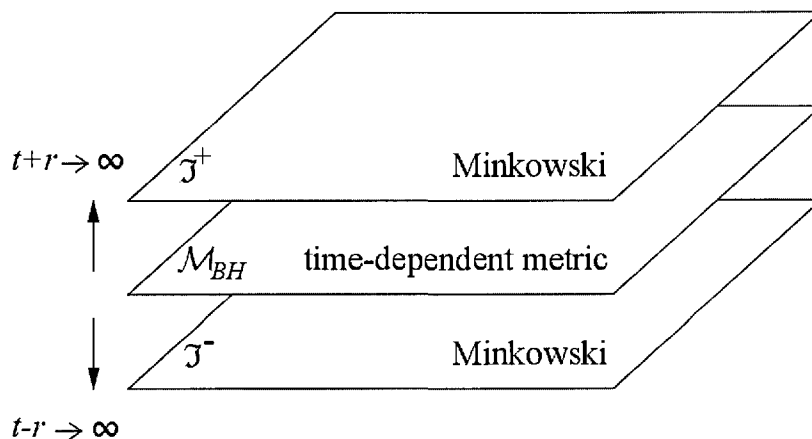


FIGURE 5. Spacetime $\mathcal{M}$ divided into three regions: $\mathcal{M}_{BH}$, and the asymptotically flat regions $\mathcal{I}^+$ and $\mathcal{I}^-$. (One space dimension suppressed.)

and we may define the same field on $\mathcal{I}^-$ as

$$\Phi = p_i \mathbf{b}_i + p_i^* \mathbf{b}_i^\dagger \quad . \quad (52)$$

The global hyperbolicity of the spacetime allows Φ to be uniquely determined at all points within the spacetime by the data on $\mathcal{I}^-$, which acts as a Cauchy surface. Thus the functions f_i must be linear combinations of the functions p_i and p_i^* :

$$f_i = \alpha_{ji} p_j + \beta_{ji} p_j^* \quad . \quad (53)$$

Then,

$$\Phi = (\alpha_{ji} p_j + \beta_{ji} p_j^*) \mathbf{a}_i + (\alpha_{ji}^* p_j^* + \beta_{ji}^* p_j) \mathbf{a}_i^\dagger \quad (54)$$

$$\Rightarrow p_j \mathbf{b}_j + p_j^* \mathbf{b}_j^\dagger = p_j (\alpha_{ji} \mathbf{a}_i + \beta_{ji}^* \mathbf{a}_i^\dagger) + p_j^* (\beta_{ji} \mathbf{a}_i + \alpha_{ji}^* \mathbf{a}_i^\dagger) \quad . \quad (55)$$

So α and β , the *Bogoliubov coefficients*, transform the annihilation and creation operators between $\mathcal{I}^+$ and $\mathcal{I}^-$:

$$\mathbf{b}_i = \alpha_{ij} \mathbf{a}_j + \beta_{ij}^* \mathbf{a}_j^\dagger \quad . \quad (56)$$

We define $|0_- \rangle$ to be the vacuum state at $\mathcal{I}^-$. Clearly, $\mathbf{a}_i |0_- \rangle = 0 \, \forall i$. However, $\mathbf{b}_i |0_- \rangle \neq 0$, since the operator $\mathbf{b}_i$ contains terms in $\mathbf{a}_j^\dagger$ which act on $|0_- \rangle$ to produce states $|j \rangle$. The interpretation of this is that an observer at $\mathcal{I}^+$ does not see the vacuum seen by an observer

at $\mathcal{I}^-$: instead the $\mathcal{I}^+$ observer sees a spectrum of particles. Such a flux of particles is caused by the initial collapse to a black hole, but time dilation at the horizon means it can be observed at late times, when the spacetime is stationary. The nature of these ‘late time’ particles is independent of the exact nature of the collapse.

3.6.2. Spectrum of Hawking Radiation. The number operator for particles seen in mode i , by an observer at $\mathcal{I}^+$, is given by $N_i^+ = \mathbf{b}_i^\dagger \mathbf{b}_i$. This gives an expectation value for the number of particles in mode i , seen at $\mathcal{I}^+$:

$$\langle N_i^+ \rangle = \langle 0_- | \mathbf{b}_i^\dagger \mathbf{b}_i | 0_- \rangle \quad (57)$$

$$\begin{aligned} &= \langle 0_- | \left(\alpha_{ij}^* \mathbf{a}_j^\dagger + \beta_{ij} \mathbf{a}_j \right) \left(\alpha_{ik} \mathbf{a}_k + \beta_{ik}^* \mathbf{a}_k^\dagger \right) | 0_- \rangle \\ &= \left(\beta_{ij} \beta_{ki}^\dagger \right) \langle 0_- | \left(\mathbf{a}_j \mathbf{a}_k^\dagger \right) | 0_- \rangle = \left(\beta_{ij} \beta_{ki}^\dagger \right) \delta_{jk} \\ \Rightarrow \langle N_i^+ \rangle &= (\beta^\dagger \beta)_{ii} \quad . \end{aligned} \quad (58)$$

Therefore, in order to determine the spectrum of particles produced, we must determine the β Bogoliubov matrix elements.

We may define the outgoing Eddington-Finkelstein (EF) coordinates (u, r, Ω) , in a similar manner to the ingoing coordinates described in §2.4. Here $u = t - r^*$ (where, again, r^* is the Regge-Wheeler radial coordinate), and the resulting Schwarzschild metric is given as

$$ds^2 = - \left(1 - \frac{2M}{r} \right) du^2 - 2drdu + r^2 d\Omega^2 \quad (59)$$

which is again defined for all $r > 0$ by analytic continuation from the $r > 2M$ domain. This may be combined with the ingoing Schwarzschild metric, eq.13, to give

$$ds^2 = - \left(1 - \frac{2M}{r} \right) dudv + r^2 d\Omega^2 \quad . \quad (60)$$

Define $U = \exp \left\{ -\frac{u}{4M} \right\}$ and $V = \exp \left\{ \frac{v}{4M} \right\}$, to give the Schwarzschild metric in Kruskal-Szekeres (KS) coordinates, (U, V, Ω) , which may again be analytically continued to all definite values of U and V :

$$ds^2 = \frac{-32M^3}{r} \exp \left\{ -\frac{r}{2M} \right\} dUdV + r^2 d\Omega^2 \quad . \quad (61)$$

These coordinates are used below, and in the labelling of Figure 6.

$\gamma_{\mathcal{H}}$ may be continued back into the past to meet $\mathcal{I}^-$ at the point $v = 0$. The affine parameter on $\mathcal{I}^-$ is v , and the geometry of Figure 6 suggests that γ meets $\mathcal{I}^-$ at the point $v = -\varepsilon$. (This result can be proved rigorously by considering the parallel transport of tangent and normal null vectors along the null geodesics: see [Hawking (1975), Townsend (1997)].) Then from eq.63, and recognizing that for $v > 0$ any outgoing ray does not reach $\mathcal{I}^+$:

$$p_{\omega} \sim \begin{cases} 0 & v > 0 \\ \exp\left\{\frac{i\omega}{\kappa} \ln(-v)\right\} & v < 0 \end{cases} . \quad (64)$$

Then the Fourier transform of p_{ω} is given as

$$\tilde{p}_{\omega} = \int_{-\infty}^0 dv \exp\left\{i\omega'v + \frac{i\omega}{\kappa} \ln(-v)\right\} \quad (65)$$

and it can be shown that ([Townsend (1997)] at pp. 127-128):

$$\tilde{p}_{\omega}(-\omega') = -\exp\left\{-\frac{\pi\omega}{\kappa}\right\} \tilde{p}_{\omega}(\omega') \quad \forall \omega' > 0 . \quad (66)$$

It is this Fourier transform of p_{ω} which allows the calculation of $\alpha_{\omega\omega'}$ and $\beta_{\omega\omega'}$ [Hawking (1975)]. Since then,

$$\alpha_{\omega\omega'} = \tilde{p}_{\omega}(\omega') \quad (67)$$

$$\beta_{\omega\omega'} = \tilde{p}_{\omega}(-\omega') = -\exp\left\{-\frac{\pi\omega}{\kappa}\right\} \tilde{p}_{\omega}(\omega') \quad (68)$$

$$\Rightarrow \beta_{\omega\omega'} = -\exp\left\{-\frac{\pi\omega}{\kappa}\right\} \alpha_{\omega\omega'} . \quad (69)$$

This result, together with the constraint on the Bogoliubov matrices that $(\alpha\alpha^{\dagger} - \beta\beta^{\dagger})_{ij} = \delta_{ij}$ [Townsend (1997)], gives

$$\delta_{ij} = \left[\exp\left\{\frac{\pi}{\kappa}(\omega_i + \omega_j)\right\} - 1 \right] \beta_{ik}\beta_{jk}^* \quad (70)$$

where ω_i gives the value of ω corresponding to mode i . Hence our result for the number operator,

$$\langle N_i^+ \rangle \propto (\beta\beta^{\dagger})_{ii} = \frac{1}{\exp\left\{\frac{2\pi\omega_i}{\kappa}\right\} - 1} . \quad (71)$$

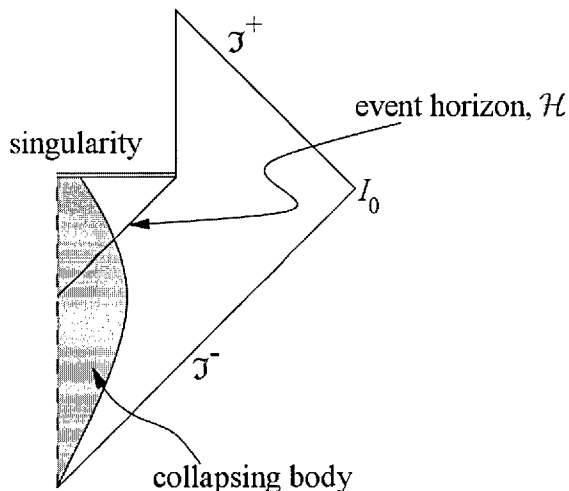


FIGURE 7. Gravitational collapse and evaporation of a black hole, leaving empty space with no singularity. (After [Hawking (1975)].)

This is a Planck distribution for black body radiation. Since the general form of such a distribution is $(\exp\{\frac{\hbar\omega}{kT}\} - 1)^{-1}$, we may make the identification of the Hawking temperature, T_H :

$$T_H = \frac{\kappa}{2\pi} \quad (72)$$

in units with $\hbar = k = 1$. (We will use such units from hereon.)

3.6.3. Further Discussion. The result for T_H may be extended to charged and rotating black holes, although the radiative spectrum will be altered as a consequence of angular momentum and charge. Details may be found in [Hawking (1975)] at pp. 213-215. It is also possible to apply a similar approach to massive particles: in this case, the frequency ω will contain a term in the rest mass energy of the particle, and hence there will be limited emission of particles for which mass, $m > \frac{\kappa}{2\pi}$. For fermions the particle distribution is found to obey Fermi-Dirac statistics, so that $\langle N_i^+ \rangle = (\beta\beta^\dagger)_{ii} = (\exp\{\frac{2\pi\omega_i}{\kappa}\} + 1)^{-1}$.

We note that an effective temperature of $\frac{\kappa}{2\pi}$ implies an entropy given by

$$S_{BH} = \frac{A}{4} \quad (73)$$

in accordance with the coefficients of the first law, eq.34.

The existence of Hawking radiation resolves many of the problems discussed in relation to classical black hole thermodynamics. The temperature of the black hole is no longer zero, removing the possibility of perfectly efficient heat exchange. Furthermore, the emission of Hawking radiation reduces the surface area of the horizon, allowing the area of individual black holes to decrease, whilst overall entropy increases.

However, further problems arise as a result of these quantum effects. In particular, the emission of radiation leads to the gradual evaporation of the black hole, a process which accelerates rapidly as the black hole reaches masses of order 10^{-5}g [Hawking (1975)], as a result of the back reaction on the metric: see Figure 7. Information which has been encoded (for example, as bits on a computer hard-drive) and thrown into the black hole is lost, to be re-emitted as thermal radiation without any information content. There are a number of reasons (thermodynamic and otherwise) to think that this situation might, in fact, be far more complicated than a simple treatment allows: Susskind provides an overview in [Susskind (2000)]. This gives rise to the *information paradox*: quantum mechanics is unitary in nature, and as such the time-dependent evolution of a system does not allow for information loss.

4. EXISTING CAUSAL SET APPROACHES TO BLACK HOLE ENTROPY.

4.1. Origins of Black Hole Entropy. Whilst existing results indicate a thermodynamic interpretation of black hole physics, the physical cause of black hole entropy is not yet fully understood. The existence of horizon degrees of freedom is suggestive of some quantum theory of spacetime and, indeed, work has been done to explain black hole entropy in terms of allowable states of strings with boundary conditions on varying configurations of Dirichlet branes (see, for example, [Strominger (1996)]). For Sorkin and Dou, the role of black hole thermodynamics is closely analogous to that of the analyses of late Nineteenth Century thermodynamics, in revealing, “the underlying atomicity and quantum nature of everyday matter and radiation” [Dou (2003)].

A continuum analysis of black hole entropy, as ‘missing information’, based on tracing out the interior modes of quantum fields, gives infinite results, unless one imposes a short distance cutoff. Choosing this cutoff to be of order the Planck length leads to an entropy result with the same order of magnitude as the black hole horizon. If, extrapolating from this cutoff to a discretized spacetime, one takes the horizon to consist of Planck-scale elements, a full statistical analysis would require a dynamical theory of those elements. However, even a kinematical analysis may provide an approximation to the entropy, by analogy with a gas at high enough temperatures to avoid quantum degeneracy, at which point one may estimate the entropy by simply counting the number of gas molecules. Causal set theory may provide one way in which to construct and analyse such a discrete spacetime. In particular, the idea of counting links which cross the horizon is appealing, through the association of those links with ‘information flow’ across the horizon: by counting such links one would thus measure some form of entanglement between regions external and internal to the horizon.

In his thesis ([Dou (1999)]) and a paper with Sorkin ([Dou (2003)]) Dou considers this proposition in the context of both collapsing and eternal, static black holes. In sections which follow (§5, etc.) we will have cause to summarize Dou’s key results, so we will not repeat them in detail at this point. However, a short summary of these results will provide an introduction to our work.

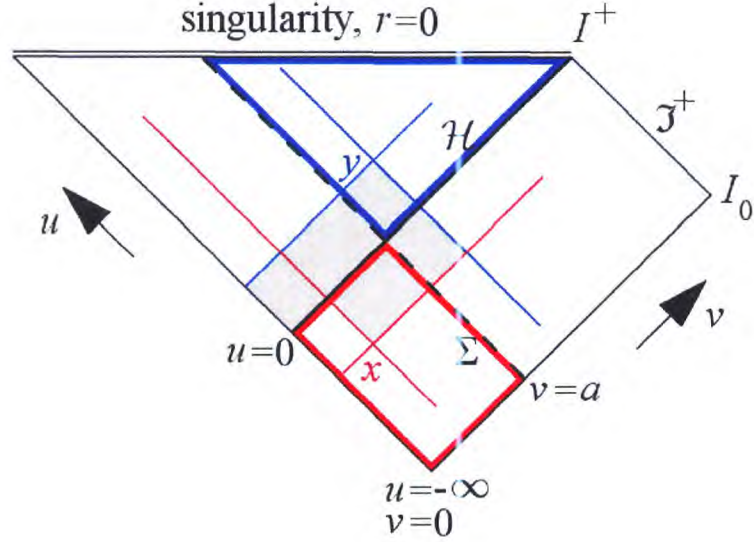


FIGURE 8. Penrose diagram showing Dou's link-counting arrangement for a static black hole, truncated from 3+1 to 1+1 dimensions, as described in the main text. x may fall anywhere within the region bounded red. y may fall anywhere within the region bounded blue. (After [Dou (1999), Dou (2003)].)

4.2. 1+1-Dimensional Static Black Hole. Figure 8 shows an eternal, spherically symmetric black hole, labelled with Kruskal-Szekeres coordinates, u and v , with $uv = (1 - \frac{r}{2M}) e^{r/2M}$, for which the horizon entropy is to be calculated. This is a truncation of the full 3+1-dimensional black hole to 1+1 dimensions, undertaken on the basis that if the result obtained pertains to links clustering around the intersection of $\mathcal{H}$ and Σ , then a return to 3+1 dimensions should give the 1+1-dimensional result multiplied by the area of the horizon (and some multiplicative factor of order one): see §4.4. Under this truncation the 3+1-dimensional metric, $ds^2 = \frac{16M^3}{r} e^{-r/2M} dudv + r^2 d\Omega^2$, is reduced to $ds^2 = \frac{16M^3}{r} e^{-r/2M} dudv$, the angular terms being discarded.

The black hole has horizon $\mathcal{H}$, which is intersected by a null hypersurface, Σ . The region of spacetime being considered, $\mathcal{M}_{BH}$, is that which could have arisen from the collapse of a spherically symmetric body to form the black hole: that is to say, we do not extend back through the period of black hole formation to $\mathcal{I}^-$.

We sprinkle random points into $\mathcal{M}_{BH}$ according to a Poisson distribution of $\rho = 1$ in appropriate units. These points are elements of the causal set $\mathcal{C}$, which is faithfully

embedded in $\mathcal{M}_{BH}$. We then consider the points x and y , elements of $\mathcal{C}$, with $x \in J^-(\mathcal{H}) \cap J^-(\Sigma)$ (the region bounded red in Figure 8), and $y \in J^+(\mathcal{H}) \cap J^+(\Sigma)$ (the region bounded blue). We require $x \prec y$ (x is linked to y), and posit that we may calculate black hole entropy by counting these linked pairs of elements, crossing from outside the horizon and to the past of Σ to inside the horizon and to the future of Σ .

A naive approach to the integral arising from such a model (i.e. with no further restrictions on x and y) is divergent, since there are an infinite number of pairs of points which are so related. Physically, the problem arises as we are ‘overcounting’ causal links. We want to consider links crossing the horizon at ‘a given time’, which corresponds to their being in some way in the vicinity of the hypersurface Σ . In order to ensure this proximity to Σ , we must impose further conditions on x and y . In particular, we choose:

$$\begin{aligned}
 x &\in J^-(\Sigma) \cap J^-(\mathcal{H}) && \text{as above} \\
 y &\in J^+(\Sigma) \cap J^+(\mathcal{H}) && \text{as above} \\
 x &\prec y && \text{i.e. } x \prec y \text{ is a link} \\
 x &\text{ maximal in } J^-(\Sigma) \cap J^-(\mathcal{H}) \\
 y &\text{ minimal in } J^+(\Sigma) \cap J^+(\mathcal{H}) \quad .
 \end{aligned}$$

We note that these are not the only possible conditions on x and y , and that other, similar conditions produce the same asymptotic behaviour of the integrals discussed below. However, the chosen conditions produce tractable integral formulations. Furthermore, as Sorkin points out ([Dou (2003)]), the exact continuation of the hypersurface Σ within the horizon should not influence the calculated entropy: and the conditions here seek to minimize the constraints on (null) Σ within the horizon.

The calculation subsequently performed is discussed in Appendix C.4, and we will not repeat it here. The counting of discrete links is extrapolated to its analytic limit, giving an integral formulation which equates the number of links to the entropy of the black hole, $S_{\text{Dou}}^{\text{static}}$. The outcome is an expectation value for the entropy:

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \quad (74)$$

where $\alpha = 2M$, with M being the mass of the black hole³.

4.3. 1+1-Dimensional Collapsing Black Hole. Dou also calculates the entropy associated with a black hole formed from a collapsing null shell of matter. This calculation is discussed in detail in the introduction to §5 and Appendix B.1. At this stage we will simply mention that the leading order, numeric term from this calculation is in agreement with eq.74:

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{a^2}\right) \quad (75)$$

taking the limit in which the collapse is pushed to future infinity⁴. Here a relates to the position of a null hypersurface intersecting the horizon. Again, §5 makes this clear.

4.4. Extension of Results Back to 3+1 Dimensions. The general argument of Dou and Sorkin is that the result above, although derived in 1+1-dimensional spacetime, would not be greatly affected by the shift to a full 4-dimensional spacetime. The dominant contribution to the calculated entropy come from elements x and y which reside near the intersection of horizon, $\mathcal{H}$, and hypersurface, Σ . In such a scenario the counting procedure is effectively restricted to spacetime volumes with near-horizon geometry which, in four dimensions, is the 1+1-dimensional geometry $\times$ the Euclidean plane. So, one would expect $\langle S \rangle$ to be proportional to the area of the horizon, as required. Furthermore, by equating the results of a causal set approach to black hole entropy with those obtained through the application of semiclassical models, it is possible to fix the fundamental scale associated with causal set kinematics. This will be discussed in more detail in §5.6.

However, as we shall see in §6, the link-counting model of Dou produces infinite values for the black hole entropy in 2+1 and higher dimensions. It is this problem which we seek to address in the rest of this thesis, providing a model of counting groups of (three) sprinkled elements which can be extended to higher dimensions whilst continuing to produce finite results.

³Dou has a correction term of $\mathcal{O}(1/\alpha)$, but a careful repetition of his calculation gives $\mathcal{O}(1/\alpha^2)$. See Appendix C.4.

⁴Again, Dou has a correction term of $\mathcal{O}(1/a)$, but we find $\mathcal{O}(1/a^2)$. See Appendix B.1.

5. LINK COUNTING IN AN ‘EXTENDED’ SPACETIME.

5.1. Introduction. Dou describes a spherically collapsing null shell of matter with stress-energy tensor given by

$$T_{\mu\mu} = \frac{M\delta(v-b)}{4\pi r^2} \quad (76)$$

with off-diagonal elements being zero. Figure 9 shows a Penrose diagram of this spacetime. The matter shell sits at $v = b$. Inside the shell, where the spacetime is Minkowski, there is a null hypersurface, Σ , with $v = a < b$, which intersects the horizon, $\mathcal{H}$. We will be interested in the limiting case in which $a \gg 1$ (in Planck units) and b tends to infinity. This, again, is a truncation of the full 3+1-dimensional black hole to 1+1 dimensions under which the 3+1-dimensional Minkowski metric, $ds^2 = -2dudv + r^2d\Omega^2$, is reduced to $ds^2 = -2dudv$, the angular terms being discarded. Defining $u = (t-r)/\sqrt{2}$ and $v = (t+r)/\sqrt{2}$ gives $\sqrt{-g} = 1$.

Dou’s integral requires the grey-bounded volume V to be free from sprinkled points, giving

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell\left(\frac{a}{b}\right) + \mathcal{F}(a, b) \quad (77)$$

where $\ell\left(\frac{a}{b}\right) = \sum_{k=1}^{\infty} \left(\frac{a}{b}\right)^k \frac{1}{k^2}$ and $\mathcal{F}(a, b) = \frac{3}{4a^2} - \frac{a}{3b^3}$, to leading orders in a and b . The details of this calculation can be found in Appendix B.1.

5.2. Reconstructing Dou’s Integral in an ‘Extended’ Spacetime: Link Counting.

Figure 10 shows a second Penrose diagram of Dou’s spacetime. The Penrose diagram has been ‘extended’ from Figure 9, which is truncated to the ‘left of’ the central ‘axis’, along which $u = v$. In Figure 10 this ‘left-hand side’ is retained to provide a more complete picture of the spacetime under consideration. The 1+1-dimensional model represented by these figures is a conformal compactification of the full, 3+1-dimensional picture. Extending this representation as we have done confers two advantages over Figure 9. Firstly, and most importantly, it enables us to consider link configurations which would otherwise be omitted, since the causal elements span the central ‘axis’. In particular, it will be seen that this removes the $\mathcal{F}(a, b)$ term of $\langle S_{\text{Dou}}^{\text{coll.}} \rangle$. Secondly, it provides a basis for a natural construction of the 2+1-dimensional model, by rotating the 1+1-dimensional picture through π radians about its central ‘axis’, such that orthogonal lines (of the horizon, null

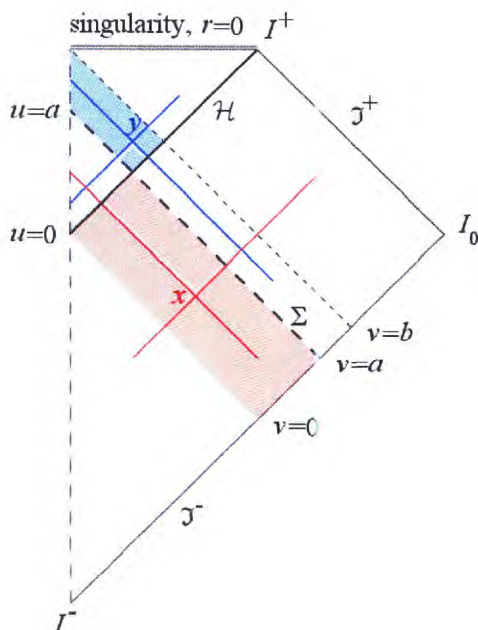


FIGURE 9. Penrose diagram (after [Dou (1999)]) showing a black hole formed from a collapsing null shell of matter. The black hole has been truncated from 3+1 to 1+1 dimensions, as discussed in the text. The matter shell sits at $v = b$. Within the shell the spacetime is Minkowski. A null hypersurface, Σ , sits at $v = a$, intersecting the horizon, $\mathcal{H}$. Two sprinkled points, x and y , are shown. x must be within the region shaded red; y within the region shade blue. The region bounded grey must be free from sprinkled points.

hypersurface, etc.) form boundaries of conical regions ('light cones'). The details of both additional link configurations and the construction of the 2+1-dimensional model will be found below.

5.2.1. Basic Conditions. Having sprinkled the elements of a causal set $\mathcal{C}$ into our spacetime, we wish to consider pairs of those elements, arranged as shown in Figure 11. The pair of elements is then restricted as follows⁵:

⁵This restriction that $y \in J^-$ (matter shell) will be used in the formulation and solution of the integrals associated with this arrangement of elements. Subsequently, taking the limit $b \rightarrow \infty$ will, in effect, remove this restriction.

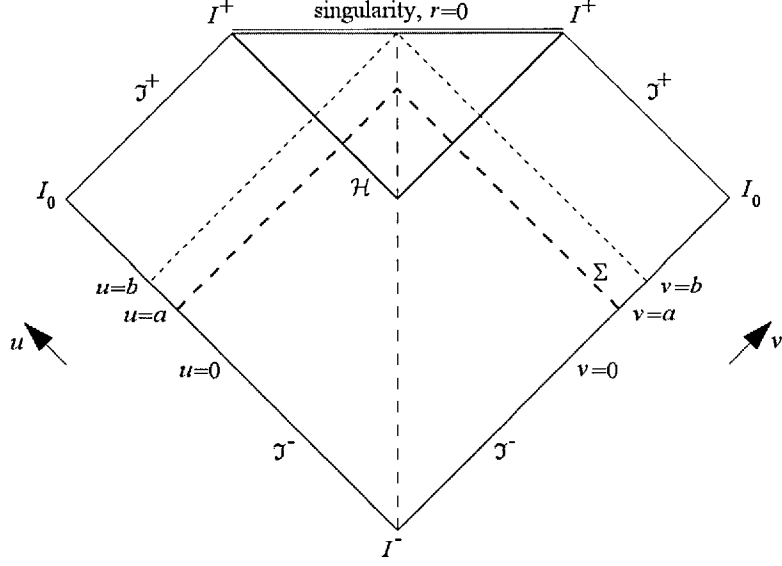


FIGURE 10. Penrose diagram showing a black hole formed from a collapsing null shell of matter. The matter shell sits at $v = b$ in the ‘right half’ and $u = b$ in the ‘left half’. Within the shell the spacetime is Minkowski. A null hypersurface, Σ , sits at $v = a$ ($u = a$), intersecting the horizon, $\mathcal{H}$.

location	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\Sigma) \cap J^+(\mathcal{H}) \cap J^-(\text{matter shell})$
causal relations	$x \prec y$
maximality/minimality	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\mathcal{H})$

Where we have used the notation $p \prec q$ to denote $p \prec q$ is a link; $p \in \mathcal{A}$ to denote $p \in \mathcal{A}$ and p is maximal in $\mathcal{A}$; and $p \in \mathcal{A}$ to denote $p \in \mathcal{A}$ and p is minimal in $\mathcal{A}$.

Note, however, that Figure 11 shows the range of x limited to the ‘right half’ of the 1+1-dimensional spacetime. This is necessary to avoid ‘double counting’ of links when we rotate about the central ‘axis’ to construct a 2+1-dimensional picture. Thus we may define

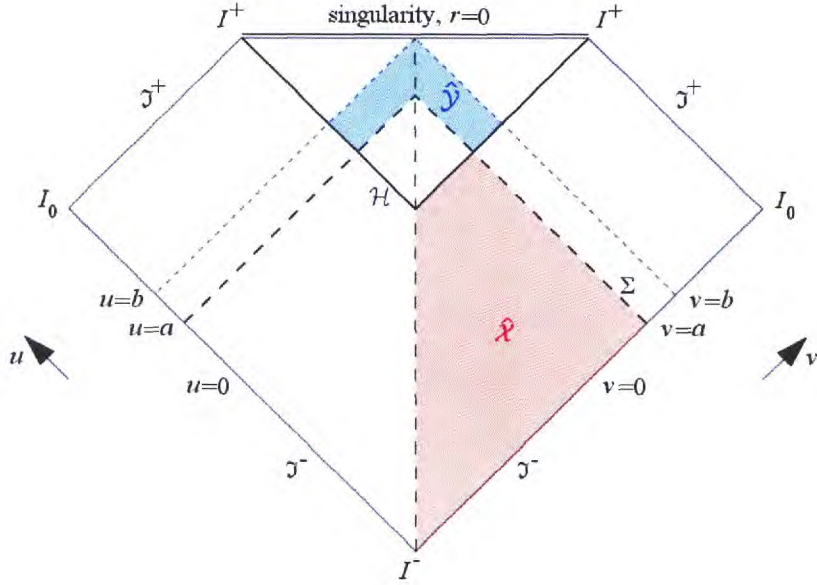


FIGURE 11. Penrose diagram showing the total ranges of locations for the two causal set elements $x \in \mathcal{C}$ and $y \in \mathcal{C}$. x must have been sprinkled into the region $\hat{\mathcal{X}}$ (shaded red) and y into $\hat{\mathcal{Y}}$ (shaded blue).

the regions $\hat{\mathcal{X}}$ and $\hat{\mathcal{Y}}$ as

$$\begin{aligned} \hat{\mathcal{X}} &= \{q \mid -\infty < v_q \leq 0 \text{ and } -\infty < u_q \leq v_q\} \\ &\quad \cup \{p \mid 0 < v_p \leq a \text{ and } -\infty < u_p \leq 0\} \end{aligned}$$

and

$$\begin{aligned} \hat{\mathcal{Y}} &= \{q \mid a \leq v_q \leq b \text{ and } 0 \leq u_q \leq v_q\} \\ &\quad \cup \{p \mid a \leq u_p \leq b \text{ and } 0 \leq v_p < u_q\} \quad . \end{aligned}$$

However, we will show that the contributions of many links with $x \in \hat{\mathcal{X}}$ and $y \in \hat{\mathcal{Y}}$ are negligible, and that we may restrict ourselves to the regions $\mathcal{X}$ and $\mathcal{Y}$ of Figure 12. Here

$$\mathcal{X} = \{p \mid 0 \leq v_p \leq a \text{ and } -\infty < u_p \leq 0\}$$

and

$$\mathcal{Y} = \{q \mid a \leq v_q \leq b \text{ and } 0 \leq u_q \leq v_q\} \quad .$$

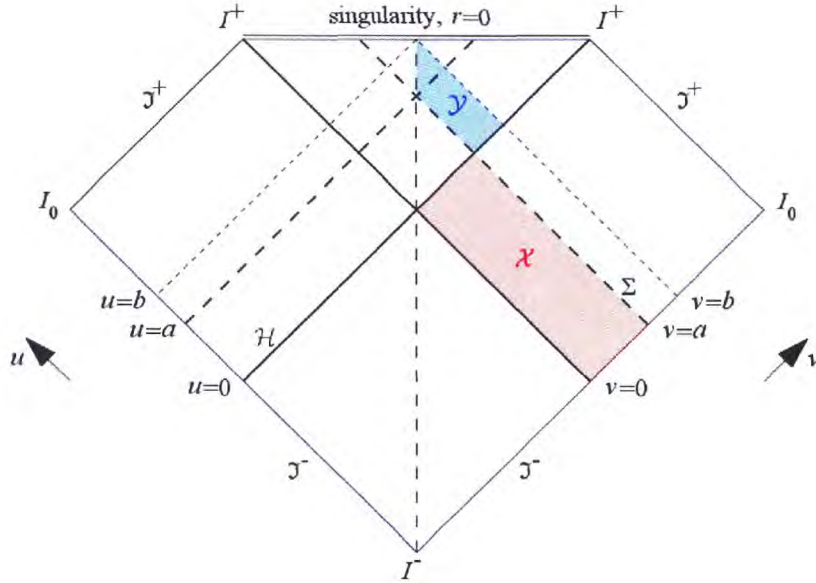


FIGURE 12. Penrose diagram showing the restricted ranges of locations for the two causal set elements $x \in \mathcal{C}$ and $y \in \mathcal{C}$. The details and justification of the restriction of these ranges is given in the main text. x must have been sprinkled into the region $\mathcal{X}$ (shaded red) and y into $\mathcal{Y}$ (shaded blue). In this and subsequent diagrams, the boundaries of $\mathcal{H}$ and Σ have been extended to $\mathcal{I}^-$ and $r = 0$, respectively, to assist in illustrating the formulation of associated integrals.

We then require

location	$x \in \mathcal{X}$
	$y \in \mathcal{Y}$
causal relations	$x \prec y$
maximality/minimality	$x \notin J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\mathcal{H})$.

5.2.2. The Poisson Distribution. The integrals which enable us to count relevant x - y pairs are formulated on the basis that the sprinkling of points of $\mathcal{C}$ within our spacetime is governed by the Poisson distribution. (We refer to ‘points’ in this discussion, rather than ‘elements’ of $\mathcal{C}$, to avoid confusion, since we will also refer to ‘volume elements’ of the spacetime.) Given a sprinkling density, ρ , the probability of finding n sprinkled points in

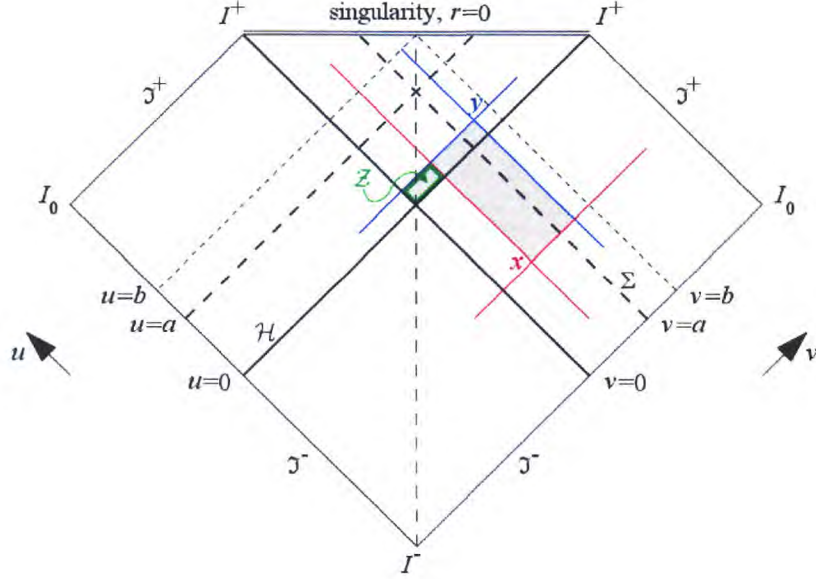


FIGURE 13. Penrose diagram showing an example of two sprinkled elements, x and y . The grey shaded area contains no sprinkled points. The area, $\mathcal{Z}$, bounded in green, is discussed in the main text.

some volume V is given by

$$\Pr_{\rho}(V, n) = \frac{(\rho V)^n e^{-\rho V}}{n!} . \quad (78)$$

(The derivation of this result can be found in Appendix B.2.)

5.2.3. Characteristic Functions. We define the following characteristic functions, in order to evaluate the discrete sum which will give us our required integral, in the infinitesimal limit. Consider volume elements ΔV_x and ΔV_y , where x and y are labels to identify each volume element within the spacetime, uniquely. Then, to simplify our notion we write χ_x for $\chi(\Delta V_x)$, etc. We define

$$\chi_x = \begin{cases} 1 & \text{if } \Delta V_x \text{ contains a sprinkled point} \\ 0 & \text{otherwise} \end{cases}$$

and similarly for χ_y . We then define

$$\bar{\chi}_x = \begin{cases} 1 & \text{if } \Delta V_x \text{ does not contain a sprinkled point} \\ 0 & \text{otherwise} \end{cases}$$

and similarly for $\bar{\chi}_y$.

We further define

$$\xi_{xy} = \begin{cases} 1 & \text{if the interval between } \Delta V_x \text{ and } \Delta V_y \text{ is empty} \\ 0 & \text{otherwise} \end{cases}$$

where the interval excludes points at ΔV_x and ΔV_y , and

$$\eta_{xy} = \begin{cases} 1 & \text{if the region } \mathcal{Z} \text{ is empty} \\ 0 & \text{otherwise} \end{cases}$$

where $\mathcal{Z}$, as shown in Figure 13, is defined as

$$\mathcal{Z} = \{q \mid 0 < v_q < v_x \text{ and } 0 < u_q < u_y\} \quad .$$

(For a definition of ‘interval’ see §1.2.2.)

The number of pairs meeting our requirement, $S_{\text{link}}^{\text{coll.}}$, is then given by the sum

$$S_{\text{link}}^{\text{coll.}} = \sum_{x \in \mathcal{X}} \chi_x \sum_{y \in \mathcal{Y}} \chi_y \eta_{xy} \xi_{xy} \quad . \quad (79)$$

We now use the fact that our points are sprinkled with a Poisson distribution, to calculate the expectation value of $S_{\text{link}}^{\text{coll.}}$, $\langle S_{\text{link}}^{\text{coll.}} \rangle$.

5.2.4. The Expectation Value of $S_{\text{link}}^{\text{coll.}}$: Discrete Sum.

$$\begin{aligned} \langle S_{\text{link}}^{\text{coll.}} \rangle &= \left\langle \sum_{x \in \mathcal{X}} \chi_x \sum_{y \in \mathcal{Y}} \chi_y \eta_{xy} \xi_{xy} \right\rangle \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \langle \chi_x \chi_y \eta_{xy} \xi_{xy} \rangle \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \langle \chi_x \rangle \langle \chi_y \rangle \langle \eta_{xy} \rangle \langle \xi_{xy} \rangle \end{aligned} \quad (80)$$

where the final step is permissible as χ_x , χ_y , η_{xy} , and ξ_{xy} are all independent variables.

Our distribution has a point within a volume element ΔV with probability p , giving

$$\langle \chi_x \rangle = 1 \cdot \Pr(\chi_x = 1) + 0 \cdot \Pr(\chi_x = 0) = p \quad (81)$$

and similarly $\langle \chi_y \rangle = p$. Clearly, $\langle \bar{\chi}_x \rangle = \langle \bar{\chi}_y \rangle = 1 - p$.

Then,

$$\langle \eta_{xy} \rangle = \Pr(\eta_{xy} = 1) = (1 - p)^{N_{xy}^{\mathcal{Z}}} \quad (82)$$

where $N_{xy}^{\mathcal{Z}}$ is the number of volume elements in the region $\mathcal{Z}$.

And,

$$\langle \xi_{xy} \rangle = \Pr(\xi_{xy} = 1) = (1 - p)^{N_{xy}^{\text{int}}} \quad (83)$$

where N_{xy}^{int} is the number of volume elements in the interval between ΔV_x and ΔV_y .

We now take V to be the space-time volume of N volume elements, that is, $V = N \cdot \Delta V$.

Then we have:

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p^2 \cdot (1 - p)^{V_{xy}/\Delta V} \quad (84)$$

where $V_{xy} = V_{xy}^{\text{int}} + V_{xy}^{\mathcal{Z}} = (N_{xy}^{\text{int}} + N_{xy}^{\mathcal{Z}}) \cdot \Delta V$.

5.2.5. *The Expectation Value of $S_{\text{link}}^{\text{coll.}}$: Integral Formulation.* We have $Np = \rho V$, but $N = V/\Delta V$. Hence $p = \rho \Delta V$ and

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} (\Delta V)^2 \rho^2 \cdot (1 - \rho \Delta V)^{V_{xy}/\Delta V} \quad (85)$$

Taking $\lim \Delta V \rightarrow dV$ we have $(1 - \rho \Delta V)^{V_{xy}/\Delta V} \rightarrow e^{-\rho V_{xy}}$. Then

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \rho^2 \int_{\mathcal{X}} \int_{\mathcal{Y}} dV^2 e^{-\rho V_{xy}} \quad (86)$$

where we have temporarily omitted factors of $\sqrt{-g}$ for clarity.

5.2.6. *Evaluating V_{xy}^{int} , $V_{xy}^{\mathcal{Z}}$ and V_{xy} .* The region of interest lies within the collapsing shell, where spacetime is Minkowski, and, as discussed above, we consider the 1+1-dimensional truncation, using $dV = \sqrt{-g} du dv = du dv$, with $u = (t - r)/\sqrt{2}$ and $v = (t + r)/\sqrt{2}$, giving $ds^2 = -2 du dv$ and $\sqrt{-g} = 1$. Then

$$V_{xy}^{\text{int}} = \int_{u_x}^{u_y} \int_{v_x}^{v_y} du dv = (v_y - v_x)(u_y - u_x) \quad , \quad (87)$$

$$V_{xy}^{\mathcal{Z}} = \int_0^{u_y} \int_0^{v_x} du dv = u_y v_x \quad . \quad (88)$$

Hence,

$$V_{xy} = -u_x(v_y - v_x) + u_y v_y \quad . \quad (89)$$

5.2.7. *Using V_{xy} in the Integral Form of $\langle S_{\text{link}}^{\text{coll.}} \rangle$.* Similarly, we need to express the integral for $\langle S_{\text{link}}^{\text{coll.}} \rangle$, eq.86, in terms of u and v , with appropriate limits. We have

$$\begin{aligned} \int_{\mathcal{X}} \dots &= \int_0^a dv_x \int_{-\infty}^0 du_x \dots \quad , \\ \int_{\mathcal{Y}} \dots &= \int_a^b dv_y \int_0^{v_y} du_y \dots \quad . \end{aligned}$$

Ordering appropriately, the integral $\langle S_{\text{link}}^{\text{coll.}} \rangle = \rho^2 \int_{\mathcal{X}} \int_{\mathcal{Y}} dV^2 e^{-\rho V_{xy}}$ gives

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \rho^2 \int_a^b dv_y \int_0^{v_y} du_y \int_0^a dv_x \int_{-\infty}^0 du_x e^{-\rho(-u_x(v_y-v_x)+u_y v_y)} \quad . \quad (90)$$

5.3. **Solving the Integral for $\langle S_{\text{link}}^{\text{coll.}} \rangle$.** (The details of this calculation can be found in Appendix B.3.)

Taking units in which $\rho = 1$ (i.e. fundamental units), we have,

$$\begin{aligned} \langle S_{\text{link}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^{v_y} du_y \int_0^a dv_x \int_{-\infty}^0 du_x e^{u_x(v_y-v_x)-u_y v_y} \\ &= \int_a^b dv_y \ln\left(\frac{v_y}{v_y-a}\right) \left(\frac{1}{v_y} - \frac{e^{-v_y^2}}{v_y}\right) \quad . \end{aligned} \quad (91)$$

Writing y for v_y and expanding the logarithm in powers of a gives

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \int_a^b dy \left(\sum_{n=1}^{\infty} \frac{a^n}{n y^{n+1}} \right) (1 - e^{-y^2}) \quad . \quad (92)$$

For $y \geq a \gg 1$, $e^{-y^2} \leq e^{-a^2} \ll 1$, and the integral reduces to,

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \left(\frac{a}{b}\right)^n \frac{1}{n^2} \quad . \quad (93)$$

Hence,

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell\left(\frac{a}{b}\right) \quad (94)$$

where $\ell\left(\frac{a}{b}\right) = \sum_{k=1}^{\infty} \left(\frac{a}{b}\right)^k \frac{1}{k^2}$.

5.4. **Additional Arrangements for Link Counting.** In §5.2.1 we increased our restrictions on x and y from $x \in \hat{\mathcal{X}}$ and $y \in \hat{\mathcal{Y}}$ to $x \in \mathcal{X}$ and $y \in \mathcal{Y}$. In so doing we have ignored some additional link arrangements, some of which (those involving $x \in \hat{\mathcal{X}} \setminus \mathcal{X}$) are also omitted in Dou's work. These arrangements seem extraneous to our counting, since they have no implication of clustering about the intersection of the null hypersurface, Σ , and

where we note that the second integral contains an additional term in the exponent, $-u_x(a - v_y)$ which ensures that x is maximal in the appropriate region, as described in the caption to Figure 15.

Again, this integral requires numerical analysis in order to demonstrate that its contributions are negligible, or zero in the large a and b limit. However, that this result is expected can be seen from a consideration of Figure 15. The requirement that both the interval between x and y and the area shaded grey (bounded by $\mathcal{H}$, Σ and the lines $u = u_x$ and $v = v_y$) be empty causes the exponential suppression of this integral, in a manner similar to that of the additional arrangements with $v_x > 0$. It is also the case that this arrangement will not be of importance in the following discussion of models involving triplets of elements: §7.

5.5. Comparison with Dou's Original Integral Formulation. It will be recalled from eq.77 that Dou's result for the entropy of the collapsing black hole can be written as

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell\left(\frac{a}{b}\right) + \mathcal{F}(a, b) \quad (97)$$

where $\ell\left(\frac{a}{b}\right) = \sum_{k=1}^{\infty} \left(\frac{a}{b}\right)^k \frac{1}{k^2}$ and $\mathcal{F}(a, b) = \frac{3}{4a^2} - \frac{a}{3b^3}$, to leading orders in a and b . In the limit as b approaches future infinity,

$$\lim_{b \rightarrow \infty} \langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{a^2}\right) \quad (98)$$

On the other hand, from eq.94 we have

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell\left(\frac{a}{b}\right) \quad (99)$$

where, again, $\ell\left(\frac{a}{b}\right) = \sum_{k=1}^{\infty} \left(\frac{a}{b}\right)^k \frac{1}{k^2}$. Now, however, there is no $\mathcal{F}(a, b)$ term. In the large b limit,

$$\lim_{b \rightarrow \infty} \langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} \quad (100)$$

The reason for this difference is clear when one examines the integrands of $\langle S_{\text{Dou}}^{\text{coll.}} \rangle$ and $\langle S_{\text{link}}^{\text{coll.}} \rangle$. Because Dou uses only 'half' of the 'full diamond' spacetime the volume required to be free of sprinkled points in order to ensure that y is minimal in $J^+(\mathcal{H})$ is artificially truncated. The extended spacetime which we use does not impose this artificial limitation. This is illustrated in Figure 16.

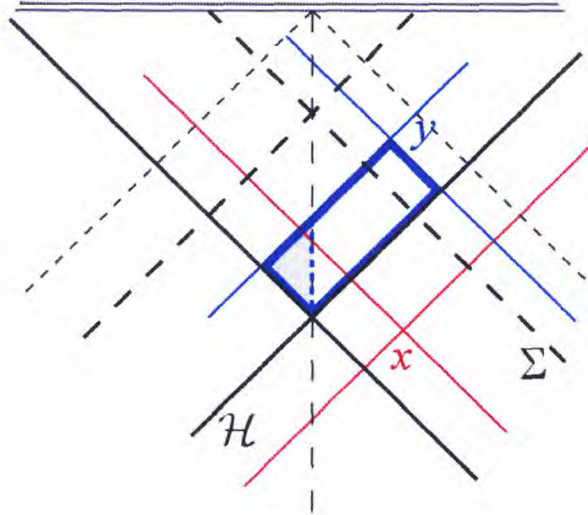


FIGURE 16. An enlarged view of the central area of the ‘extended’ spacetime. In order for y to be minimal in $J^+(\mathcal{H})$ the region bounded in solid blue must be free from sprinkled points. In Dou’s original model this region is truncated, ending at the dotted blue line and thus omitting the area shaded grey.

The consequence of Dou’s truncation is an expression for the volume (in the exponential) which includes a term $-u_y^2/2$, this being necessary to adjust for the omission of the grey triangular region in Figure 16. It is this term which leads to the $\mathcal{F}(a, b)$ correction in $\langle S_{\text{Dou}}^{\text{coll.}} \rangle$. No such term exists in the case of $\langle S_{\text{link}}^{\text{coll.}} \rangle$ and, as a consequence, the final result does not include a $\mathcal{F}(a, b)$ correction.

5.6. Summary of the ‘Extended’ Spacetime. We have shown that extending the spacetime under consideration into a full ‘diamond’ does not affect the link-counting model of Dou, except to remove the $\mathcal{F}(a, b)$ correction term. It is submitted that this alone would justify the extension to the full ‘diamond’, as the additional region considered (the grey triangle of Figure 16) must be empty to ensure that $x-y$ is a link, and to exclude it in 1+1-dimensions would clearly invalidate any extension to higher dimensions. The full ‘diamond’ spacetime provides a solid basis for any extension to higher dimensions. In the

following section we ‘rotate’ our 1+1-dimensional picture to demonstrate a failing with the link-counting model which leads to infinities arising in higher dimensional calculations.

For the moment, however, let us suppose that these failings do not exist, or, as is actually the case, that they can be removed by an extension to the link-counting model. This being the case, we may use our results, together with those derived from the semiclassical models of black hole entropy, to fix the fundamental length scale associated with causal set kinematics. In eq.91, and subsequent equations, we have been working in units in which $\rho = 1$, that is, in units in which length is measured in terms of some fundamental length scale, l_f . This l_f is of order the Planck length, $l_P = \sqrt{G\hbar}$. The exact relationship between the two can be determined as follows.

From semiclassical arguments we have

$$S_{BH} = \frac{A}{4l_P^2} \quad . \quad (101)$$

Let us suppose that the full, 3+1-dimensional calculation for the causal set case gives simply the 1+1-dimensional result above multiplied by the horizon area. Then,

$$\langle S_{\text{link}}^{3+1} \rangle = \frac{\pi^2}{6} \frac{A}{l_f^2} \quad (102)$$

where A/l_f^2 gives the horizon area in fundamental units. We can then equate eq.101 and eq.102 to give

$$\frac{l_f}{l_P} = \pi \sqrt{\frac{2}{3}} \quad . \quad (103)$$

We emphasise that the ratio given in eq.103 is merely illustrative: even if the link-counting model were to give the correct value for $\langle S \rangle$, the return to 3+1 dimensions would introduce other factors, albeit of order 1, which would alter the exact value of l_f/l_P .

6. HIGHER DIMENSIONS: FAILURE OF THE LINK-COUNTING MODEL.

6.1. Failure of the Link-Counting Model. Figure 17 shows the failure of the link-counting model when dealing with more than one spatial dimension. In 1+1 dimensions the entropy integral is rendered finite by the e^{-V} term appearing in the integrand. This term has its physical origin in the requirement that the interval between the two points x and y be free from sprinkled points, that is, that x and y are linked. In 1+1 dimensions V quickly becomes very large as either of x or y move away from the point of intersection of $\mathcal{H}$ and Σ .

However, in 2+1 dimensions (or higher) the intersection of $\mathcal{H}$ and the boundary of the future light-cone of x , $J^+(x)$, is no longer a point but, rather, a curve. This added degree of freedom allows the existence of sprinkled points, y , asymptotically close to $\mathcal{H} \cap J^+(x)$, and arbitrarily far from the intersection of $\mathcal{H}$ and Σ . For these ‘distant’ y , the interval between x and y remains small and highly likely to be free from additional sprinkled points. This is reflected in the analytic limit, as the e^{-V} term in the integrand does not ‘suppress’ the contribution of such points to the integral.

In summary: the link-counting model does not prevent the existence of arbitrarily large numbers of x - y links in 2+1 and higher dimensions. As a consequence, higher-dimensional calculations on this basis will give infinite values for the entropy of the black hole [Dou (2004)].

6.2. Resolving the Problem: Using Triplets of Elements in Higher Dimensions.

Consider the inclusion of an additional sprinkled element in our spacetime, which is causally related to y but not to x , and is to the future of the horizon $\mathcal{H}$. Then, with the addition of further conditions described in §7, the resulting 2+1-dimensional picture is shown in Figure 18. The causal relationships of x , y and z require y to fall within the shaded ‘pseudo-cone’ formed by the intersection of the future light-cones of z and x . This requirement prevents sprinkled points y from being found arbitrarily close to the intersection of $\mathcal{H}$ and the surface of the future light-cone of x . This ensures that the x - y interval increases without limit as y ‘moves away from’ $\mathcal{H} \cap \Sigma$, thus avoiding the infinities arising from the arrangements discussed in relation to Figure 17. In order to further consider this model we discuss its 1+1-dimensional truncation in the following sections.

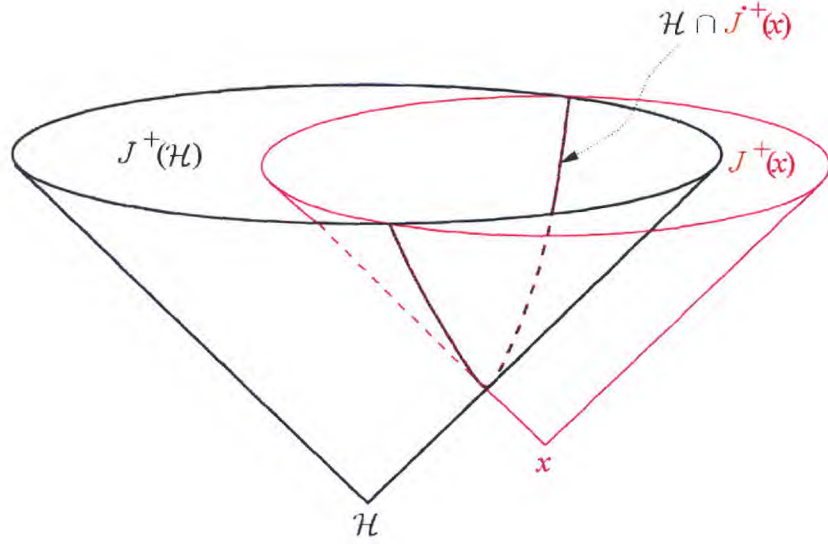


FIGURE 17. Spacetime diagram showing the problem arising when Dou's link-counting model is extended to 2+1 dimensions. By considering sprinkled elements y (not shown) which asymptotically approach the intersection of $\mathcal{H}$ and the boundary of the future light-cone of x , an arbitrarily large number of x - y links can be found.

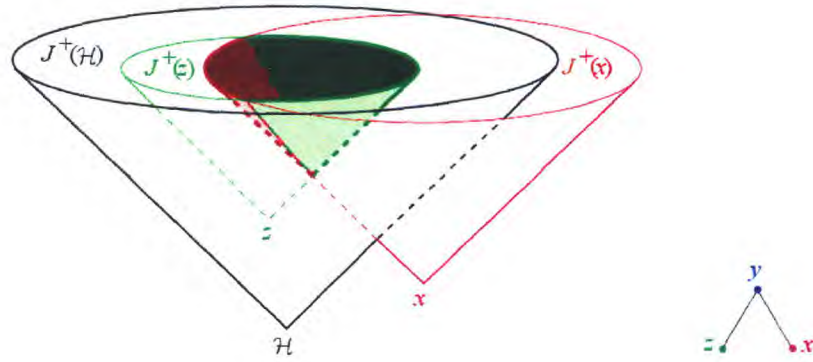


FIGURE 18. Spacetime diagram (left) showing the proposed solution to the problem described above. The restriction of y to the region bounded by the intersection of the future light-cones of x and z (here in solid shading) prevents y 's being arbitrarily close to the intersection of $\mathcal{H}$ and the surface of the future light-cone of x . The Hasse diagram (right) shows the causal structure of x , y and z : we will refer to this construction as the Λ -triplet, recalling this structure.

7. THE Λ -TRIPLET: COLLAPSING SHELL MODEL.7.1. Constructing the Integral for the Λ -Triplet.

7.1.1. *Basic Conditions.* Having sprinkled the elements of a causal set $\mathcal{C}$ into our spacetime, we wish to consider triplets of those elements, arranged as shown in Figure 19 and Figure 20. The coloured lines on this diagram show the ‘light cones’ of the points x , y , and z . The elements are restricted as follows:

location	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\Sigma) \cap J^+(\mathcal{H}) \cap J^-(\text{matter shell})$
	$z \in J^-(\Sigma) \cap J^+(\mathcal{H})$
causal relations	$x \checkmark y$
	$z \checkmark y$
	$\Rightarrow x \not\prec z \quad (\text{i.e. } x \text{ spacelike to } z.)$
maximality/minimality	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \text{ minimal-but-one in } J^+(\mathcal{H})$
	$z \in J^-(\Sigma) \cap J^+(\mathcal{H})$

Where, again, we have used the notation $p \checkmark q$ to denote $p \prec q$ is a link; $p \in \mathcal{A}$ to denote $p \in \mathcal{A}$ and p is maximal in $\mathcal{A}$; and $p \in \mathcal{A}$ to denote $p \in \mathcal{A}$ and p is minimal in $\mathcal{A}$.

We restrict, as before, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, thus excluding only contributions which we will show to be negligible. Thus,

$$\begin{aligned} \text{location} \quad x &\in \mathcal{X} \\ y &\in \mathcal{Y} \end{aligned}$$

where, again,

$$\mathcal{X} = \{p \mid 0 \leq v_p \leq a \text{ and } -\infty < u_p \leq 0\}$$

and now

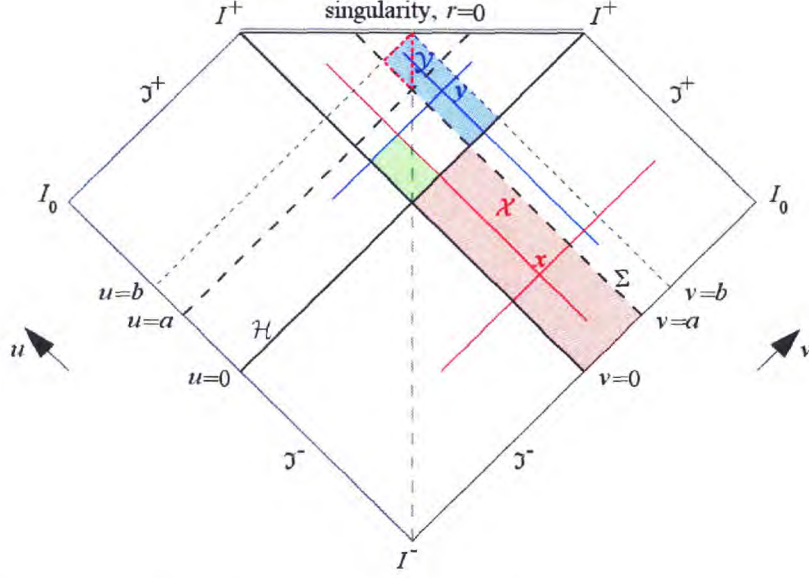


FIGURE 19. Penrose diagram showing the Λ -triplet arrangement. x may fall anywhere within the region $\mathcal{X}$ (shaded red). y may fall anywhere within the region $\mathcal{Y}$ (shaded blue). For a given choice of x and y , the third element, z , may fall anywhere within the region shaded green. Note that the upper bound of u_z is a , even if $u_y > a$. The small triangular region bounded in broken red is not required by considerations of rotational symmetry, but simplifies mathematical analysis and does not affect the results obtained: see Appendix C.1.

$$\mathcal{Y} = \{q \mid a \leq v_q \leq b \text{ and } 0 \leq u_q \leq b\}$$

with all other conditions remaining unchanged.

Note that for $\mathcal{Y}$ we have shifted the upper limit of $u_q \forall q \in \mathcal{Y}$, to give $\{q \mid 0 < u_q < b\}$ rather than the $\{q \mid 0 < u_q < v_q\}$ of the link-counting model. The additional region covered by this change is shown as the small triangular region bounded in broken red of Figure 19. Whilst this change is not required by considerations of rotational symmetry, it simplifies mathematical analysis and does not affect the results obtained: see Appendix C.1.

The condition that y is minimal-but-one in $J^+(\mathcal{H})$, ensures that the only point to the causal past of y , in that region, is z .

In Figure 20 these conditions require that the area shaded grey is free of causal elements, and that the area shaded green is free of causal elements but for one at z .

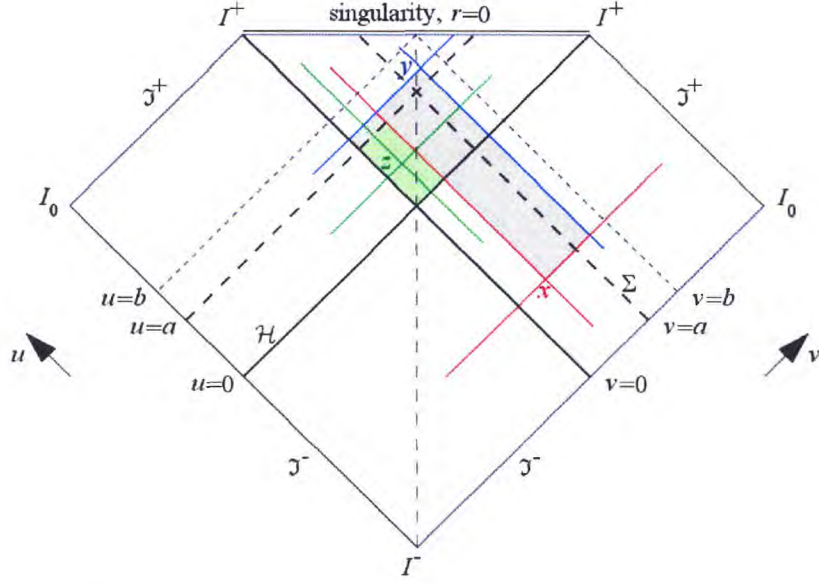


FIGURE 20. Penrose diagram showing a specific choice of x , y , and z . The region shaded grey must be free from sprinkled points. The area shaded green can contain only a single sprinkled element, z .

7.1.2. *Characteristic Functions.* We define our characteristic functions as for the link-counting model, in order to evaluate the discrete sum which will give us our required integral, in the infinitesimal limit. Additionally, we define χ_z by analogy with χ_x and χ_y , and

$$\eta_{xyz} = \begin{cases} 1 & \text{if } J^+(\mathcal{H}) \cap J^-(\Delta V_y) \cap J^+(\Delta V_x)^c \cap (\Delta V_z)^c \text{ is empty} \\ 0 & \text{otherwise} \end{cases}.$$

That is, η_{xyz} is 1 if the green-shaded region of Figure 19⁶ is empty, or empty but for a point at ΔV_z . Otherwise, η_{xyz} is 0.

To simplify our notation, we define

$$\tilde{\mathcal{Z}} = J^+(\mathcal{H}) \cap J^-(\Delta V_y) \cap J^+(\Delta V_x)^c.$$

⁶The green-shaded region of Figure 20 has an upper bound of $u = a$ and shows the limit of $\mathcal{Z}$, the area into which the point z is actually sprinkled.

We note that,

$$\eta_{xyz} = \prod_{\substack{\Delta V_q \in \tilde{\mathcal{Z}} \\ \Delta V_q \neq \Delta V_z}} \bar{\chi}_q \quad . \quad (104)$$

The number of triplets meeting our requirement, $S_{\Lambda\text{-triplet}}^{\text{coll.}}$, is then given by the sum

$$S_{\Lambda\text{-triplet}}^{\text{coll.}} = \sum_{x \in \mathcal{X}} \chi_x \sum_{y \in \mathcal{Y}} \chi_y \xi_{xy} \sum_{z \in \mathcal{Z}} \chi_z \eta_{xyz} \quad (105)$$

where $\mathcal{Z}$ is the region shaded green in the figure.

(Note that whilst the product formulation of η_{xyz} in eq.104 covers volume elements in the region $\tilde{\mathcal{Z}}$, we only sum over elements z in the region $\mathcal{Z}$.)

7.1.3. The Expectation Value of $S_{\Lambda\text{-triplet}}^{\text{coll.}}$.

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle &= \left\langle \sum_{x \in \mathcal{X}} \chi_x \sum_{y \in \mathcal{Y}} \chi_y \xi_{xy} \sum_{z \in \mathcal{Z}} \chi_z \eta_{xyz} \right\rangle \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \sum_{z \in \mathcal{Z}} \langle \chi_x \chi_y \chi_z \xi_{xy} \eta_{xyz} \rangle \\ &= \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \sum_{z \in \mathcal{Z}} \langle \chi_x \rangle \langle \chi_y \rangle \langle \chi_z \rangle \langle \xi_{xy} \rangle \langle \eta_{xyz} \rangle \end{aligned} \quad (106)$$

where the final step is permissible as χ_x , χ_y , χ_z , ξ_{xy} , and η_{xyz} are all independent variables.

As before, $\langle \chi_x \rangle = \langle \chi_y \rangle = \langle \chi_z \rangle = p$; $\langle \bar{\chi}_x \rangle = \langle \bar{\chi}_y \rangle = \langle \bar{\chi}_z \rangle = 1 - p$; $\langle \xi_{xy} \rangle = (1 - p)^{N_{xy}^{\text{int}}}$ where N_{xy}^{int} is the number of volume elements in the interval between ΔV_x and ΔV_y .

Then,

$$\langle \eta_{xyz} \rangle = \text{Pr}(\eta_{xyz} = 1) = (1 - p)^{N_{xy}^{\tilde{\mathcal{Z}}} - 1} \quad (107)$$

where $N_{xy}^{\tilde{\mathcal{Z}}}$ is the number of volume elements in the region $\tilde{\mathcal{Z}}$, and the additional ‘-1’ term in the exponent occurs because we require the existence of a single sprinkled element at ΔV_z

Again, taking V to be the space-time volume of N volume elements, that is, $V = N \cdot \Delta V$.

Then we have:

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \sum_{z \in \mathcal{Z}} p^3 \cdot (1 - p)^{V_{xy}/\Delta V} \cdot (1 - p)^{-1} \quad (108)$$

where $V_{xy} = V_{xy}^{\text{int}} + V_{xy}^{\tilde{\mathcal{Z}}} = (N_{xy}^{\text{int}} + N_{xy}^{\tilde{\mathcal{Z}}}) \cdot \Delta V$

giving

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \sum_{z \in \mathcal{Z}} (\Delta V)^3 \rho^3 \cdot (1 - \rho \Delta V)^{V_{xy}/\Delta V} \cdot (1 - \rho \Delta V)^{-1} \quad (109)$$

and taking $\lim \Delta V \rightarrow dV$

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \rho^3 \int_{\mathcal{X}} \int_{\mathcal{Y}} \int_{\mathcal{Z}} dV^3 e^{-\rho V_{xy}} \quad . \quad (110)$$

7.1.4. *Evaluating V_{xy}^{int} , $V_{xy}^{\tilde{\mathcal{Z}}}$ and V_{xy} .* In the region in which we are interested spacetime is Minkowski, and we consider the 1 + 1-dimensional case, using $dV = \sqrt{-g} du dv = du dv$.

Then

$$V_{xy}^{\text{int}} = \int_{u_x}^{u_y} \int_{v_x}^{v_y} du dv = (v_y - v_x)(u_y - u_x) \quad , \quad (111)$$

$$V_{xy}^{\tilde{\mathcal{Z}}} = \int_0^{u_y} \int_0^{v_x} du dv = u_y v_x \quad . \quad (112)$$

Hence,

$$V_{xy} = -u_x(v_y - v_x) + u_y v_y \quad (113)$$

which is unchanged from the link-counting model, eq.89.

7.1.5. *Using V_{xy} in the Integral Form of $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$.* We have

$$\begin{aligned} \int_{\mathcal{X}} \dots &= \int_0^a dv_x \int_{-\infty}^0 du_x \dots \quad , \\ \int_{\mathcal{Y}} \dots &= \int_a^b dv_y \int_0^b du_y \dots \quad . \end{aligned}$$

We now have an integral $\int_{\mathcal{Z}} du_z dv_z$. We note that neither u_z nor v_z appear in the exponent of the integrand. More importantly, the bounds of the integral over these measures is dependent on the value of u_y if we are to satisfy both the causality restriction $z \prec y$ and the requirement that $z \in \mathcal{Z}$. The causality restriction $y \prec z$ requires that $u_z < u_y$. The restriction that $z \in \mathcal{Z}$ requires that $u_z \leq a$. For $u_y < a$, the requirements $u_z < u_y$ and $u_z \leq a \Rightarrow u_z < u_y$. For $u_y > a$, the requirements $u_z < u_y$ and $u_z \leq a \Rightarrow u_z \leq a$.

$$\int_{\mathcal{Z}} = \begin{cases} \int_0^{v_x} dv_z \int_0^{u_y} du_z & \text{if } 0 \leq u_y < a \\ \int_0^{v_x} dv_z \int_0^a du_z & \text{otherwise} \end{cases} \quad .$$

Ordering appropriately, the integral is given as

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle &= \rho^3 \int_a^b dv_y \int_0^a du_y \int_0^a dv_x \int_{-\infty}^0 du_x \int_0^{v_x} dv_z \int_0^{u_y} du_z e^{-\rho V_{xy}} \\ &\quad + \rho^3 \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x \int_0^{v_x} dv_z \int_0^a du_z e^{-\rho V_{xy}} . \end{aligned} \quad (114)$$

Performing the integrals over u_z and v_z and working in units where $\rho = 1$ gives

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^a du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &\quad + \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x a e^{u_x(v_y - v_x) - u_y v_y} . \end{aligned} \quad (115)$$

7.2. Solving the Integral for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$. Here we give the key elements of the solution to the integral formulation of $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$. The full details of this calculation can be found in Appendix C.2.

7.2.1. Rewriting the Integral. We rewrite the integral for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$ as

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &\quad - \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x (u_y - a) e^{u_x(v_y - v_x) - u_y v_y} . \end{aligned} \quad (116)$$

7.2.2. ‘Removing’ the Second Term. Let us define the first term of $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$ to be $\langle S_{\Lambda\text{-triplet}}^1 \rangle$ and the second term to be $\langle S_{\Lambda\text{-triplet}}^2 \rangle$. Again, see Appendix C.2.

Then

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^2 \rangle &= \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x (u_y - a) e^{u_x(v_y - v_x) - u_y v_y} \\ &= \int_a^b dy \left[-\frac{ae^{-ay}}{y^2} + \frac{ae^{-by}}{y^2} + a(b-a) \frac{e^{-by}}{y} + \frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-ay} \right. \\ &\quad \left. - \frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-by} - (b-a) e^{-by} \right] \end{aligned} \quad (117)$$

It can be shown that each of the terms in this integral is exponentially suppressed in a and/or b as $a \rightarrow \infty$, $b \rightarrow \infty$, with a/b fixed and small.

This gives our final result,

$$\langle S_{\Lambda\text{-triplet}}^2 \rangle \sim 0 \quad (118)$$

for the values of a and b with which we are concerned.

7.2.3. Solving the Remaining Integral.

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^1 \rangle &= \int_a^b dv_y \int_0^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &= \int_a^b dy \left[\frac{a}{y^2} e^{-by} + \frac{ab}{y} e^{-by} - \frac{a}{y^2} - \frac{e^{-by}}{y} \ln \left(\frac{y}{y-a} \right) \right. \\ &\quad \left. - b e^{-by} \ln \left(\frac{y}{y-a} \right) + \frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\ &= \langle S_{\Lambda 1}^1 \rangle + \langle S_{\Lambda 2}^1 \rangle + \langle S_{\Lambda 3}^1 \rangle + \langle S_{\Lambda 4}^1 \rangle + \langle S_{\Lambda 5}^1 \rangle + \langle S_{\Lambda 6}^1 \rangle \quad . \end{aligned} \quad (119)$$

Again, it can be shown that only two terms of this integral are not exponentially suppressed (Appendix C.2):

$$\begin{aligned} \langle S_{\Lambda 3}^1 \rangle &= \int_a^b dy \left[\frac{-a}{y^2} \right] \\ &= \frac{a}{b} - 1 \quad . \end{aligned} \quad (120)$$

and

$$\begin{aligned} \langle S_{\Lambda 6}^1 \rangle &= \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\ &= \langle S_{\text{link}}^{\text{coll.}} \rangle \\ &= \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \quad . \end{aligned} \quad (121)$$

7.2.4. Final Solution for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$.

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - 1 + \ell' \left(\frac{a}{b} \right) \quad (122)$$

where $\ell' \left(\frac{a}{b} \right) = - \sum_{n=2}^{\infty} \left(\frac{a}{b} \right)^n \frac{1}{n^2}$.

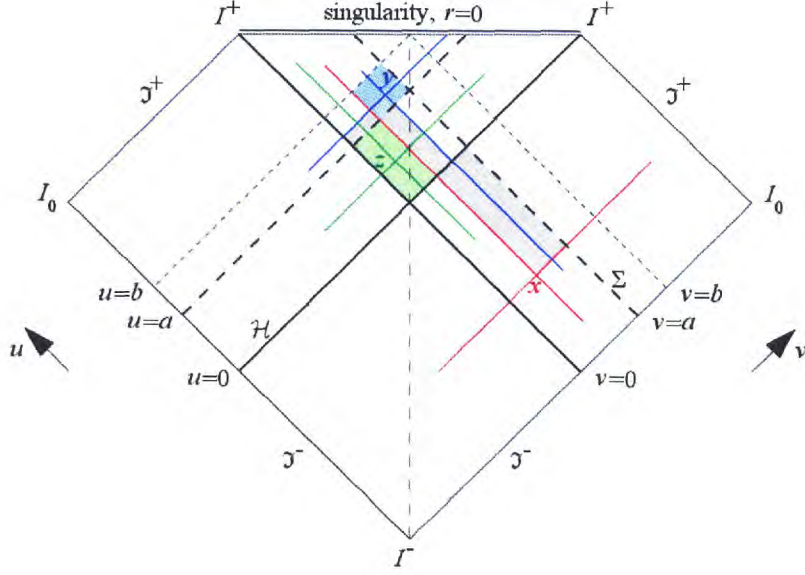


FIGURE 21. Penrose diagram showing possible additional arrangements for the triplet case. The range for x is the same as Figure 19 (unshaded in this diagram). y may range over the region $\mathcal{Y}$ (shaded blue and blue/grey), with $a < u_y < b$ and $v_x < v_y < a$. z is restricted to the region $0 < u_z < a$ and $0 < v_z < v_x$ (shaded green). The areas shaded grey and blue/grey must be free from sprinkled points.

Or, to lowest order in (a/b) ,

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - 1 + \mathcal{O}\left(\left(\frac{a}{b}\right)^2\right) . \quad (123)$$

We note that the final correction term in a/b is of one higher order than the correction term in the result of Dou and Sorkin.

7.3. Additional Arrangements for Triplet Counting. Unlike the link-counting case (see §5), the Λ -triplet does not possess additional arrangements for which $v_x < 0$, as this would not allow for $x \not\prec z$ (that is, $\forall x$ with $v_x < 0 \nexists z$ such that $z \notin J^+(x)$). However, there remains the case of the arrangement shown in Figure 21. Here, again, this arrangement does not carry with it the idea of ‘clustering’ around the intersection of $\mathcal{H}$ and Σ , with which we are concerned (and is not permitted in the case of the z - and l -triplets of §9 and §10). We have

$$\begin{aligned}
\langle S_{\Lambda\text{-triplet}}^{\text{coll.}+} \rangle &= \int_a^b du_y \int_0^{v_y} dv_x \int_0^a dv_y \int_{-\infty}^0 du_x \\
&\quad \times \int_0^{v_x} dv_z \int_0^a du_z e^{u_x(a-v_x)-u_y v_y} \\
&< a^2 \int_0^a dv_y \sum_{n=1}^{\infty} \frac{v_y^{n-1}}{n a^n} (e^{-a v_y} - e^{-b v_y})
\end{aligned} \tag{124}$$

as can be seen by comparison with eq.95, taking the upper limit of the av_x factor arising from the integrals over du_z and dv_z . See the discussion in §5.4.1 for further details.

8. THE Λ -TRIPLET: STATIC MODEL.

8.1. Introduction. We have thus far restricted ourselves, in §5, §6 and §7, to a consideration of the collapsing black hole model proposed by Dou. However, the infinities arising in the link-counting case (as described in §6) also occur in the case of the static black hole, to which we now turn.

8.2. Constructing the Integral for the Static-Case Λ -Triplet.

8.2.1. Basic Conditions. The elements are restricted in the same way as the collapsing case (see Figure 22 and Figure 23):

location	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\Sigma) \cap J^+(\mathcal{H})$
	$z \in J^-(\Sigma) \cap J^+(\mathcal{H})$
causal relations	$x \prec y$
	$z \prec y$
	$\Rightarrow x \not\prec z \quad (\text{i.e. } x \text{ spacelike to } z.)$
maximality/minimality	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \text{ minimal-but-one in } J^+(\mathcal{H})$
	$z \in J^-(\Sigma) \cap J^+(\mathcal{H}) \quad .$

8.2.2. Characteristic functions. We perform the same analysis as for the collapsing shell, defining characteristic functions χ_x , etc., ξ_{xy} and η_{xyz} as previously.

The number of triplets meeting our requirement, $S_{\Lambda\text{-triplet}}^{\text{static}}$, is again given by the sum

$$S_{\Lambda\text{-triplet}}^{\text{static}} = \sum_{x \in \mathcal{X}} \chi_x \sum_{y \in \mathcal{Y}} \chi_y \xi_{xy} \sum_{z \in \mathcal{Z}_{xy}} \chi_z \eta_{xyz} \quad . \quad (125)$$

Sprinkling with a Poisson distribution enables us to calculate the expectation value of

$$S_{\Lambda\text{-triplet}}^{\text{static}}, \left\langle S_{\Lambda\text{-triplet}}^{\text{static}} \right\rangle .$$

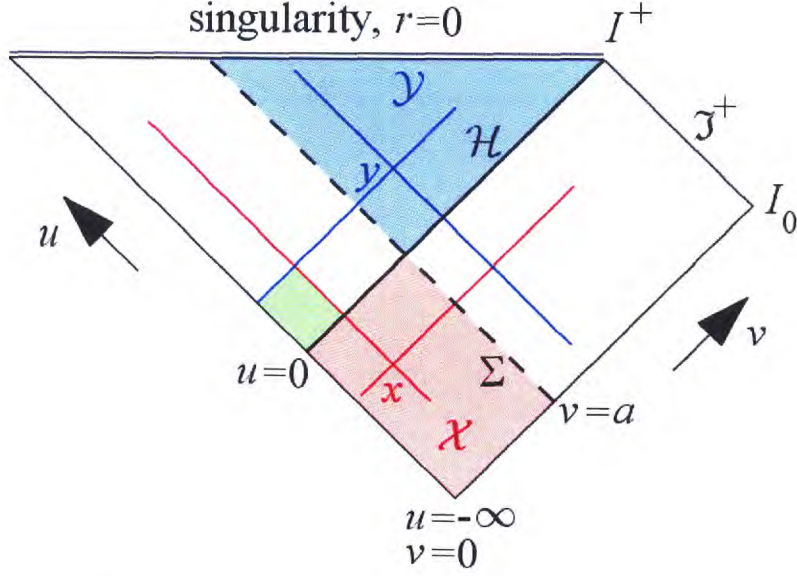


FIGURE 22. Penrose diagram showing the Λ -triplet arrangement for a static black hole. x may fall anywhere within the region $\mathcal{X}$ (shaded red). y may fall anywhere within the region $\mathcal{Y}$ (shaded blue). For a given choice of x and y , the third element, z , may fall anywhere within the region shaded green.

8.2.3. The Expectation Value of $S_{\Lambda\text{-triplet}}^{\text{static}}$

$$\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle = \rho^3 \int_{\mathcal{X}} \int_{\mathcal{Y}} \int_{\mathcal{Z}_{xy}} dV^3 e^{-\rho V_{xy}} \quad (126)$$

where V_{xy} is defined as previously.

8.2.4. *Evaluating V_{xy}^{int} , $V_{xy}^{\mathcal{Z}}$ and V_{xy} .* We consider 1+1-dimensional truncated spacetime, and use $dV = \sqrt{-g}dudv = \frac{16M^3}{r} e^{-r/2M} dudv$, where $uv = (1 - \frac{r}{2M}) e^{r/2M}$. Then

$$V_{xy}^{\text{int}} = 16M^3 \int_{u_x}^{u_y} \int_{v_x}^{v_y} \frac{e^{-r/2M}}{r} dudv, \quad (127)$$

$$V_{xy}^{\mathcal{Z}} = 16M^3 \int_0^{u_y} \int_0^{v_x} \frac{e^{-r/2M}}{r} dudv. \quad (128)$$

Now, in general,

$$16M^3 \int_{u_p}^{u_q} \int_{v_s}^{v_t} \frac{e^{-r/2M}}{r} dudv = r_{pt}^2 - r_{qt}^2 - r_{ps}^2 + r_{qs}^2, \quad (129)$$

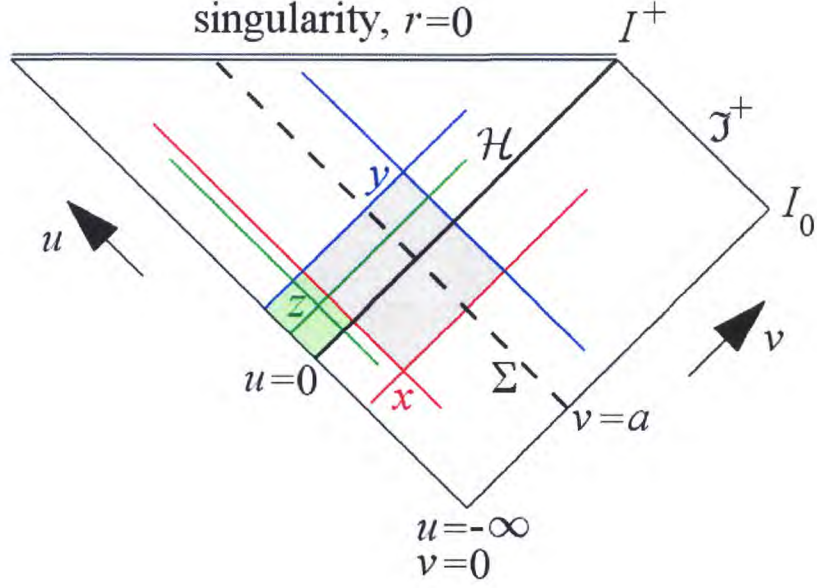


FIGURE 23. Penrose diagram showing a specific choice of x , y , and z . The region shaded grey must be free from sprinkled points. The area shaded green can contain only a single sprinkled element, z .

where, for example, $u_p v_t = \left(1 - \frac{t_{pt}}{2M}\right) e^{r_{pt}/2M}$. See Appendix C.3. Thus,

$$V_{xy}^{\text{int}} = r_{xy}^2 - r_{yy}^2 - r_{xx}^2 + r_{yx}^2, \quad (130)$$

$$V_{xy}^{\mathcal{Z}} = r_{0x}^2 - r_{yx}^2 - r_{00}^2 + r_{y0}^2 = (2M)^2 - r_{yx}^2 \quad (131)$$

where we have used the fact that, if $u_p = 0$ or $u_q = 0$ or $v_r = 0$ or $v_s = 0$, then $r^2 = (2M)^2$.

This gives the result

$$V_{xy} = (2M)^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2. \quad (132)$$

8.2.5. *Using V_{xy} in the Integral Form of $\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle$.* In a similar fashion to the above volume integrals, we need to express the integral for $\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle$ in terms of u and v , with appropriate limits. Omitting the $\sqrt{-g}$ terms for the moment, and using $uv = 1$ at $r = 0$,

we have

$$\begin{aligned}\int_{\mathcal{X}} &= \int_0^a dv_x \int_{-\infty}^0 du_x \quad , \\ \int_{\mathcal{Y}} &= \int_a^\infty dv_y \int_0^{1/v_y} du_y \quad , \\ \int_{\mathcal{Z}_{xy}} &= \int_0^{v_x} dv_z \int_0^{u_y} du_z \quad .\end{aligned}$$

Including the $\sqrt{-g}$ terms, and ordering appropriately, the integral, eq.126 gives

$$\begin{aligned}\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle &= (16M^3)^3 \rho^3 \int_{-\infty}^0 du_x \int_0^a dv_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \int_0^{v_x} dv_z \\ &\times \int_0^{u_y} du_z \frac{e^{-r_{xx}/2M}}{r_{xx}} \frac{e^{-r_{yy}/2M}}{r_{yy}} \frac{e^{-r_{zz}/2M}}{r_{zz}} e^{-\rho((2M)^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \quad .\end{aligned}\quad (133)$$

But now,

$$16M^3 \int_0^{v_x} dv_z \int_0^{u_y} du_z \frac{e^{-r_{zz}/2M}}{r_{zz}} = V_{xy}^Z = (2M)^2 - r_{yx}^2 \quad (134)$$

so

$$\begin{aligned}\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle &= (16M^3)^2 \rho^3 \int_{-\infty}^0 du_x \int_0^a dv_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \\ &\times \frac{e^{-r_{xx}/2M}}{r_{xx}} \frac{e^{-r_{yy}/2M}}{r_{yy}} e^{-\rho((2M)^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \left((2M)^2 - r_{yx}^2 \right) \quad .\end{aligned}\quad (135)$$

8.3. Solving the Integral for $\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle$.

8.3.1. *Dou's Approach to Solving $\langle S_{\text{Dou}}^{\text{static}} \rangle$.* Here we summarize the approach taken by Dou in evaluating his integral, formulated using only two points, x and y , as given in his doctoral thesis. A more detailed version of this summary can be found in Appendix C.4. For Dou,

$$\begin{aligned}\langle S_{\text{Dou}}^{\text{static}} \rangle &= (16M^3)^2 \int_0^a dv_x \int_{-\infty}^0 du_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \\ &\times \frac{e^{-r_{xx}/2M}}{r_{xx}} \frac{e^{-r_{yy}/2M}}{r_{yy}} e^{-(4M^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)}\end{aligned}\quad (136)$$

in units for which $\rho = 1$. (We have replaced Dou's original reference to the position of the hypersurface Σ , $v = v_0$, with our own, $v = a$, which corresponds more closely to the collapsing shell case.)

Taking $u_x = u_x(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $v_x = v_x(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $u_y = u_y(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $v_y = v_y(r_{xx}, r_{xa}, r_{xy}, r_{yy})$ and introducing the appropriate Jacobian and limits gives

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha e^{r_{yy}^2} dr_{yy} \int_\alpha^\infty dr_{xy} \frac{r_{xy}}{r_{xy} - \alpha} e^{-r_{xy}^2} \int_\alpha^{r_{xy}} \frac{r_{xa}}{r_{xa} - \alpha} dr_{xa} \int_\alpha^{r_{xa}} e^{r_{xx}^2} dr_{xx} \quad (137)$$

writing $\alpha \equiv 2M$.

Let $x = r_{xy}$, $y = r_{xa}$, $z = r_{xx}$. Then,

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha e^{r_{yy}^2} dr_{yy} I(\alpha) \quad (138)$$

with

$$I(\alpha) = \int_\alpha^\infty dx \frac{x}{x - \alpha} e^{-x^2} \int_\alpha^x dy \frac{y}{y - \alpha} \int_\alpha^y dz e^{z^2} \quad (139)$$

Series expansions allow eq.139 to be written as

$$I(\alpha) = \frac{\alpha}{2} \left(\sum_{n=1}^\infty \frac{1}{n^2} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) - \mathcal{O}\left(\frac{1}{\alpha^4}\right) + \dots \right) \quad (140)$$

which is then substituted into eq.138 to give, after solving the integral over dr_{yy} using a further series expansion,

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \quad (141)$$

to leading order in α .

8.3.2. Application to the Static Case Λ -Triplet Integral. Again, we take units in which $\rho = 1$, and write $\alpha \equiv 2M$. The triplet integral may be split into two terms, thus:

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle &= (16M^3)^2 \alpha^2 \int \int \int \int \frac{e^{-r_{xx}/\alpha}}{r_{xx}} \frac{e^{-r_{yy}/\alpha}}{r_{yy}} e^{-(\alpha^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \\ &\quad - (16M^3)^2 \int \int \int \int r_{yx}^2 \frac{e^{-r_{xx}/\alpha}}{r_{xx}} \frac{e^{-r_{yy}/\alpha}}{r_{yy}} e^{-(\alpha^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \end{aligned} \quad (142)$$

where $\int \int \int \int$ represents $\int_{-\infty}^0 du_x \int_0^\alpha dv_x \int_\alpha^\infty dv_y \int_0^{1/v_y} du_y$.

The first of these terms is a multiple of Dou's original integral, and hence we have:

$$\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle = \alpha^2 \langle S_{\text{Dou}}^{\text{static}} \rangle - \langle S_{yx}^{\text{static}} \rangle \quad (143)$$

where

$$\langle S_{yx}^{\text{static}} \rangle = (16M^3)^2 \int \int \int \int r_{yx}^2 \frac{e^{-r_{xx}/\alpha}}{r_{xx}} \frac{e^{-r_{yy}/\alpha}}{r_{yy}} e^{-(\alpha^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \quad (144)$$

where, again, $\int \int \int \int$ represents $\int_{-\infty}^0 du_x \int_0^a dv_x \int_a^\infty dv_y \int_0^{1/v_y} du_y$.

Using Dou's method, eq.144 can be rewritten as

$$\begin{aligned} \langle S_{yx}^{\text{static}} \rangle = & 4e^{-\alpha^2} \int_0^\alpha e^{r_{yy}^2} dr_{yy} \int_\alpha^\infty dr_{xy} r_{yx}^2 \frac{r_{xy}}{r_{xy} - \alpha} e^{-r_{xy}^2} \\ & \times \int_\alpha^{r_{xy}} \frac{r_{xa}}{r_{xa} - \alpha} dr_{xa} \int_\alpha^{r_{xa}} e^{r_{xx}^2} dr_{xx} \quad . \end{aligned} \quad (145)$$

8.4. Analytic Intractability of the Static Case Λ -Triplet Integral. Unfortunately, eq.145 is not susceptible to the approach outlined above, due to the presence of an r_{yx}^2 term in the integrand, without a corresponding integral over dr_{yx} . Furthermore, the definition of r_{yx} means it cannot be redefined in terms of the appropriate measures:

$$\left(1 - \frac{r_{yx}}{\alpha}\right) e^{r_{yx}/\alpha} = \frac{(\alpha - r_{yy})(\alpha - r_{xx})}{\alpha(\alpha - r_{xy})} e^{(r_{yy} + r_{xx} - r_{xy})/\alpha} \quad . \quad (146)$$

The implicit nature of r_{yx} necessitates a numerical approach, which may adopt two modes of attack. In the first, a numerical analysis of eq.145 may be undertaken. Alternatively, a direct simulation of sprinkling into the appropriate spacetime may be executed, without the formulation of intermediate, analytic descriptions. In this latter instance, the stochastic processes of the causal set's embedding are performed directly, with counting procedures applied directly to links thus created: there is no stage at which the embedding is taken to the Poisson limit, $\Delta V \rightarrow 0$. In fact, there are many similarities between the direct approach and a numerical treatment of eq.145 which, as a consequence of its complexity, appears to be suitable for a treatment using Monte Carlo methods.

Further consideration of the Λ -triplet indicated that strenuous efforts to solve this integral, by either means, were not justified (see §9). It should also be noted that the region of spacetime with which we are concerned in this instance is not one for which sprinkling techniques have been devised. Thus far, numerical models have been constructed in Minkowski spacetimes or, at most, those which are conformally flat.

At this point we would merely observe that the similarities between $\langle S_{\text{Dou}}^{\text{static}} \rangle$ and $\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle$, taken together with the results for the collapsing shell case, above, strongly suggest that the addition of a further element to the static, link-counting model will leave the 1+1-dimensional case unchanged, as desired. Subsequent work, detailed in §10, confirms this correlation for a different triplet arrangement, the ‘ l -triplet’. Additionally, the l -triplet work demonstrates how a term similar in form to the $\alpha^2 \langle S_{\text{Dou}}^{\text{static}} \rangle$ term of eq.143 can cancel with a component of a second term similar to $\langle S_{yx}^{\text{static}} \rangle$. This removes the apparent dependency on α , to leave a 1+1-dimensional solution which is a pure number to leading order, as required, and leads further support to the assertion that a numerical solution of $\langle S_{\Lambda\text{-triplet}}^{\text{static}} \rangle$ would be in agreement with $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$.

9. FAILURE OF THE Λ -TRIPLET AND SOME POSSIBLE SOLUTIONS.

9.1. Introduction. Whilst the Λ -triplet introduced in §7 solves the problem arising with Dou's original link-counting model, as described in §6, further investigation has shown that it does not ensure the finiteness of any entropy result calculated through a causal set approach. The exact nature of the outstanding problem is discussed below, together with some suggested resolutions.

9.2. Statement of the Problem. It will be recalled that, in Figure 18, the introduction of an additional sprinkled element, z , and the restriction of y to the region $J^+(x) \cap J^+(z)$ prevents y from asymptotically approaching $J^+(x) \cap \mathcal{H}$, and hence removes the infinite number of x - y links which would otherwise arise. However, Figure 24 illustrates a problem which, it turns out, remains when one extends the Λ -triplet model to 2+1 (or higher) dimensions. We now turn from the position of y as it 'moves' further into the future of x , and consider x as it 'moves' further into the past of y . Initially, we will restrict ourselves to consideration of the collapsing shell case: the same problem arises in the static case but the resolution discussed in this section does not remove the r_{yx} term in the static-case integrand (see eq.145), and is not, therefore, a fruitful approach to an analytic solution. §10 will resolve this remaining issue.

In the Λ -triplet, z 'enforces' an ever increasing spacetime interval between x and y as y 'moves' into the future, further away from the intersection of $\mathcal{H}$ and Σ : Figure 18. This increasing interval exponentially suppresses the contribution of any such y . Suppose now that, whilst y remains close to the $\mathcal{H}$ - Σ intersection, x 'moves' away into the past. x must, of course, remain within the causal pasts of both y and Σ . However, in a similar manner to $J^+(x) \cap \mathcal{H}$, there exists the intersection $J^-(y) \cap \Sigma$. Furthermore, z does not bound x away from Σ , as x is *spacelike* to z . Thus, x can asymptotically approach $J^-(y) \cap \Sigma$ as it 'moves' further into the past, simultaneously keeping the x - y interval small, and thus avoiding any exponential suppression in the analytic counting of Λ -triplets. As with the link-counting case, this results in an infinite number of Λ -triplets which satisfy the conditions imposed on x , y and z . Consequently, the associated expectation value for the black-hole entropy will be infinite when the Λ -triplet model is extended to 2+1 or higher dimensions.

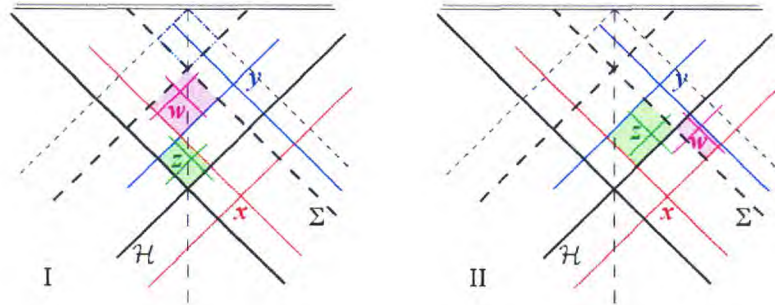


FIGURE 25. Penrose diagrams showing two possible arrangements of four elements. (The spacetime of the black hole is shown truncated for clarity.) In case (I) an extra element, w , has been added to bound x away from Σ . In case (II) w mirrors z , which has been moved so that it is now causally related to both x and y . z now bounds x away from Σ . The details of these diagrams can be found discussed in the main text.

9.3. Using a ‘Quadruplet’ of Elements. Clearly, we need to ensure that x is bounded away from Σ , just as y is bounded away from $\mathcal{H}$. There are a couple of ways in which this might be achieved through the addition of a fourth element, w , as shown in Figure 25.

In case (I), an element w has been added, to fall within the region shaded light-purple. x is then required to be maximal-but-one in $J^-(\Sigma)$, with the maximal element in that region being w . In this way, x is bounded away from Σ , just as the requirement that y be minimal-but-one in $J^+(\mathcal{H})$ ensures that y is bounded away from $\mathcal{H}$. (Note that w is in the causal past of Σ ($w \in J^-(\Sigma)$), but not that of y ($w \notin J^-(y)$), which requires an upper bound on u_y , $u_y < a$.)

In case (II), it is the shifting of z , such that it is now in the causal future of x , which bounds x away from Σ . w has been added as a ‘mirror’ of z , to produce an aesthetically pleasing ‘diamond’ formation of sprinkled elements.

At this juncture, rather than introduce further complexity by pressing forward with such four-element models, it is appropriate to move on and consider a triplet arrangement which is, in fact, case (II) of Figure 25, without the addition of w . Somewhat for want of a better sobriquet, we will refer to this arrangement as the z -triplet.

9.4. The z -Triplet. Figure 26 shows the z -triplet arrangement, in which z is within both the causal past of y and the causal future of x . We now require that x is maximal-but-one

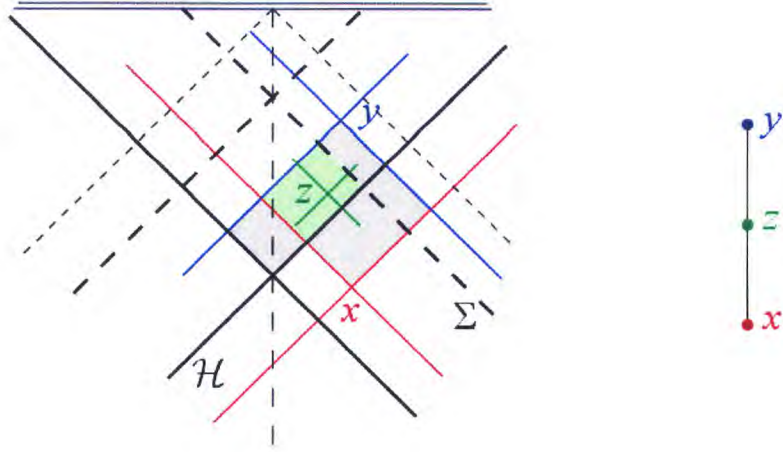


FIGURE 26. Penrose diagram (left) showing a second possible arrangement of three elements, the z -triplet. z is now within both the causal past of y and the causal future of x , thus bounding both y away from $\mathcal{H}$, and x away from Σ . The area shaded grey must be free from sprinkled elements. The area shaded green must contain one and only one element. The Hasse diagram (right) shows the causal structure of x , y and z . Note that only links are shown (i.e. the causal connection between x and y is not shown).

in the causal past of Σ . Such an arrangement ensures that x and y are bounded away from $\mathcal{H}$ and Σ , respectively. The elements are restricted as follows:

location	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	$y \in J^+(\Sigma) \cap J^+(\mathcal{H})$
	$z \in J^-(\Sigma) \cap J^+(\mathcal{H})$
causal relations	$x \prec y$ (Note that this is no longer a link.)
	$z \prec y$ (i.e. z remains linked to y .)
	$\Rightarrow x \prec z$ (i.e. x is now linked to z .)
maximality/minimality	$x \in J^-(\Sigma) \cap J^-(\mathcal{H})$
	y minimal-but-one in $J^+(\mathcal{H})$

We restrict, as before, $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, (thus excluding only contributions which we have shown to be negligible) where $\mathcal{X} = \{p \mid 0 < v_p < a \text{ and } -\infty < u_p < 0\}$ and $\mathcal{Y} = \{q \mid a < v_q < b \text{ and } 0 < u_q < v_q\}$.

The integral for the expectation value of the entropy $S_{z\text{-triplet}}^{\text{coll.}}$, $\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle$, having performed the integrals over du_z and dv_z , is given as

$$\begin{aligned} \langle S_{z\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^a du_y \int_0^a dv_x \int_{-\infty}^0 du_x (a - v_x) u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &\quad + \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x (a - v_x) a e^{u_x(v_y - v_x) - u_y v_y} \\ &= \langle S_{z\text{-triplet}}^1 \rangle + \langle S_{z\text{-triplet}}^2 \rangle \quad . \end{aligned} \quad (147)$$

Writing y for v_y , and x for u_y , and performing the integrals over du_x and dv_x gives

$$\begin{aligned} \langle S_{z\text{-triplet}}^1 \rangle &= a \int_a^b dy \int_0^a dx x e^{-xy} \ln \left(\frac{y}{y-a} \right) \\ &\quad - \int_a^b dy \int_0^a dx x e^{-xy} \left[y \ln \left(\frac{y}{y-a} \right) - a \right] \\ &= a \int_a^b dy \ln \left(\frac{y}{y-a} \right) \left[\frac{1}{y^2} - \frac{e^{-ay}}{y^2} - \frac{ae^{-ay}}{y} \right] \\ &\quad - \int_a^b dy \left[y \ln \left(\frac{y}{y-a} \right) - a \right] \left[\frac{1}{y^2} - \frac{e^{-ay}}{y^2} - \frac{ae^{-ay}}{y} \right] \end{aligned} \quad (148)$$

and

$$\begin{aligned} \langle S_{z\text{-triplet}}^2 \rangle &= a^2 \int_a^b dy \int_a^b dx e^{-xy} \ln \left(\frac{y}{y-a} \right) \\ &\quad - a \int_a^b dy \int_a^b dx e^{-xy} \left[y \ln \left(\frac{y}{y-a} \right) - a \right] \\ &= a^2 \int_a^b dy \ln \left(\frac{y}{y-a} \right) \left[\frac{e^{-ay}}{y} - \frac{e^{-by}}{y} \right] \\ &\quad - a \int_a^b dy \left[y \ln \left(\frac{y}{y-a} \right) - a \right] \left[\frac{e^{-ay}}{y} - \frac{e^{-by}}{y} \right] \quad . \end{aligned} \quad (149)$$

Disregarding terms in e^{-ay} and e^{-by} , which are exponentially suppressed for large a :

$$\begin{aligned}
\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle &= a \int_a^b dy \frac{1}{y^2} \ln \left(\frac{y}{y-a} \right) - \int_a^b dy \frac{1}{y} \ln \left(\frac{y}{y-a} \right) + a \int_a^b dy \frac{1}{y^2} \quad (150) \\
&= \int_a^b dy \left[\frac{a}{y^2} \sum_{n=1}^{\infty} \frac{a^n}{ny^n} - \frac{1}{y} \sum_{m=1}^{\infty} \frac{a^m}{my^m} + \frac{a}{y^2} \right] \\
&= \int_a^b dy \left[\sum_{n=2}^{\infty} \frac{a^n}{(n-1)y^{n+1}} - \sum_{m=2}^{\infty} \frac{a^m}{my^{m+1}} \right] \\
&= \left[-\sum_{n=2}^{\infty} \frac{a^n}{n(n-1)y^n} + \sum_{m=2}^{\infty} \frac{a^m}{m^2 y^m} \right]_a^b \\
&= -\sum_{n=2}^{\infty} \frac{a^n}{n(n-1)b^n} + \sum_{m=2}^{\infty} \frac{a^m}{m^2 b^m} + \sum_{n=2}^{\infty} \frac{1}{n(n-1)} - \sum_{m=2}^{\infty} \frac{1}{m^2} \\
&= \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n(n-1)} \right) \left(\frac{a}{b} \right)^n + 1 - \left(\frac{\pi^2}{6} - 1 \right) .
\end{aligned}$$

Thus, our final result:

$$\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle = 2 - \frac{\pi^2}{6} + \ell'' \left(\frac{a}{b} \right) \quad (151)$$

where $\ell'' \left(\frac{a}{b} \right) = \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n(n-1)} \right) \left(\frac{a}{b} \right)^n$. Again, the lowest order corrective term is of order 2 (being $-\frac{1}{2} \left(\frac{a}{b} \right)^2$).

Appendix D provides confirmation of this calculation, making use of some results from §7.

The results for the z -triplet suggest that a triplet of elements can be constructed which *does* avoid infinities in the calculation of entropy. However, it will be recalled that in the case of the Λ -triplet, the integral formulated for the static black hole was found to be analytically intractable (§8.4). This was a consequence of an r_{yx} term arising in the integrand, which cannot be expressed in terms of the appropriate measures. In fact, this is true of all the arrangements discussed in this section, which have a third element in $J^-(\Sigma) \cap J^+(\mathcal{H})$: this element is unavoidably bounded in regions defined by v_x and u_y , leading to an r_{yx} term in any integral formulated to cover the static case. Thus, whilst the z -triplet succeeds in providing a finite result for the collapsing case, any extension to a static black hole will require numerical analysis.

One arrangement which does not fall into this category is the ‘ l -triplet’ introduced in §10. Here the third element is found in $J^-(\mathcal{H}) \cap J^+(\Sigma)$, and its bounding region is defined by v_y and u_x . This leads to an integral over r_{xy} in the static case, which is analytically solvable.

10. THE l -TRIPLET: COLLAPSING AND STATIC CASES.

10.1. Introduction. As discussed in §9 we turn now to a triplet arrangement in which the ‘third’ element is found in $J^-(\mathcal{H}) \cap J^+(\Sigma)$, with its bounding region defined by v_y and u_x . This leads to an integral over r_{xy} in the static case, which should prove to be analytically solvable. We refer to this triplet as the linear- or l -triplet.

We follow §7 and §8, dealing first with the collapsing case, and then the static case.

10.2. The Collapsing Case.

10.2.1. Introduction. To recap: our original Λ -triplet model aimed to solve the link-counting problem shown in Figure 17 by restricting y to the region bounded by the intersection of the future light-cones of x and z (Figure 18). However, Figure 24 showed that the Λ -triplet model fails, as x remains free to approach asymptotically the intersection of Σ and the surface of the past light-cone of y .

The z -triplet arrangement of §9 bounded y away from $\mathcal{H}$, solving the link-counting problem, and x away from Σ , solving the problem with the Λ -triplet model. The l -triplet also solves both problems, but does so by bounding y away from Σ , and x away from $\mathcal{H}$. Hence, this approach also ensures that arbitrarily large numbers of x - y links are not found in 2+1 and higher dimensions.

10.2.2. The Expectation Value of $S_{l\text{-triplet}}^{\text{coll.}}$. We proceed as previously, noting that the integral over $du_z dv_z$ is now given as $u_x(a - v_y)$:

$$\begin{aligned}
\langle S_{l\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x \int_{u_x}^0 du_z \int_a^{v_y} dv_z e^{u_x(v_y - v_x) - u_y v_y} \quad (152) \\
&= \int_a^b dv_y (a - v_y) \int_0^b du_y e^{-u_y v_y} \int_0^a dv_x \int_{-\infty}^0 du_x u_x e^{u_x(v_y - v_x)} \\
&= \int_a^b dv_y (a - v_y) \int_0^b du_y e^{-u_y v_y} \int_0^a dv_x \left(\frac{-1}{(v_y - v_x)^2} \right) \\
&= \int_a^b dv_y (a - v_y) \int_0^b du_y e^{-u_y v_y} \left(\frac{1}{a - v_y} + \frac{1}{v_y} \right) \\
&= a \int_a^b dv_y \int_0^b du_y \frac{e^{-u_y v_y}}{v_y} \quad (\text{over})
\end{aligned}$$

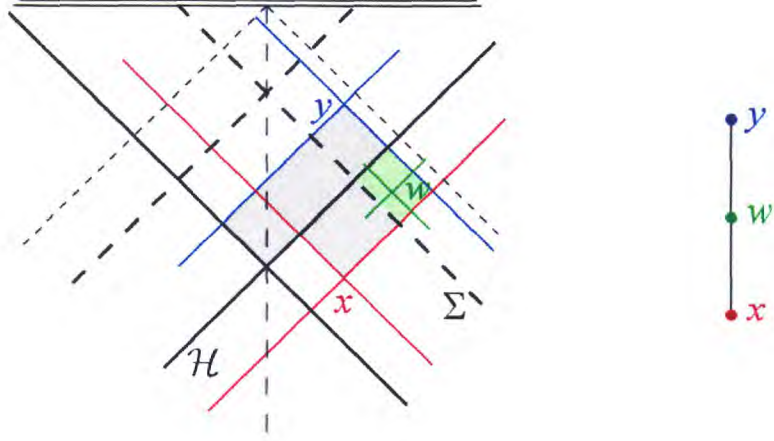


FIGURE 27. Penrose diagram (left) showing the arrangement of three elements in the l -triplet, in the case of a collapsing black hole. w is within both the causal past of y and the causal future of x , but now bounds y away from Σ , and x away from $\mathcal{H}$. The area shaded grey must be free from sprinkled elements. The area shaded green must contain one and only one element. The Hasse diagram (right) shows the causal structure of x , y and w . Note that only links are shown (i.e. the causal connection between x and y is not shown).

$$\begin{aligned}
 &= a \int_a^b dv_y \left(\frac{1 - e^{-bv_y}}{v_y^2} \right) && \text{(cont.)} \\
 &= a \left(-\frac{1}{b} + \frac{1}{a} \right) \quad \text{giving} \\
 \langle S_{l\text{-triplet}}^{\text{coll.}} \rangle &= 1 - \frac{a}{b} . && (153)
 \end{aligned}$$

neglecting the term in $e^{-bv_y} < e^{-ab}$. Unlike $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$ (eq.122) the expression for $\langle S_{l\text{-triplet}}^{\text{coll.}} \rangle$ does not contain an $\ell'(\frac{a}{b})$ of $\mathcal{O}\left(\left(\frac{a}{b}\right)^2\right)$ and higher. Mathematically, this is a consequence of the term in the integrand in u_x now being $u_x e^{-u_x(v_y - v_x)}$, rather than the simple $e^{-u_x(v_y - v_x)}$ of, say, $\langle S_{\text{link}}^{\text{coll.}} \rangle$. This ensures that performing the integrals over du_x and dv_x no longer results in a logarithmic integrand, thus avoiding a correction term in a power series.

As with the Λ -triplet, the contributions from additional arrangements will be negligible.

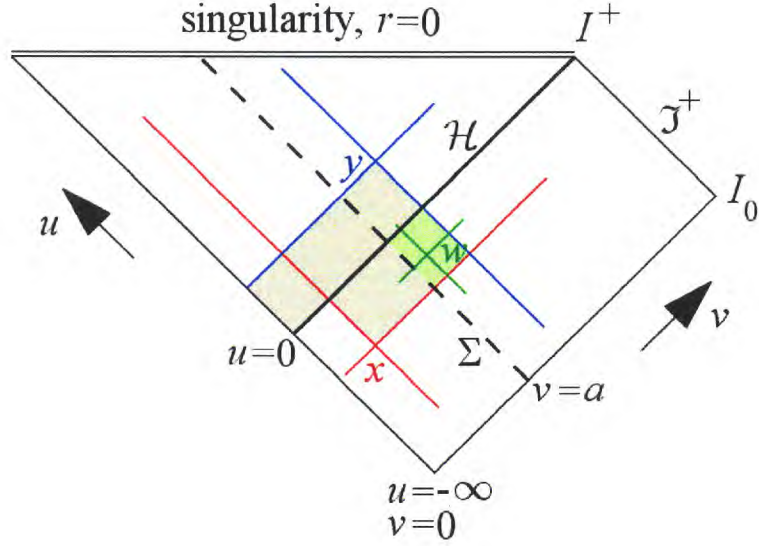


FIGURE 28. Penrose diagram showing the arrangement of three elements in the l -triplet, in the case of a static black hole. w is within both the causal past of y and the causal future of x , but now bounds y away from Σ , and x away from $\mathcal{H}$. The area shaded grey must be free from sprinkled elements. The area shaded green must contain one and only one element.

10.3. The Static Case. The integral is the same as in the static-case Λ -triplet (eq.135), but now

$$\begin{aligned}
 V_{xy}^{\mathcal{Z}} &= 16M^3 \int_{u_x}^0 du_z \int_a^{v_y} dv_z \frac{e^{-r/2M}}{r} \\
 &= r_{xy}^2 - r_{0y}^2 - r_{xa}^2 + r_{0a}^2 \\
 &= r_{xy}^2 - r_{xa}^2 .
 \end{aligned} \tag{154}$$

Then,

$$\begin{aligned}
 \langle S_{l\text{-triplet}}^{\text{static}} \rangle &= (16M^3)^2 \int_{-\infty}^0 du_x \int_0^a dv_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \\
 &\quad \times (r_{xy}^2 - r_{xa}^2) \frac{e^{-r_{xx}/2M}}{r_{xx}} \frac{e^{-r_{yy}/2M}}{r_{yy}} e^{-(4M^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)}
 \end{aligned} \tag{155}$$

in units for which $\rho = 1$.

We split $\langle S_{l\text{-triplet}}^{\text{static}} \rangle$ into $\langle S_{xy}^{\text{static}} \rangle$ and $\langle S_{xa}^{\text{static}} \rangle$. Then, writing z for r_{xx} , y for r_{xa} , and x for r_{xy} ,

$$\langle S_{xy}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha dr_{yy} e^{r_{yy}^2} \int_\alpha^\infty dx \frac{x^3}{x-\alpha} e^{-x^2} \int_\alpha^x dy \frac{y}{y-\alpha} \int_\alpha^y dz e^{z^2} \quad (156)$$

$$\text{and } \langle S_{xa}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha dr_{yy} e^{r_{yy}^2} \int_\alpha^\infty dx \frac{x}{x-\alpha} e^{-x^2} \int_\alpha^x dy \frac{y^3}{y-\alpha} \int_\alpha^y dz e^{z^2} \quad (157)$$

with $\alpha \equiv 2M$.

10.3.1. *Solving the Integral for $\langle S_{xy}^{\text{static}} \rangle$.* Following Dou's original analysis of the link-counting, static-case integral (see Appendix C.4.4), we have

$$\begin{aligned} I^{xy}(\alpha) &= \int_\alpha^\infty dx \frac{x^3}{x-\alpha} e^{-x^2} \int_\alpha^x dy \frac{y}{y-\alpha} \int_\alpha^y dz e^{z^2} \\ &= \int_\alpha^\infty dx x^3 e^{-x^2} \left[\frac{\alpha}{1 \cdot 0!} \left(\frac{\frac{x}{\alpha} + 1}{2} \right) + \frac{\alpha^3}{3 \cdot 1!} \left(\frac{\left(\frac{x}{\alpha}\right)^3 + \left(\frac{x}{\alpha}\right)^2 + \left(\frac{x}{\alpha}\right) + 1}{4} \right) + \dots \right] \\ &= \frac{\alpha}{1 \cdot 0!} \left(\frac{J_2 + J_1}{2} \right) \\ &\quad + \frac{\alpha^3}{3 \cdot 1!} \left(\frac{J_4 + J_3 + J_2 + J_1}{4} + \frac{J_3 + J_2 + J_1}{3} + \frac{J_2 + J_1}{2} \right) + \dots \end{aligned} \quad (158)$$

For our integral,

$$J_n = \frac{1}{\alpha^{n-1}} \int_\alpha^\infty x^{n+2} e^{-x^2} dx \quad (159)$$

with the recursion relation

$$\begin{aligned} J_n &= \frac{1}{\alpha^{n-1}} \left(\left[-x^{n+1} \frac{1}{2} e^{-x^2} \right]_\alpha^\infty + \int_\alpha^\infty dx (n+1) x^n \frac{1}{2} e^{-x^2} \right) \\ &= \frac{\alpha^2}{2} e^{-\alpha^2} + \frac{n+1}{2\alpha^2} J_{n-2} \end{aligned} \quad (160)$$

The integrals for J_{2n+1} are exactly solvable:

$$J_{2n+1} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^{n+1} \frac{(n+1)!}{(\alpha^2)^m (n+1-m)!} \right) \quad (161)$$

The solutions for J_{2n} are of the form

$$J_{2n} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^n \frac{(2n+1)!!}{2^m (\alpha^2)^m (2n+1-2m)!!} + \frac{(2n+1)!!}{2^{n+1} \alpha (\alpha^2)^n} \left(e^{\alpha^2} \sqrt{\pi} \text{Erfc}(\alpha) \right) \right) \quad (162)$$

where $\text{Erfc}(\alpha) = 1 - \frac{2}{\sqrt{\pi}} \int_0^\alpha e^{-t^2} dt$. We are interested in the large α limit, under which

$$\begin{aligned} e^{\alpha^2} \sqrt{\pi} \left(1 - \frac{2}{\sqrt{\pi}} \int_0^\alpha e^{-t^2} dt \right) &= \frac{1}{\alpha} - \frac{1}{2\alpha^3} + \frac{3 \cdot 1}{2^2 \alpha^5} - \frac{5 \cdot 3 \cdot 1}{2^3 \alpha^7} + \dots \\ &= \sum_{p=0}^{\infty} \frac{(-1)^p (2p+1)!!}{(2p+1) 2^p \alpha^{2p+1}}. \end{aligned} \quad (163)$$

This gives our expression for J_{2n} as

$$\begin{aligned} J_{2n} &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^n \frac{(2n+1)!!}{2^m (\alpha^2)^m (2n+1-2m)!!} \right. \\ &\quad \left. + (2n+1)!! \sum_{p=0}^{\infty} \frac{(-1)^p (2p+1)!!}{(2p+1) 2^{p+n+1} \alpha^{2p+2n+2}} \right). \end{aligned} \quad (164)$$

We write

$$I^{xy}(\alpha) = \frac{\alpha^2}{2} e^{-\alpha^2} (I_0 + I_2 + I_4 + I_6 + \dots) \quad (165)$$

where I_{2k} sums terms of order $\frac{1}{\alpha^{2k}}$.

So, for I_{2k} the components of J_{2n+1} and J_{2n} with which we are concerned are given by

$$J_{2n+1}|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{2^k (n+1)!}{(2\alpha^2)^k (n+1-k)!} \right) & \forall k \leq n+1 \\ 0 & \text{otherwise} \end{cases} \quad (166)$$

$$\text{and } J_{2n}|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n+1)!!}{(2\alpha^2)^k (2n+1-2k)!!} \right) & \forall k \leq n \\ \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n+1)!! (-1)^{k-n-1} (2k-2n-1)!!}{(2k-2n-1) 2^k (\alpha^2)^k} \right) & k > n \end{cases}. \quad (167)$$

These expressions can be simplified as follows. For $k = 0$,

$$J_{2n+1}|_0 = J_{2n}|_0 = \frac{\alpha^2}{2} e^{-\alpha^2}. \quad (168)$$

For $k \geq 1$,

$$J_{2n+1}|_k = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{2^k (n+1)(n)(n-1)\dots(n+1-(k-1))}{2^k \alpha^{2k}} \right) \quad (169)$$

$$\text{and } J_{2n}|_k = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n+1)(2n-1)(2n-3)\dots(2n+1-(2k-2))}{2^k \alpha^{2k}} \right). \quad (170)$$

Note that for $k > n+1$ the $(n+1-(n+2-1))$ term in the numerator ensures $J_{2n+1}|_k = 0$. For $k > n$, the numerator of $J_{2n}|_k$ contains the necessary negative terms to generate the required $(2n+1)!! (-1)^{k-n-1} (2k-2n-3)!!$.

Furthermore, the expressions for $J_{2n+1}|_k$ and $J_{2n}|_k$ are identical. For even j ,

$$J_j|_k = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(j+1)(j-1)(j-3) \dots (j+1-(2k-2))}{2^k \alpha^{2k}} \right) \quad (171)$$

whilst for odd j ,

$$\begin{aligned} J_j|_k &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{2^k \left(\frac{j-1}{2} + 1\right) \left(\frac{j-1}{2}\right) \left(\frac{j-1}{2} - 1\right) \dots \left(\frac{j-1}{2} + 1 - (k-1)\right)}{2^k \alpha^{2k}} \right) \\ &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(j+1)(j-1)(j-3) \dots (j+1-(2k-2))}{2^k \alpha^{2k}} \right) \end{aligned} \quad (172)$$

giving the general result

$$J_j|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} & k = 0 \\ \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(j+1)(j-1)(j-3) \dots (j+1-(2k-2))}{2^k \alpha^{2k}} \right) & k \geq 1 \end{cases} \quad (173)$$

from which we derive

$$I_{2k} = \begin{cases} \alpha e^{\alpha^2} & k = 0 \\ \frac{1}{2^k \alpha^{2k}} \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1)n!} \sum_{u=2}^{2n+2} \frac{1}{u} \\ \quad \times \sum_{j=1}^u (j+1)(j-1)(j-3) \dots (j+1-(2k-2)) & k \geq 1 \end{cases} \quad (174)$$

Performing this calculation generates

$$\begin{aligned} I_0 &= \alpha e^{\alpha^2} , \\ I_2 &= \alpha e^{\alpha^2} \left(\frac{1}{4} + \frac{5}{4\alpha^2} \right) , \\ I_4 &= \alpha e^{\alpha^2} \left(\frac{1}{9} + \frac{43}{72\alpha^2} + \frac{3}{8\alpha^4} \right) , \\ &\dots \end{aligned}$$

10.3.2. Solving the Integral for $\langle S_{xa}^{\text{static}} \rangle$. We can perform a similar calculation to determine the contribution of $\langle S_{xa}^{\text{static}} \rangle$. The full details of the calculation can be found in Appendix E. For this integral,

$$\begin{aligned}
I^{xa}(\alpha) &= \int_{\alpha}^{\infty} dx \frac{x}{x-\alpha} e^{-x^2} \int_{\alpha}^x dy \frac{y^3}{y-\alpha} \int_{\alpha}^y dz e^{z^2} \\
&= \frac{\alpha}{1 \cdot 0!} \left(\frac{J_4 + J_3 + J_2 + J_1}{4} \right) \\
&\quad + \frac{\alpha^3}{3 \cdot 1!} \left(\frac{J_6 + \dots + J_1}{6} + \frac{J_5 + \dots + J_1}{5} + \frac{J_4 + \dots + J_1}{4} \right) + \dots
\end{aligned} \tag{175}$$

with

$$J_n = \frac{\alpha^2}{\alpha^{n-1}} \int_{\alpha}^{\infty} x^n e^{-x^2} dx \quad . \tag{176}$$

The integrals for J_{2n+1} are exactly solvable:

$$J_{2n+1} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^n \frac{n!}{(\alpha^2)^m (n-m)!} \right) \quad . \tag{177}$$

The solutions for J_{2n} (expanding $\text{Erfc}(\alpha)$ as before) are of the form

$$\begin{aligned}
J_{2n} &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^{n-1} \frac{(2n-1)!!}{2^m (\alpha^2)^m (2n-1-2m)!!} \right. \\
&\quad \left. + (2n-1)!! \sum_{p=0}^{\infty} \frac{(-1)^p (2p+1)!!}{(2p+1) 2^{p+n} \alpha^{2p+2n}} \right) \quad .
\end{aligned} \tag{178}$$

Again,

$$I^{xa}(\alpha) = \frac{\alpha^2}{2} e^{-\alpha^2} (I_0 + I_2 + I_4 + I_6 + \dots) \tag{179}$$

where I_{2k} sums terms of order $\frac{1}{\alpha^{2k}}$.

For I_{2k} the components of J_j with which we are concerned are now given by

$$J_j|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} & k = 0 \\ \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(j-1)(j-3)\dots(j-1-(2k-2))}{2^k \alpha^{2k}} \right) & k \geq 1 \end{cases} \tag{180}$$

from which we derive

$$I_{2k} = \begin{cases} \alpha e^{\alpha^2} & k = 0 \\ \frac{1}{2^k \alpha^{2k}} \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1)n!} \sum_{u=4}^{2n+4} \frac{1}{u} \\ \quad \times \sum_{j=1}^u (j-1)(j-3)\dots(j-1-(2k-2)) & k \geq 1 \end{cases} \quad . \tag{181}$$

Performing this calculation generates

$$\begin{aligned} I_0 &= \alpha e^{\alpha^2} , \\ I_2 &= \alpha e^{\alpha^2} \left(\frac{1}{4} + \frac{3}{4\alpha^2} \right) , \\ I_4 &= \alpha e^{\alpha^2} \left(\frac{1}{9} + \frac{31}{72\alpha^2} + \frac{1}{8\alpha^4} \right) , \\ &\dots \end{aligned}$$

10.3.3. *The Expectation Value of $S_{l\text{-triplet}}^{\text{static}}$.* Our complete integral is given as

$$\langle S_{l\text{-triplet}}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha e^{y^2} dy [I^{xy}(\alpha) - I^{xa}(\alpha)] . \quad (182)$$

For large α , the integral component can be solved as a series in $\frac{1}{\alpha}$:

$$\begin{aligned} e^{-\alpha^2} \int_0^\alpha e^{y^2} dy &= \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \frac{(2n+1)!!}{(2n+1)} \left(\frac{1}{\alpha} \right)^{2n+1} \\ &= \frac{1}{2\alpha} + \frac{1}{4\alpha^3} + \mathcal{O}\left(\frac{1}{\alpha^5}\right) \end{aligned} \quad (183)$$

Then

$$I^{xy}(\alpha) - I^{xa}(\alpha) = \frac{\alpha^2 e^{-\alpha^2}}{2} [(I_0^{xy} - I_0^{xa}) + (I_2^{xy} - I_2^{xa}) + (I_4^{xy} - I_4^{xa}) + \dots] \quad (184)$$

with eq.174 and eq.312 giving

$$\begin{aligned} (I_0^{xy} - I_0^{xa}) &= 0 , \\ (I_2^{xy} - I_2^{xa}) &= \alpha e^{\alpha^2} \left(\frac{1}{2\alpha^2} \right) , \\ (I_4^{xy} - I_4^{xa}) &= \alpha e^{\alpha^2} \left(\frac{1}{6\alpha^2} + \frac{1}{4\alpha^4} \right) , \\ (I_6^{xy} - I_6^{xa}) &= \alpha e^{\alpha^2} \left(\frac{1}{12\alpha^2} + \frac{5}{48\alpha^4} - \frac{3}{16\alpha^6} \right) , \\ (I_8^{xy} - I_8^{xa}) &= \alpha e^{\alpha^2} \left(\frac{1}{20\alpha^2} + \frac{7}{120\alpha^4} - \frac{29}{240\alpha^6} + \frac{3}{8\alpha^8} \right) , \\ (I_{10}^{xy} - I_{10}^{xa}) &= \alpha e^{\alpha^2} \left(\frac{1}{30\alpha^2} + \frac{3}{80\alpha^4} - \frac{43}{480\alpha^6} + \frac{61}{192\alpha^8} - \frac{75}{64\alpha^{10}} \right) , \\ &\dots \end{aligned}$$

so

$$\begin{aligned}
 I^{xy}(\alpha) - I^{xa}(\alpha) &= \frac{\alpha^3}{2} \left[\frac{1}{\alpha^2} \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \dots \right) + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right] \\
 &= \frac{\alpha^3}{2} \left[\frac{1}{\alpha^2} \left(\sum_{n=1}^{\infty} \frac{(n-1)!}{(n+1)!} \right) + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right] \\
 &= \frac{\alpha^3}{2} \left[\frac{1}{\alpha^2} + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right] .
 \end{aligned} \tag{185}$$

Using these results,

$$\begin{aligned}
 \langle S_{l\text{-triplet}}^{\text{static}} \rangle &= 4e^{-\alpha^2} \int_0^\alpha e^{y^2} dy [I^{xy}(\alpha) - I^{xa}(\alpha)] \\
 &= 4 \left[\frac{1}{2\alpha} + \frac{1}{4\alpha^3} + \mathcal{O}\left(\frac{1}{\alpha^5}\right) \right] \frac{\alpha^3}{2} \left[\frac{1}{\alpha^2} + \mathcal{O}\left(\frac{1}{\alpha^4}\right) \right] \\
 &= 4 \frac{\alpha^3}{2} \left(\frac{1}{2\alpha^3} + \mathcal{O}\left(\frac{1}{\alpha^5}\right) \right)
 \end{aligned} \tag{186}$$

which gives our final result

$$\langle S_{l\text{-triplet}}^{\text{static}} \rangle = 1 + \mathcal{O}\left(\frac{1}{\alpha^2}\right) . \tag{187}$$

10.4. Collapsing and Static Cases Compared. To leading order, the expectation values for the collapsing ($S_{l\text{-triplet}}^{\text{coll}}$, eq.153) and static ($S_{l\text{-triplet}}^{\text{static}}$, eq.187) cases are numerically equal: $\langle S_{l\text{-triplet}}^{\text{coll}} \rangle = \langle S_{l\text{-triplet}}^{\text{static}} \rangle = 1$ (+correction terms). This situation replicates that of Dou's original thesis, in which $\langle S_{\text{Dou}}^{\text{coll}} \rangle = \langle S_{\text{Dou}}^{\text{static}} \rangle = \frac{\pi^2}{6}$ (+correction terms). (Indeed, it may be noted that I^{xy} and I^{xa} both have $\frac{\pi^2}{6}$ as their leading-order terms. These terms are cancelled when $I^{xy} - I^{xa}$ is calculated.)

We may draw the same conclusion as Dou: that by pushing the collapse away to infinity (that is, taking the limit $b \rightarrow \infty$) it is possible to work in a flat spacetime regime and obtain results which are valid in the Schwarzschild, static case. Furthermore, this agreement between collapsing and static cases in the l -triplet model adds further support to the conjecture that such agreement may be found in both Λ - and z -triplet cases. We will discuss this further in §11.

11. SUMMARY AND CONCLUSION.

11.1. Summary. In his thesis ([Dou (1999)]) and subsequent paper with Sorkin ([Dou (2003)]), Dou proposed that the entropy of a black hole might be calculated by counting the number of causal links crossing its event horizon. Dou applied this link-counting model (§4) to 1+1-dimensional models of both static and collapsing black holes. The entropy calculated in this way is given by

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \mathcal{O}\left(\frac{a}{b}\right) + \mathcal{O}\left(\frac{1}{a^2}\right) \quad (188)$$

for the collapsing black hole, and

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \quad (189)$$

for the static black hole.

In order to further our analysis, to ensure we cover all link arrangements and to provide a framework for extension to 2+1 and higher dimensions, we extended the spacetime used for the black hole models, in §5. Repeating the calculation for the collapsing black hole in this extended spacetime gave an expectation value for the entropy of

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \mathcal{O}\left(\frac{a}{b}\right) \quad (190)$$

The extended spacetime removed the spurious correction term of leading order $1/a^2$ which appeared in $\langle S_{\text{Dou}}^{\text{coll.}} \rangle$.

§6 showed that the link-counting model, whilst working in 1+1 dimensions, would give infinite results in higher dimensions. The introduction of a third sprinkled element, z , to the x and y elements of the link-counting model, together with appropriate causal relations, serves to bound y away from $\mathcal{H}$ and remove this potential infinity. This Λ -triplet results in an integral for the static black hole which cannot be solved analytically (§8). However, the collapsing case can be solved (§7), to give a leading order result of

$$\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - 1 - \mathcal{O}\left(\frac{a}{b}\right)^2 \quad (191)$$

A reconsideration of the Λ -triplet led to the conclusion that whilst it removes infinities arising from y asymptotically approaching $\mathcal{H}$ in the future of Σ , it does not prevent x from

asymptotically approaching Σ to the past of $\mathcal{H}$: §9. A number of solutions were proposed, the simplest of which involves a linear ‘ z -triplet’. In the collapsing case the z -triplet gives an entropy (to leading order) of

$$\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle = 2 - \frac{\pi^2}{6} - \mathcal{O}\left(\frac{a}{b}\right)^2 . \quad (192)$$

However, the z -triplet still produces an integral for the static black hole which is analytically intractable.

Finally, we applied the approaches of the foregoing sections to a second linear triplet, the l -triplet, which solves the infinities of the Λ -triplet and, furthermore, can produce an analytic result for both collapsing and static black holes (§10). The l -triplet gives collapsing and static entropies of

$$\langle S_{l\text{-triplet}}^{\text{coll.}} \rangle = 1 - \frac{a}{b} \quad (193)$$

$$\text{and } \langle S_{l\text{-triplet}}^{\text{static}} \rangle = 1 + \mathcal{O}\left(\frac{1}{\alpha^2}\right) . \quad (194)$$

respectively.

11.2. Conclusion. Causal set theory provides an interesting approach to the calculation of black hole entropy. It provides a natural cut-off scale, inherent in the structure of spacetime itself. It is currently the only approach which can calculate an entropy for both *collapsing*, and static black holes. Indeed, the collapsing black hole case provides the most tractable approach to performing causal set calculations, and both link-counting and l -triplet models support Dou’s original assertion that the results obtained in the collapsing case can be considered valid in the static case, and vice-versa. That is, in the collapsing case one may push the collapse to future infinity, and obtain results in flat spacetime which are the same as the Schwarzschild case. This is of particular importance when one considers the need to extend any model to 2+1 and higher dimensions, with a concomitant increase in the complexity of the integrals involved. For now, it is possible to state, again echoing Dou, that the existence of the foregoing solutions to the 1+1-dimensional model suggests that, in higher dimensions, one will find entropy proportional to the area of the black hole horizon, as desired.

As §6 and §9 have shown, there are constraints on the arrangements of causal set elements which will permit these higher-dimensional calculations to provide finite results. Furthermore, it is not evident which arrangements (say, linear triplet or quadruplets) carry more weight than others. The link-counting model provided the most basic unit to count, involving only two elements which spanned the black hole horizon, and, as such, possessed a certain elegance. Moving to three elements, the Λ -triplet carried with it a certain physical perspective: two elements, one outside the horizon and one inside, both to the past of the hypersurface Σ , become, in a sense ‘entangled’ at a third element, inside the horizon and to the future of Σ . Once these two models have been abandoned, the linear triplets, the z -triplet and l -triplet, are perhaps the next ‘best’ choices.

The l -triplet also carries with it some continued concept of ‘entanglement’. A single element, to the past of both horizon and hypersurface is causally related to two elements, both to the future of the hypersurface, one inside the horizon and one outside. It is as though some event external to the black hole becomes associated with a pair of events, themselves linked, one inside and one outside the black hole. But ‘event’ is a somewhat misleading noun to use in this context. It is more the case that the spacetime is sprinkled with causal elements which endow it with the *potential* to contain events forming a l -triplet. In this sense, the entropy is intrinsic to the ‘fabric’ of the spacetime, not to any events occurring on the spacetime background.

At the end of the day, the results in this thesis suggest that one *can* model black hole entropy in a discrete spacetime, through the counting of particular arrangements of causal set elements, localized around the intersection of black hole horizon and hypersurface. Link counting is insufficient, as is the Λ -triplet. The l -triplet, however, shows how both static *and* collapsing black holes can have an entropy calculated for them. Furthermore, returning to our discussion in §5.6, putting $S_{BH} = A/4l_P^2 = A/l_f^2$ for the l -triplet would suggest that our fundamental length scale l_f is given as $l_f \sim 2l_P$, (where $\sim$ is used to indicate that the exact relationship between fundamental and Planck lengths will be dependent on additional multiplicative factors arising in the return from 1+1 to 3+1 dimensions).

Future work seems likely to involve recourse to numerical methods and simulations, most likely in the collapsing case, where a Minkowski metric can be used. Additionally, it will be recalled that §9.3 discussed arrangements of four causal set elements. There seems nothing

to preclude these types of arrangements from contributing to the black hole entropy. Indeed, it may be that all conceivable n -element, finite-entropy arrangements play some part in the calculation. If this is the case then one would need to sum over n to obtain a result and it is to be hoped that the contributions of large- n arrangements are suppressed, by the increasing complexity and number of regions required to be free of elements in order to ensure link relationships. Thus, despite the sum over n , a finite result may remain.

In the meantime, after an interesting journey through many possibilities and increasing complexity, it is rather heartening that, for the l -triplet at least, in both static and collapsing cases, the answer is simply... 1.

APPENDIX A. CARTER-PENROSE DIAGRAMS

A.1. Introduction. Carter-Penrose diagrams depict spacetime in such a way as to allow its global causal structure to be evident, ascribing finite coordinate values to asymptotic areas. A brief introduction to key concepts is given here, based on [Misner (1973)]. Further details can be found in [Townsend (1997)] at pp. 40 ff., and [Hawking (1973)] at pp. 118 ff. Figure 29, shows the Carter-Penrose diagram formed by the action of the mapping Φ_{cp} on the coordinates of Minkowski spacetime, $\mathcal{M}_\eta$: the spherical symmetry of $\mathcal{M}_\eta$ means that the angular components of the metric are unaffected, and the corresponding diagrams suppress the angular coordinates of the spacetime, accordingly.

A.2. Spacetime Regions. We may identify the following asymptotic regions of a general spacetime, $\mathcal{M}$, but note that not all regions exist for all topologies, $\{\mathcal{M}\}$. Defining distance and time parameters r and t , respectively:

I^+	<i>future timelike infinity</i> : region to which timelike lines extend; $t \rightarrow \infty$, r remains finite
I^-	<i>past timelike infinity</i> : region from which timelike lines extend; $t \rightarrow -\infty$, r remains finite
I_0	<i>spacelike infinity</i> : region to which spacelike hypersurfaces extend; $r \rightarrow \infty$, t remains finite
$\mathcal{I}^+$	<i>future null infinity</i> : region to which outgoing null lines extend; $t + r \rightarrow \infty$, $t - r$ remains finite
$\mathcal{I}^-$	<i>past null infinity</i> : region from which incoming null line extend; $t - r \rightarrow -\infty$, $t + r$ remains finite

These regions are shown in Figure 29. Note that Φ_{cp} maps these asymptotic areas to finite values of the parameters U and V (which are *not* Kruskal-Szekeres coordinates, in this appendix).

A.3. Example: Minkowski Spacetime. Consider the Minkowski metric, in spherical polar coordinates:

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2 \quad . \quad (195)$$

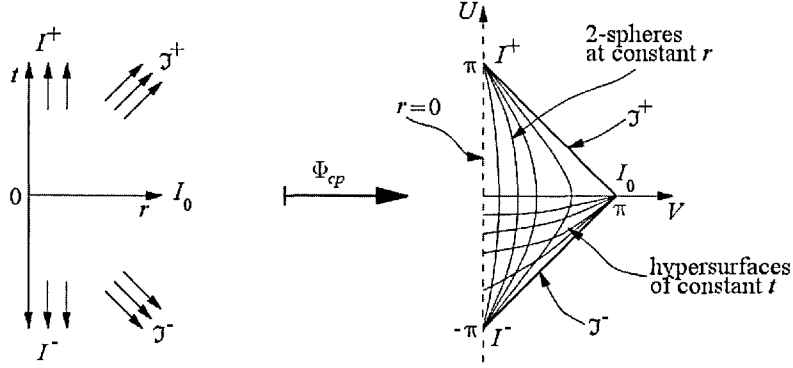


FIGURE 29. Conformal compactification of Minkowski spacetime, to give corresponding Carter-Penrose diagram. (After [Hawking (1973), Misner (1973), Townsend (1997)].)

Then, applying the mappings

$$t + r = \tan \frac{1}{2} (U + V) \quad \text{and} \quad t - r = \tan \frac{1}{2} (U - V) \quad (196)$$

gives

$$ds^2 = \frac{-dU^2 + dV^2}{4 \cos^2 \frac{1}{2} (U + V) \cos^2 \frac{1}{2} (U - V)} + r^2 d\Omega^2 \quad . \quad (197)$$

If we restrict the ranges of U and V , then Φ_{cp} is bijective, and we may define the inverse mappings

$$U + V = 2 \arctan (t + r) \quad \text{and} \quad U - V = 2 \arctan (t - r) \quad . \quad (198)$$

It is then possible to examine the behaviour of the regions defined in Appendix A.2 under Φ_{cp} . $I^\pm$ and I_0 appear at the points shown in Figure 29.

For $\mathcal{I}^+$:

$$\begin{aligned} \lim_{t+r \rightarrow \infty} [U] &= \lim_{t+r \rightarrow \infty} [\arctan (t + r) + \arctan (t - r)] \\ &= \frac{\pi}{2} + \arctan (t - r) \quad , \end{aligned} \quad (199)$$

$$\begin{aligned} \lim_{t+r \rightarrow \infty} [V] &= \lim_{t+r \rightarrow \infty} [\arctan (t + r) - \arctan (t - r)] \\ &= \frac{\pi}{2} - \arctan (t - r) \quad , \end{aligned} \quad (200)$$

for finite $t - r$.

Given the range of U and V , this is the equation of a line segment, $U = \pi - V$, running from I^+ to I_0 . Similarly, $\mathcal{I}^-$ is the line segment joining I^- and I_0 .

A.4. Other Features of Figure 29.

- The mapping Φ_{cp} is conformal, maintaining local angles.
- Whilst the points at infinity do not exist as part of the original manifold, they are added as part of the mapping operation, rendering the Carter-Penrose representation topologically compact.
- The topology of the Carter-Penrose representation is $\mathbb{R}$ (time) $\times S^3$ (space). The dimension of the representation is reduced when projected onto a $\mathbb{R}^2$ surface, such that each point represents a spacelike 2-sphere (with the exception of points on $V = 0$, and points added at $I^\pm, I_0$).
- Radial null geodesics continue to make angles of 45° with the vertical axis after mapping by Φ_{cp} . Non-radial null geodesics make angles of less than 45° .

APPENDIX B. CALCULATIONS FOR LINK COUNTING.

B.1. **The Expectation Value of $S_{\text{Dou}}^{\text{coll.}}$.** We have,

$$\begin{aligned}
\langle S_{\text{Dou}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^{v_y} du_y \int_{-\infty}^0 du_x \int_0^a dv_x e^{-u_y v_y + (v_y - v_x)u_x + u_y^2/2} \\
&= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y + u_y^2/2} \int_0^a dv_x \left[\frac{1}{v_y - v_x} e^{(v_y - v_x)u_x} \right]_{-\infty}^0 \text{ with } v_y - v_x > 0 \\
&= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y + u_y^2/2} \int_0^a dv_x \frac{1}{v_y - v_x} \\
&= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y + u_y^2/2} \left[\ln \left(\frac{v_y}{v_y - v_x} \right) \right]_0^a \\
&= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y + u_y^2/2} \ln \left(\frac{v_y}{v_y - a} \right) .
\end{aligned} \tag{201}$$

Then let $x = v_y$ and $y = v_y - u_y$, so

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \int_a^b dx \ln \left(\frac{x}{x - a} \right) e^{-x^2/2} \int_0^x dy e^{y^2/2} . \tag{202}$$

Now, $a \leq x \leq b$ and for $a \gg 1$ we have

$$e^{-x^2/2} \int_0^x dy e^{y^2/2} = \frac{\frac{d}{dy} \int_0^x dy e^{y^2/2}}{\frac{d}{dx} e^{x^2/2}} \tag{203}$$

by l'Hôpital's rule, so

$$e^{-x^2/2} \int_0^x dy e^{y^2/2} \sim \frac{1}{x} .$$

Expanding in a gives

$$\begin{aligned}
\frac{1}{x} \ln \left(\frac{x}{x - a} \right) &= \frac{1}{x} \ln x - \frac{1}{x} \ln (x - a) \\
&= \frac{1}{x} \ln x - \frac{1}{x} \left[\ln x - \frac{a}{x} - \frac{a^2}{2x^2} - \frac{a^3}{3x^3} - \dots \right] \\
&= \frac{a}{x^2} + \frac{a^2}{2x^3} + \frac{a^3}{3x^4} + \dots
\end{aligned} \tag{204}$$

Then

$$\begin{aligned}
 \int_a^b \left[\frac{a}{x^2} + \frac{a^2}{2x^3} + \frac{a^3}{3x^4} + \dots \right] dx &= \left[-\frac{a}{x} - \frac{a^2}{2 \cdot 2x^2} - \frac{a^3}{3 \cdot 3x^3} - \dots \right]_a^b \\
 &= \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{k=1}^{\infty} \left(\frac{a}{b} \right)^k \frac{1}{k^2} \\
 &= \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right)
 \end{aligned} \tag{205}$$

where $\ell \left(\frac{a}{b} \right) = \sum_{k=1}^{\infty} \left(\frac{a}{b} \right)^k \frac{1}{k^2}$.

However, our use of l'Hôpital's rule ignores correction terms arising from the fact that our limit for x is a , which is large, but not infinite. A more accurate expansion gives

$$e^{-x^2/2} \int_0^x dy e^{y^2/2} = \frac{1}{x} + \frac{1}{x^3} + \frac{3}{x^5} + \dots \tag{206}$$

(cf eq.183).

We have dealt with the first term. The second gives a contribution

$$\begin{aligned}
 \int_a^b dx \frac{1}{x^3} \ln \left(\frac{x}{x-a} \right) &= \left[-\frac{1}{4x^2} - \frac{1}{2ax} + \frac{1}{2a^2} \ln \left(\frac{x}{x-a} \right) - \frac{1}{2x^2} \ln \left(\frac{x}{x-a} \right) \right]_a^b \\
 &= -\frac{1}{4b^2} - \frac{1}{2ab} + \frac{1}{2a^2} \left(\frac{a}{b} + \frac{1}{2} \left(\frac{a}{b} \right)^2 + \dots \right) \\
 &\quad - \frac{1}{2b^2} \left(\frac{a}{b} + \frac{1}{2} \left(\frac{a}{b} \right)^2 + \dots \right) + \frac{3}{4a^2} \\
 &= \frac{3}{4a^2} - \frac{1}{3} \frac{a}{b^3} + \dots
 \end{aligned} \tag{207}$$

Combining eq.205 and eq.207 gives our result:

$$\langle S_{\text{Dou}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) + \mathcal{F}(a, b) \tag{208}$$

where $\ell \left(\frac{a}{b} \right) = \sum_{k=1}^{\infty} \left(\frac{a}{b} \right)^k \frac{1}{k^2}$ and $\mathcal{F}(a, b) = \frac{3}{4a^2} - \frac{a}{3b^3}$, to leading orders in a and b .

B.2. The Poisson Distribution. The Poisson distribution can be defined as follows. Consider dividing a volume V into N elements, each of volume ΔV . A point is placed in each volume element independently, with probability p : $\Pr(\text{point in volume } \Delta V) = p$. Hence, in some volume V we expect $\frac{V}{\Delta V} p$ points, in a total of $N = \frac{V}{\Delta V}$ volume elements. Then, for an average sprinkling density of ρ , we have $\rho V = Np$ and hence $p = \frac{\rho V}{N} = \rho \Delta V$. The Poisson distribution is the continuum limit, $\Delta V \rightarrow 0$, whilst keeping Np and V fixed.

The chance of finding n points in N volume elements (that is, n ‘full’ volume elements, and $N - n$ ‘empty’ volume elements) is given by:

$$\Pr_p(n | N) = \frac{N!}{n!(N-n)!} p^n (1-p)^{N-n} \quad . \quad (209)$$

In those N volume elements, we expect to find $s = Np$ points, so

$$\Pr_{s/N}(n | N) = \frac{N!}{n!(N-n)!} \frac{s^n}{N^n} \left(1 - \frac{s}{N}\right)^{N-n} \quad . \quad (210)$$

Taking the limit $N \rightarrow \infty$ gives the probability of finding n points in some volume, given an expectation value of s points in that volume:

$$\begin{aligned} \Pr_s(n) &= \lim_{N \rightarrow \infty} \frac{N(N-1)\dots(N-n+1)}{n!} \frac{s^n}{N^n} \left(1 - \frac{s}{N}\right)^N \left(1 - \frac{s}{N}\right)^{-n} \\ &= 1 \cdot \frac{s^n}{n!} \cdot e^{-s} \cdot 1 \\ &= \frac{s^n e^{-s}}{n!} \quad . \end{aligned} \quad (211)$$

We have $s = Np = \rho V$, and hence $\lim N \rightarrow \infty$ is equivalent to $\lim \Delta V \rightarrow 0$, for fixed V . Hence, given a sprinkling density, ρ , the probability of finding n sprinkled points in some volume V is given by

$$\Pr_\rho(V, n) = \frac{(\rho V)^n e^{-\rho V}}{n!} \quad . \quad (212)$$

B.3. Calculating the Expectation Value of $S_{\text{link}}^{\text{coll.}}$. Taking units in which $\rho = 1$ (i.e. fundamental units), we have,

$$\begin{aligned} \langle S_{\text{link}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^{v_y} du_y \int_0^a dv_x \int_{-\infty}^0 du_x e^{u_x(v_y - v_x) - u_y v_y} \\ &= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y} \int_0^a dv_x \left[\frac{1}{v_y - v_x} e^{u_x(v_y - v_x)} \right]_{-\infty}^0 \\ &= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y} \int_0^a dv_x \frac{1}{v_y - v_x} \\ &= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y} \left[\ln \left(\frac{v_y}{v_y - v_x} \right) \right]_0^a \\ &= \int_a^b dv_y \int_0^{v_y} du_y e^{-u_y v_y} \ln \left(\frac{v_y}{v_y - a} \right) \quad (\text{over}) \end{aligned} \quad (213)$$

$$\begin{aligned}
&= \int_a^b dv_y \ln \left(\frac{v_y}{v_y - a} \right) \left[-\frac{1}{v_y} e^{-u_y v_y} \right]_0^{v_y} \quad (\text{cont.}) \\
&= \int_a^b dv_y \ln \left(\frac{v_y}{v_y - a} \right) \left(\frac{1}{v_y} - \frac{e^{-v_y^2}}{v_y} \right) .
\end{aligned}$$

Writing y for v_y and expanding the logarithm in powers of a gives

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \int_a^b dy \left(\sum_{n=1}^{\infty} \frac{a^n}{n y^{n+1}} \right) (1 - e^{-y^2}) . \quad (214)$$

For $y \geq a \gg 1$, $e^{-y^2} \leq e^{-a^2} \ll 1$, and the integral reduces to,

$$\begin{aligned}
\langle S_{\text{link}}^{\text{coll.}} \rangle &= \int_a^b dy \left(\sum_{n=1}^{\infty} \frac{a^n}{n y^{n+1}} \right) \\
&= \left[\sum_{n=1}^{\infty} -\frac{a^n}{n^2 y^n} \right]_a^b \\
&= \sum_{n=1}^{\infty} \frac{1}{n^2} - \sum_{n=1}^{\infty} \left(\frac{a}{b} \right)^n \frac{1}{n^2} .
\end{aligned} \quad (215)$$

Hence,

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \quad (216)$$

where $\ell \left(\frac{a}{b} \right) = \sum_{k=1}^{\infty} \left(\frac{a}{b} \right)^k \frac{1}{k^2}$.

B.4. Calculating the Expectation Value of $S_{\text{link}}^{\text{coll.}+}$. We have,

$$\begin{aligned}
\langle S_{\text{link}}^{\text{coll.}+} \rangle &= \int_a^b du_y \int_0^{v_y} dv_x \int_0^a dv_y \int_{-\infty}^0 du_x e^{u_x(a-v_x)-u_y v_y} \\
&= \int_a^b du_y \int_0^{v_y} dv_x \int_0^a dv_y e^{-u_y v_y} \left[-\frac{1}{a-v_x} e^{u_x(a-v_x)} \right]_{-\infty}^0 \\
&= \int_a^b du_y \int_0^a dv_y e^{-u_y v_y} \int_0^{v_y} dv_x \frac{1}{a-v_x} \\
&= \int_0^a dv_y \int_a^b du_y \ln \left(\frac{a}{a-v_y} \right) e^{-u_y v_y} \\
&= \int_0^a dv_y \ln \left(\frac{a}{a-v_y} \right) \frac{1}{v_y} (e^{-av_y} - e^{-bv_y}) \\
&= \int_0^a dv_y \left(\frac{v_y}{a} + \frac{1}{2} \left(\frac{v_y}{a} \right)^2 + \dots \right) \frac{1}{v_y} (e^{-av_y} - e^{-bv_y})
\end{aligned} \quad (217)$$

giving

$$\langle S_{\text{link}}^{\text{coll.}+} \rangle = \int_0^a dv_y \sum_{n=1}^{\infty} \frac{v_y^{n-1}}{na^n} (e^{-av_y} - e^{-bv_y}) \quad . \quad (218)$$

APPENDIX C. CALCULATIONS FOR THE Λ -TRIPLET.

C.1. Extending Region $\mathcal{Y}$ such that $0 < u_q < b \ \forall q \in \mathcal{Y}$. We consider the effect on $\langle S_{\text{link}}^{\text{coll.}} \rangle$ of extending the region in which sprinkled points y may be found, to include the triangular region bounded in broken red of Figure 19, such that we have $\int_0^b du_y$ rather than $\int_0^{v_y} du_y$ in our integral. Then,

$$\begin{aligned} \langle S_{\text{link}}^{\text{coll.}} \rangle &= \int_a^b dy \ln \left(\frac{y}{y-a} \right) \left[-\frac{1}{y} e^{-xy} \right]_0^b \\ &= \int_a^b dy \ln \left(\frac{y}{y-a} \right) \left(\frac{1}{y} - \frac{e^{-by}}{y} \right) \end{aligned} \quad (219)$$

Again for, $y \geq a \gg 1$, $e^{-by} \leq e^{-ba} \ll 1$, and the integral is unchanged as

$$\langle S_{\text{link}}^{\text{coll.}} \rangle = \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \quad (220)$$

The same is true of subsequent triplet integrals.

C.2. Solving the Integral for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$.

C.2.1. Rewriting the Integral. We rewrite the integral for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$ as

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y-v_x)-u_y v_y} \\ &\quad - \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x (u_y - a) e^{u_x(v_y-v_x)-u_y v_y} \end{aligned} \quad (221)$$

C.2.2. ‘Removing’ the Second Term. Let us define the first term of eq.221 to be $\langle S_{\Lambda\text{-triplet}}^1 \rangle$ and the second term to be $\langle S_{\Lambda\text{-triplet}}^2 \rangle$.

Then

$$\begin{aligned} \langle S_{\Lambda\text{-triplet}}^2 \rangle &= \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x (u_y - a) e^{u_x(v_y-v_x)-u_y v_y} \\ &= \int_a^b dv_y \int_0^{b-a} du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y-v_x)-(u_y+a)v_y} \\ &= \int_a^b dv_y \int_0^{b-a} du_y u_y e^{-(u_y+a)v_y} [v_y \ln(v_y) - a - v_y \ln(v_y - a)] \quad (\text{over}) \end{aligned} \quad (222)$$

$$\begin{aligned}
&= \int_a^b dy \left[-a + y \ln \left(\frac{y}{y-a} \right) \right] \int_0^{b-a} dx x e^{-(x+a)y} \quad (\text{with } y = v_y, x = u_y) \quad (\text{cont.}) \\
&= \int_a^b dy \left[-a + y \ln \left(\frac{y}{y-a} \right) \right] \left[-\frac{e^{(x+a)y} (1+xy)}{y^2} \right]_0^{b-a} \\
&= \int_a^b dy \left[-a + y \ln \left(\frac{y}{y-a} \right) \right] \left[\frac{e^{-ay}}{y^2} - \frac{e^{-by} (1+(b-a)y)}{y^2} \right] \\
&= \int_a^b dy \left[-\frac{ae^{-ay}}{y^2} + \frac{ae^{-by}}{y^2} + a(b-a) \frac{e^{-by}}{y} \right. \\
&\quad \left. + \frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-ay} - \frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-by} - \ln \left(\frac{y}{y-a} \right) (b-a) e^{-by} \right] \\
&= \langle S_{\Lambda 1}^2 \rangle + \langle S_{\Lambda 2}^2 \rangle + \langle S_{\Lambda 3}^2 \rangle + \langle S_{\Lambda 4}^2 \rangle + \langle S_{\Lambda 5}^2 \rangle + \langle S_{\Lambda 6}^2 \rangle
\end{aligned}$$

We treat each of these integrals in turn.

$$\langle S_{\Lambda 1}^2 \rangle$$

$$\langle S_{\Lambda 1}^2 \rangle = \int_a^b dy \left[-\frac{ae^{-ay}}{y^2} \right] . \quad (223)$$

Then,

$$\begin{aligned}
|\langle S_{\Lambda 1}^2 \rangle| &< e^{-a^2} \int_a^b dy \left[\frac{a}{y^2} \right] \\
&< e^{-a^2} \left(1 - \frac{a}{b} \right) \\
&\sim 0 \text{ for large } a, \text{ small } a/b
\end{aligned} \quad (224)$$

where ‘ ~ 0 ’ implies exponential suppression in a and/or b as $a \rightarrow \infty$, $b \rightarrow \infty$, with a/b fixed and small.

$$\langle S_{\Lambda 2}^2 \rangle$$

$$\langle S_{\Lambda 2}^2 \rangle = \int_a^b dy \left[\frac{ae^{-by}}{y^2} \right] . \quad (225)$$

Then,

$$\begin{aligned}
 |\langle S_{\Lambda 2}^2 \rangle| &< e^{-ab} \int_a^b dy \left[\frac{a}{y^2} \right] \\
 &< e^{-ab} \left(1 - \frac{a}{b} \right) \\
 &\sim 0 \text{ for large } a, \text{ small } a/b \quad .
 \end{aligned} \tag{226}$$

$\langle S_{\Lambda 3}^2 \rangle$

$$\langle S_{\Lambda 3}^2 \rangle = \int_a^b dy \left[a(b-a) \frac{e^{-by}}{y} \right] \quad . \tag{227}$$

Then,

$$\begin{aligned}
 |\langle S_{\Lambda 3}^2 \rangle| &< e^{-ab} a(b-a) \int_a^b dy \left[\frac{1}{y} \right] \\
 &< e^{-ab} a(b-a) \ln \left(\frac{b}{a} \right) \\
 &\sim 0 \text{ for } a > 1, \text{ large } b \quad .
 \end{aligned} \tag{228}$$

$\langle S_{\Lambda 4}^2 \rangle$

$$\langle S_{\Lambda 4}^2 \rangle = \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-ay} \right] \quad . \tag{229}$$

Then,

$$\begin{aligned}
 |\langle S_{\Lambda 4}^2 \rangle| &< e^{-a^2} \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\
 &< e^{-a^2} \langle S_{\text{link}}^{\text{coll.}} \rangle \\
 &< e^{-a^2} \left(\frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \right) \\
 &\sim 0 \text{ for large } a, \text{ small } a/b \quad .
 \end{aligned} \tag{230}$$

$\langle S_{\Lambda 5}^2 \rangle$

$$\langle S_{\Lambda 5}^2 \rangle = \int_a^b dy \left[-\frac{1}{y} \ln \left(\frac{y}{y-a} \right) e^{-by} \right] \quad . \tag{231}$$

Then,

$$\begin{aligned}
|\langle S_{\Lambda 5}^2 \rangle| &< e^{-ab} \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\
&< e^{-ab} \langle S_{\text{link}}^{\text{coll.}} \rangle \\
&< e^{-ab} \left(\frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \right) \\
&\sim 0 \text{ for large } a, \text{ small } a/b \quad .
\end{aligned} \tag{232}$$

$\langle S_{\Lambda 6}^2 \rangle$

$$\langle S_{\Lambda 6}^2 \rangle = \int_a^b -dy \left[\ln \left(\frac{y}{y-a} \right) (b-a) e^{-by} \right] \tag{233}$$

Then,

$$\begin{aligned}
|\langle S_{\Lambda 6}^2 \rangle| &< (b-a) e^{-ab} \int_a^b dy \ln \left(\frac{y}{y-a} \right) \\
&< (b-a) e^{-ab} \left[y \ln \left(\frac{y}{y-a} \right) + a \ln (y-a) \right]_a^b \\
&< (b-a) e^{-ab} \left[b \ln \left(\frac{b}{b-a} \right) + a \ln (b-a) - \lim_{y \rightarrow a} \ln \left(y^y \frac{(y-a)^a}{(y-a)^y} \right) \right] \\
&< (b-a) e^{-ab} \left[a + a \ln \left(\frac{b-a}{a} \right) \right] \quad (\text{using } b \ln \left(\frac{b}{b-a} \right) \sim a \text{ for } b \gg a) \\
&\sim 0 \text{ for } a > 1, \text{ large } b \quad .
\end{aligned} \tag{234}$$

This gives our final result,

$$\langle S_{\Lambda\text{-triplet}}^2 \rangle \sim 0 \tag{235}$$

for the values of a and b with which we are concerned.

C.2.3. *Solving the Remaining Integral.*

$$\begin{aligned}
\langle S_{\Lambda\text{-triplet}}^1 \rangle &= \int_a^b dv_y \int_0^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x v_x u_y e^{u_x(v_y - v_x) - u_y v_y} \\
&= \int_a^b dv_y \int_0^b du_y u_y e^{-u_y v_y} \int_0^a dv_x \frac{v_x}{v_y - v_x} \\
&= \int_a^b dv_y \int_0^b du_y u_y e^{-u_y v_y} [-v_x - v_y \ln(v_y - v_x)]_0^a \\
&= \int_a^b dv_y \int_0^b du_y u_y e^{-u_y v_y} [v_y \ln(v_y) - a - v_y \ln(v_y - a)] \\
&= \int_a^b dy \left[-a + y \ln\left(\frac{y}{y-a}\right) \right] \int_0^b dx x e^{-xy} \quad (\text{with } y = v_y, x = u_y) \\
&= \int_a^b dy \left[-a + y \ln\left(\frac{y}{y-a}\right) \right] \left[-\frac{1}{y^2} e^{-xy} (1 + xy) \right]_0^b \\
&= \int_a^b dy \left[-a + y \ln\left(\frac{y}{y-a}\right) \right] \left[-\frac{1}{y^2} e^{-by} (1 + by) + \frac{1}{y^2} \right] \\
&= \int_a^b dy \left[\frac{a}{y^2} e^{-by} + \frac{ab}{y} e^{-by} - \frac{a}{y^2} - \frac{e^{-by}}{y} \ln\left(\frac{y}{y-a}\right) \right. \\
&\quad \left. - be^{-by} \ln\left(\frac{y}{y-a}\right) + \frac{1}{y} \ln\left(\frac{y}{y-a}\right) \right] \\
&= \langle S_{\Lambda 1}^1 \rangle + \langle S_{\Lambda 2}^1 \rangle + \langle S_{\Lambda 3}^1 \rangle + \langle S_{\Lambda 4}^1 \rangle + \langle S_{\Lambda 5}^1 \rangle + \langle S_{\Lambda 6}^1 \rangle .
\end{aligned} \tag{236}$$

We treat each of these integrals in turn.

$$\langle S_{\Lambda 1}^1 \rangle$$

$$\langle S_{\Lambda 1}^1 \rangle = \int_a^b dy \left[\frac{a}{y^2} e^{-by} \right] . \tag{237}$$

Then,

$$\begin{aligned}
|\langle S_{\Lambda 1}^1 \rangle| &< e^{-ab} \int_a^b dy \left[\frac{a}{y^2} \right] \\
&< e^{-ab} \left(1 - \frac{a}{b} \right) \\
&\sim 0 \text{ for large } a, \text{ small } a/b .
\end{aligned} \tag{238}$$

$$\langle S_{\Lambda 2}^1 \rangle$$

$$\langle S_{\Lambda 2}^1 \rangle = \int_a^b dy \left[\frac{ab}{y} e^{-by} \right] . \tag{239}$$

Then,

$$\begin{aligned}
 |\langle S_{\Lambda 2}^1 \rangle| &< e^{-ab} \int_a^b dy \left[\frac{ab}{y} \right] \\
 &< ab e^{-ab} \ln \left(\frac{b}{a} \right) \\
 &\sim 0 \text{ for } a > 1, \text{ large } b \quad .
 \end{aligned} \tag{240}$$

$\langle S_{\Lambda 3}^1 \rangle$

$$\begin{aligned}
 \langle S_{\Lambda 3}^1 \rangle &= \int_a^b dy \left[\frac{-a}{y^2} \right] \\
 &= \frac{a}{b} - 1 \quad .
 \end{aligned} \tag{241}$$

$\langle S_{\Lambda 4}^1 \rangle$

$$\langle S_{\Lambda 4}^1 \rangle = \int_a^b dy \left[-\frac{e^{-by}}{y} \ln \left(\frac{y}{y-a} \right) \right] \quad . \tag{242}$$

Then,

$$\begin{aligned}
 |\langle S_{\Lambda 4}^1 \rangle| &< e^{-ab} \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\
 &< e^{-ab} \langle S_{\text{link}}^{\text{coll.}} \rangle \\
 &< e^{-ab} \left(\frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \right) \\
 &\sim 0 \text{ for large } a, \text{ small } a/b \quad .
 \end{aligned} \tag{243}$$

$\langle S_{\Lambda 5}^1 \rangle$

$$\langle S_{\Lambda 5}^1 \rangle = \int_a^b dy \left[-be^{-by} \ln \left(\frac{y}{y-a} \right) \right] \quad . \tag{244}$$

Then (see eq.234),

$$\begin{aligned}
 |\langle S_{\Lambda 5}^1 \rangle| &< b e^{-ab} \int_a^b dy \left[\ln \left(\frac{y}{y-a} \right) \right] \\
 &< b e^{-ab} \left[y \ln \left(\frac{y}{y-a} \right) + a \ln (y-a) \right]_a^b \\
 &< b e^{-ab} \left[b \ln \left(\frac{b}{b-a} \right) + a \ln (b-a) - a \ln (a) \right] \\
 &\sim 0 \text{ for } a > 1, \text{ large } b \quad .
 \end{aligned} \tag{245}$$

$$\langle S_{\Lambda 6}^1 \rangle$$

$$\begin{aligned}
 \langle S_{\Lambda 6}^1 \rangle &= \int_a^b dy \left[\frac{1}{y} \ln \left(\frac{y}{y-a} \right) \right] \\
 &= \langle S_{\text{link}}^{\text{coll.}} \rangle \\
 &= \frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \quad .
 \end{aligned} \tag{246}$$

C.2.4. *Final Solution for $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$.*

$$\begin{aligned}
 \langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle = \langle S_{\Lambda\text{-triplet}}^1 \rangle &= \langle S_{\Lambda 3}^1 \rangle + \langle S_{\Lambda 6}^1 \rangle \\
 &= \frac{\pi^2}{6} + \left(\frac{a}{b} - 1 \right) - \ell \left(\frac{a}{b} \right) \\
 &= \frac{\pi^2}{6} - 1 - \sum_{n=2}^{\infty} \left(\frac{a}{b} \right)^n \frac{1}{n^2} \\
 &= \frac{\pi^2}{6} - 1 + \ell' \left(\frac{a}{b} \right)
 \end{aligned} \tag{247}$$

where $\ell' \left(\frac{a}{b} \right) = - \sum_{n=2}^{\infty} \left(\frac{a}{b} \right)^n \frac{1}{n^2}$.

C.3. Solution of General Volume Integral of Appropriate Form for Static Black Hole Case. We expand on the discussion in [Dou (1999)]. Consider the integral

$$V = 16M^3 \int_{u_p}^{u_q} \int_{v_s}^{v_t} \frac{e^{-r/2M}}{r} du dv \quad . \tag{248}$$

We make the substitution $v \rightarrow r(uv)$, under which $dv \rightarrow \frac{-r}{(2M)^2 u} e^{r/2M} dr$. Then

$$\begin{aligned} V &= -4M \int_{u_p}^{u_q} \int_{r(uv_s)}^{r(uv_t)} \frac{du}{u} dr \\ &= -4M \int_{u_p}^{u_q} \left[\frac{r(uv_t)}{u} - \frac{r(uv_s)}{u} \right] du = I_1 - I_2 \quad . \end{aligned} \quad (249)$$

To evaluate the first term of eq.249, I_1 , we make the further substitution $u \rightarrow r(uv_t) = r_t$, under which $du \rightarrow \frac{-r_t}{(2M)^2 v_t} e^{r_t/2M} dr_t$, giving

$$\begin{aligned} I_1 &= \frac{1}{M} \int_{r_{pt}}^{r_{qt}} \frac{r_t^2}{1 - r_t/2M} dr_t \\ &= \left[2M \left(-2r_t - \frac{r_t^2}{2M} - 4M \ln(-2M + r_t) \right) \right]_{r_{pt}}^{r_{qt}} \\ &= 4M(r_{pt} - r_{qt}) + (r_{pt}^2 - r_{qt}^2) + 8M^2 [\ln(-2M + r_{pt}) - \ln(-2M + r_{qt})] \quad . \end{aligned} \quad (250)$$

Now,

$$\frac{u_q}{u_p} = \frac{u_q v_t}{u_p v_t} = \frac{1 - r_{qt}/2M}{1 - r_{pt}/2M} \frac{e^{r_{qt}/2M}}{e^{r_{pt}/2M}} = \frac{-2M + r_{qt}}{-2M + r_{pt}} \frac{e^{r_{qt}/2M}}{e^{r_{pt}/2M}} \quad (251)$$

so

$$\begin{aligned} \ln(-2M + r_{pt}) - \ln(-2M + r_{qt}) &= -\ln\left(\frac{-2M + r_{qt}}{-2M + r_{pt}}\right) \\ &= -\frac{1}{2M} (r_{pt} - r_{qt}) - \ln\left(\frac{u_q}{u_p}\right) \quad . \end{aligned} \quad (252)$$

Substituting into the equation for I_1 gives

$$I_1 = 4M(r_{pt} - r_{qt}) + (r_{pt}^2 - r_{qt}^2) + 8M^2 \left(-\frac{1}{2M} (r_{pt} - r_{qt}) \right) - 8M^2 \ln\left(\frac{u_q}{u_p}\right) \quad . \quad (253)$$

Similarly for I_2 , by putting $v_t \rightarrow v_s$ in the above, we get

$$I_2 = r_{ps}^2 - r_{qs}^2 - 8M^2 \ln\left(\frac{u_q}{u_p}\right) \quad (254)$$

and hence,

$$V = I_1 + I_2 = r_{pt}^2 - r_{qt}^2 - r_{ps}^2 + r_{qs}^2 \quad . \quad (255)$$

In summary, we have the result:

$$16M^3 \int_{u_p}^{u_q} \int_{v_s}^{v_t} \frac{e^{-r/2M}}{r} du dv = r_{pt}^2 - r_{qt}^2 - r_{ps}^2 + r_{qs}^2 \quad . \quad (256)$$

C.4. Dou's Approach to Solving $\langle S_{\text{Dou}}^{\text{static}} \rangle$. We examine the approach taken by Dou in evaluating his integral, formulated using only two points, x and y , as given in his doctoral thesis:

$$\begin{aligned} \langle S_{\text{Dou}}^{\text{static}} \rangle &= (16M^3)^2 \int_0^a dv_x \int_{-\infty}^0 du_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \\ &\quad \times \frac{e^{-r_{xx}/2M}}{r_{xx}} \frac{e^{-r_{yy}/2M}}{r_{yy}} e^{-(4M^2 + r_{xy}^2 - r_{xx}^2 - r_{yy}^2)} \end{aligned} \quad (257)$$

in units for which $\rho = 1$. (We have replaced Dou's original reference to the position of the hypersurface Σ , $v = v_0$, with our own, $v = a$, which corresponds more closely to the collapsing shell case.)

C.4.1. Substitution. We take $u_x = u_x(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $v_x = v_x(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $u_y = u_y(r_{xx}, r_{xa}, r_{xy}, r_{yy})$, $v_y = v_y(r_{xx}, r_{xa}, r_{xy}, r_{yy})$ to give the Jacobian

$$J = \begin{vmatrix} u_{y,r_{xx}} & u_{y,r_{xa}} & u_{y,r_{xy}} & u_{y,r_{yy}} \\ v_{y,r_{xx}} & v_{y,r_{xa}} & v_{y,r_{xy}} & v_{y,r_{yy}} \\ u_{x,r_{xx}} & u_{x,r_{xa}} & u_{x,r_{xy}} & u_{x,r_{yy}} \\ v_{x,r_{xx}} & v_{x,r_{xa}} & v_{x,r_{xy}} & v_{x,r_{yy}} \end{vmatrix} . \quad (258)$$

Now, $r_{xa} = f(u_x)$ alone, hence $u_x = f^{-1}(r_{xa})$, for some function f . This means that

$$du_x = \frac{\partial u_x}{\partial r_{xx}} dr_{xx} + \frac{\partial u_x}{\partial r_{yy}} dr_{yy} + \frac{\partial u_x}{\partial r_{xa}} dr_{xa} + \frac{\partial u_x}{\partial r_{xy}} dr_{xy} = \frac{\partial u_x}{\partial r_{xa}} dr_{xa} \quad .$$

Then, for example, $r_{xx} = g(u_x, v_x) \Rightarrow v_x = h(r_{xa}, r_{xx})$ for some functions g and h . So, $\frac{\partial v_x}{\partial r_{yy}} = \frac{\partial v_x}{\partial r_{xy}} = 0$. The Jacobian reduces to

$$\begin{aligned} J &= \begin{vmatrix} u_{y,r_{xx}} & u_{y,r_{xa}} & u_{y,r_{xy}} & u_{y,r_{yy}} \\ 0 & v_{y,r_{xa}} & v_{y,r_{xy}} & 0 \\ 0 & u_{x,r_{xa}} & 0 & 0 \\ v_{x,r_{xx}} & v_{x,r_{xa}} & 0 & 0 \end{vmatrix} \\ &= u_{y,r_{yy}} v_{y,r_{xy}} u_{x,r_{xa}} v_{x,r_{xx}} \quad . \end{aligned} \quad (259)$$

Then, for example,

$$\begin{aligned} \frac{\partial(u_y v_y)}{\partial r_{yy}} &= \frac{\partial u_y}{\partial r_{yy}} v_y + \frac{\partial v_y}{\partial r_{yy}} u_y = \frac{\partial u_y}{\partial r_{yy}} v_y \\ \Rightarrow u_{y,r_{yy}} &= \frac{1}{v_y} \frac{\partial}{\partial r_{yy}} \left[\left(1 - \frac{r_{yy}}{2M}\right) e^{r_{yy}/2M} \right] \\ &= -\frac{1}{v_y} \frac{r_{yy} e^{r_{yy}/2M}}{(2M)^2} \end{aligned} \quad (260)$$

and similarly for $v_{y,r_{xy}}, u_{x,r_{xa}}, v_{x,r_{xx}}$.

Hence

$$J = \frac{1}{(2M)^8} \frac{1}{u_x v_y} r_{xy} e^{r_{xy}/2M} \frac{1}{u_x a} r_{xa} e^{r_{xa}/2M} r_{yy} e^{r_{yy}/2M} r_{xx} e^{r_{xx}/2M} \quad (261)$$

but

$$\frac{1}{u_x v_y} = \left(\frac{2M}{2M - r_{xy}} \right) e^{-r_{xy}/2M} \quad (262)$$

and similarly for $u_x a$. Thus, our final result:

$$J = \frac{1}{(2M)^6} \frac{r_{xy}}{r_{xy} - 2M} \frac{r_{xa}}{r_{xa} - 2M} r_{xx} e^{r_{xx}/2M} r_{yy} e^{r_{yy}/2M} . \quad (263)$$

C.4.2. *Limits.* With the above substitution we have

$$\int_0^a dv_x \int_{-\infty}^0 du_x \int_a^\infty dv_y \int_0^{1/v_y} du_y \rightarrow \int dr_{yy} \int dr_{xy} \int dr_{xa} \int dr_{xx} .$$

We may calculate the limits for the post-substitution integral using the inequalities (a) $0 < v_x < a$, (b) $-\infty < u_x < 0$, (c) $a < v_y < \infty$, (d) $0 < u_y < 1/v_y$ and the relationship $u_p v_q = (1 - r_{pq}/2M) \exp\{r_{pq}/2M\}$.

From (d) we have $0 < u_y v_y < 1$ and hence $0 < r_{yy} < 2M$.

Combining (a)-(d) gives

$$\begin{aligned} 0 < v_x < a < v_y < \infty , \\ -\infty < u_x < 0 < u_y < 1/v_y . \end{aligned}$$

Then $-\infty < u_x v_y < 0$ and hence $2M < r_{xy} < \infty$.

And $u_x v_y < u_x a < 0$ giving $2M < r_{xa} < r_{xy}$.

And $u_x a < u_x v_x < 0$ giving $2M < r_{xx} < r_{xa}$.

C.4.3. *Final Integral.* With the above Jacobian and limits, Dou's integral becomes

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha e^{r_{yy}^2} dr_{yy} \int_\alpha^\infty dr_{xy} \frac{r_{xy}}{r_{xy} - \alpha} e^{-r_{xy}^2} \int_\alpha^{r_{xy}} \frac{r_{xa}}{r_{xa} - \alpha} dr_{xa} \int_\alpha^{r_{xa}} e^{r_{xx}^2} dr_{xx} \quad (264)$$

writing $\alpha \equiv 2M$.

C.4.4. *Evaluating $\langle S_{\text{Dou}}^{\text{static}} \rangle$.* For ease of expression, let $x = r_{xy}$, $y = r_{xa}$, $z = r_{xx}$. Then

$$I(\alpha) = \int_\alpha^\infty dx \frac{x}{x - \alpha} e^{-x^2} \int_\alpha^x dy \frac{y}{y - \alpha} \int_\alpha^y dz e^{z^2} \quad (265)$$

Then let

$$\begin{aligned} I_1(y) &= \int_\alpha^y dz e^{z^2} = \int_\alpha^y dz \sum_{n=0}^\infty \frac{z^{2n}}{n!} \\ &= \sum_{n=0}^\infty \frac{y^{2n+1} - \alpha^{2n+1}}{(2n+1) \cdot n!} \\ &= \frac{y - \alpha}{1 \cdot 0!} + \frac{y^3 - \alpha^3}{3 \cdot 1!} + \frac{y^5 - \alpha^5}{5 \cdot 2!} + \dots \end{aligned} \quad (266)$$

Then let

$$\begin{aligned} I_2(y) &= \frac{y}{y - \alpha} \cdot I_1(y) = \frac{y}{1 \cdot 0!} + \frac{y}{y - \alpha} \cdot \frac{1}{3 \cdot 1!} \cdot [(y - \alpha)(y^2 + \alpha y + \alpha^2)] + \dots \\ &= \frac{y}{1 \cdot 0!} + \frac{y}{3 \cdot 1!} \cdot [y^2 + \alpha y + \alpha^2] + \dots \end{aligned} \quad (267)$$

In general the n^{th} term of $I_2(y)$ is given by

$$I_2(y)|_n = \frac{y}{(2n+1) \cdot n!} \sum_{p=0}^{2n} \alpha^p y^{2n-p} \quad (268)$$

and so we have

$$I_2(y) = \sum_{n=0}^\infty \frac{1}{(2n+1) \cdot n!} \sum_{p=0}^{2n} \alpha^p y^{2n-p+1} \quad (269)$$

Finally, define $I_3(x)$ as

$$\begin{aligned}
 I_3(x) &= \frac{1}{x-\alpha} \int_{\alpha}^x dy I_2(y) \\
 &= \frac{1}{x-\alpha} \int_{\alpha}^x dy \sum_{n=0}^{\infty} \frac{1}{(2n+1) \cdot n!} \sum_{p=0}^{2n} \alpha^p y^{2n-p+1} \\
 &= \frac{1}{x-\alpha} \sum_{n=0}^{\infty} \frac{1}{(2n+1) \cdot n!} \sum_{p=0}^{2n} \alpha^p \left[y^{2n-p+2} \cdot \frac{1}{2n-p+2} \right]_{\alpha}^x \\
 &= \frac{1}{x-\alpha} \sum_{n=0}^{\infty} \sum_{p=0}^{2n} \frac{1}{(2n+1) \cdot n!} \cdot \alpha^p \cdot \frac{1}{2n-p+2} \cdot (x^{2n-p+2} - \alpha^{2n-p+2})
 \end{aligned} \tag{270}$$

and so

$$\begin{aligned}
 I_3(x) &= \frac{1}{x-\alpha} \left[\sum_{p=0}^0 \frac{1}{(2-p) \cdot 1 \cdot 0!} \cdot \alpha^p \cdot (x^{2-p} - \alpha^{2-p}) \right. \\
 &\quad + \sum_{p=0}^2 \frac{1}{(4-p) \cdot 1 \cdot 1!} \cdot \alpha^p \cdot (x^{4-p} - \alpha^{4-p}) \\
 &\quad + \sum_{p=0}^4 \frac{1}{(6-p) \cdot 1 \cdot 2!} \cdot \alpha^p \cdot (x^{6-p} - \alpha^{6-p}) \\
 &\quad \left. + \sum_{p=0}^6 \frac{1}{(8-p) \cdot 1 \cdot 3!} \cdot \alpha^p \cdot (x^{8-p} - \alpha^{8-p}) + \dots \right] \\
 &= \frac{1}{x-\alpha} \left[\frac{1}{2 \cdot 1 \cdot 0!} \cdot (x^2 - \alpha^2) \right. \\
 &\quad + \frac{1}{4 \cdot 3 \cdot 1!} \cdot (x^4 - \alpha^4) + \frac{1}{3 \cdot 3 \cdot 1!} \cdot \alpha \cdot (x^3 - \alpha^3) + \frac{1}{2 \cdot 3 \cdot 1!} \cdot \alpha^2 \cdot (x^2 - \alpha^2) \\
 &\quad + \frac{1}{6 \cdot 5 \cdot 2!} \cdot (x^6 - \alpha^6) + \frac{1}{5 \cdot 5 \cdot 2!} \cdot \alpha \cdot (x^5 - \alpha^5) + \dots + \frac{1}{2 \cdot 5 \cdot 2!} \cdot \alpha^4 \cdot (x^2 - \alpha^2) \\
 &\quad \left. + \frac{1}{8 \cdot 7 \cdot 3!} \cdot (x^8 - \alpha^8) + \dots + \frac{1}{2 \cdot 7 \cdot 3!} \cdot \alpha^6 \cdot (x^2 - \alpha^2) + \dots \right] \\
 &= \frac{\alpha}{1 \cdot 0!} \cdot \left(\frac{(x/\alpha) + 1}{2} \right) \\
 &\quad + \frac{\alpha^3}{3 \cdot 1!} \cdot \left(\frac{(x/\alpha)^3 + (x/\alpha)^2 + (x/\alpha) + 1}{4} \right. \\
 &\quad \left. + \frac{(x/\alpha)^2 + (x/\alpha) + 1}{3} + \frac{(x/\alpha) + 1}{2} \right) + \dots
 \end{aligned} \tag{271}$$

In general, the n^{th} term of $I_3(x)$ is given by

$$I_3(x)|_n = \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{q=0}^{p-1} \left(\frac{x}{\alpha}\right)^q \quad (272)$$

and hence

$$I_3(x) = \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{q=0}^{p-1} \left(\frac{x}{\alpha}\right)^q \quad (273)$$

This gives the following expression for $I(\alpha)$:

$$I(\alpha) = \int_{\alpha}^{\infty} dx \left[x e^{-x^2} \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{q=0}^{p-1} \left(\frac{x}{\alpha}\right)^q \right] \quad (274)$$

The $n = 0$ term is given by

$$\begin{aligned} I(\alpha)|_{n=0} &= \int_{\alpha}^{\infty} dx x e^{-x^2} \cdot \frac{\alpha}{1 \cdot 0!} \cdot \frac{1}{2} \cdot \left(1 + \frac{x}{\alpha}\right) \\ &= \frac{\alpha}{1 \cdot 0!} \cdot \frac{1}{2} \cdot \int_{\alpha}^{\infty} dx x e^{-x^2} + \frac{\alpha}{1 \cdot 0!} \cdot \frac{1}{2} \cdot \int_{\alpha}^{\infty} dx \frac{x^2}{\alpha} e^{-x^2} \\ &= \frac{\alpha}{1 \cdot 0!} \left(\frac{J_1 + J_2}{2} \right) \end{aligned} \quad (275)$$

where $J_1 = \int_{\alpha}^{\infty} dx x e^{-x^2}$ and $J_2 = \int_{\alpha}^{\infty} dx \frac{x^2}{\alpha} e^{-x^2}$.

More generally, the n^{th} term is given by

$$\begin{aligned} I(\alpha)|_n &= \int_{\alpha}^{\infty} dx \left[x e^{-x^2} \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{q=0}^{p-1} \left(\frac{x}{\alpha}\right)^q \right] \\ &= \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \int_{\alpha}^{\infty} dx \left[e^{-x^2} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{q=0}^{p-1} \frac{x^{q+1}}{\alpha^q} \right] \\ &= \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{r=1}^p \int_{\alpha}^{\infty} dx \left[\frac{x^r}{\alpha^{r-1}} e^{-x^2} \right] \end{aligned} \quad (276)$$

and thus

$$I(\alpha) = \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1) \cdot n!} \sum_{p=2}^{2n+2} \frac{1}{p} \sum_{r=1}^p J_r \quad (277)$$

where we define $J_r = \frac{1}{\alpha^{r-1}} \int_{\alpha}^{\infty} dx x^r e^{-x^2}$. A recursion relation for J_r can be defined:

$$\begin{aligned} J_r &= \frac{1}{\alpha^{r-1}} \int_{\alpha}^{\infty} dx x^{r-1} x e^{-x^2} \\ &= \frac{1}{\alpha^{r-1}} \left[\left[x^{r-1} \cdot -\frac{1}{2} e^{-x^2} \right]_{\alpha}^{\infty} - \int_{\alpha}^{\infty} dx (r-1) x^{r-2} \cdot -\frac{1}{2} e^{-x^2} \right] \\ &= \frac{1}{2} e^{-\alpha^2} + \frac{r-1}{2\alpha^2} J_{r-2} \quad \forall r \geq 3 \end{aligned} \quad (278)$$

with $J_1 = \int_{\alpha}^{\infty} dx x e^{-x^2} = \frac{1}{2} e^{-\alpha^2}$.

Terms with odd r can be solved exactly:

$$J_{2t+1} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^t \frac{t!}{(\alpha^2)^{m+1} (t-m)!} \right) . \quad (279)$$

Terms with even r have:

$$J_{2t} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^{t-1} \frac{(2t-1)!!}{2^m (\alpha^2)^{m+1} (2t-1-2m)!!} + \frac{(2t-1)!!}{2^t \alpha (\alpha^2)^t} \left(e^{\alpha^2} \sqrt{\pi} \text{Erfc}(\alpha) \right) \right) \quad (280)$$

Where $\text{Erfc}(\alpha) = 1 - \frac{2}{\sqrt{\pi}} \int_0^{\alpha} e^{-\beta^2} d\beta$. We are interested in the large α limit, under which

$$\begin{aligned} e^{\alpha^2} \sqrt{\pi} \left(1 - \frac{2}{\sqrt{\pi}} \int_0^{\alpha} e^{-\beta^2} d\beta \right) &= \frac{1}{\alpha} - \frac{1}{2\alpha^3} + \frac{3 \cdot 1}{2^2 \alpha^5} - \frac{5 \cdot 3 \cdot 1}{2^3 \alpha^7} + \dots \\ &= \sum_{q=0}^{\infty} \frac{(-1)^q (2q+1)!!}{(2q+1) 2^q \alpha^{2q+1}} . \end{aligned} \quad (281)$$

This gives our expression for J_{2t} as

$$\begin{aligned} J_{2t} &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^{t-1} \frac{(2t-1)!!}{2^m (\alpha^2)^{m+1} (2t-1-2m)!!} \right. \\ &\quad \left. + (2t-1)!! \sum_{q=0}^{\infty} \frac{(-1)^q (2q+1)!!}{(2q+1) 2^{q+t} \alpha^{2q+2t+1}} \right) . \end{aligned} \quad (282)$$

We then rewrite eq.277 as

$$I(\alpha) = \frac{\alpha^2}{2} e^{-\alpha^2} (I_0 + I_2 + I_4 + I_6 + \dots) \quad (283)$$

where I_{2k} sums terms of order $\frac{1}{\alpha^{2k}}$. The exact form of each I_{2k} can be determined from eq.279 and eq.282. This procedure is carried out in full in §10, using very similar constructions of J_r , and we will not repeat the detail here. The resulting sequence of I_{2k} is given

by Dou as beginning:

$$\begin{aligned}
I_0 &= \alpha e^{\alpha^2} , \\
I_2 &= \alpha e^{\alpha^2} \left(\frac{1}{4} + \frac{1}{4\alpha^2} \right) , \\
I_4 &= \alpha e^{\alpha^2} \left(\frac{1}{9} + \frac{7}{72\alpha^2} - \frac{1}{8\alpha^4} \right) , \\
I_6 &= \alpha e^{\alpha^2} \left(\frac{1}{16} + \frac{5}{96\alpha^2} - \frac{1}{12\alpha^4} + \frac{3}{16\alpha^6} \right) , \\
I_8 &= \alpha e^{\alpha^2} \left(\frac{1}{25} + \frac{13}{400\alpha^2} - \frac{143}{2400\alpha^4} + \frac{17}{96\alpha^6} - \frac{15}{32\alpha^8} \right) , \\
&\dots
\end{aligned}$$

The leading order term of $I(\alpha)$ is given as

$$I(\alpha) = \frac{\alpha}{2} \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\alpha\pi^2}{12} . \quad (284)$$

Our complete integral is given as

$$\langle S_{\text{Dou}}^{\text{static}} \rangle = 4e^{-\alpha^2} \int_0^\alpha e^{y^2} dy I(\alpha) . \quad (285)$$

The integral component can be solved as a series in $\frac{1}{\alpha}$:

$$\begin{aligned}
e^{-\alpha^2} \int_0^\alpha e^{y^2} dy &= \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \frac{(2n+1)!!}{(2n+1)} \left(\frac{1}{\alpha} \right)^{2n+1} \\
&= \frac{1}{2\alpha} + \frac{1}{4\alpha^3} + \mathcal{O}\left(\frac{1}{\alpha^5}\right) .
\end{aligned} \quad (286)$$

This gives our final result

$$\begin{aligned}
\langle S_{\text{Dou}}^{\text{static}} \rangle &= 4 \left[\frac{1}{2\alpha} + \frac{1}{4\alpha^3} + \mathcal{O}\left(\frac{1}{\alpha^5}\right) \right] \left[\frac{\alpha\pi^2}{12} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) \right] \\
&= \frac{\pi^2}{6} + \mathcal{O}\left(\frac{1}{\alpha^2}\right) .
\end{aligned} \quad (287)$$

APPENDIX D. CALCULATIONS FOR THE z -TRIPLET.

D.1. Solving the Integral for $\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle$ Using $\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle$. The integral for the expectation value of the entropy $S_{z\text{-triplet}}^{\text{coll.}}$, $\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle$, is given as

$$\begin{aligned} \langle S_{z\text{-triplet}}^{\text{coll.}} \rangle &= \int_a^b dv_y \int_0^a du_y \int_0^a dv_x \int_{-\infty}^0 du_x (a - v_x) u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &\quad + \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x (a - v_x) a e^{u_x(v_y - v_x) - u_y v_y} \\ &= -\langle S_{\Lambda\text{-triplet}}^{\text{coll.}} \rangle + \langle S'_{z\text{-triplet}} \rangle \end{aligned} \quad (288)$$

where

$$\begin{aligned} \langle S'_{z\text{-triplet}} \rangle &= \int_a^b dv_y \int_0^a du_y \int_0^a dv_x \int_{-\infty}^0 du_x a u_y e^{u_x(v_y - v_x) - u_y v_y} \\ &\quad + \int_a^b dv_y \int_a^b du_y \int_0^a dv_x \int_{-\infty}^0 du_x a^2 e^{u_x(v_y - v_x) - u_y v_y} \\ &= \langle S'^{1}_{z\text{-triplet}} \rangle + \langle S'^{2}_{z\text{-triplet}} \rangle \end{aligned} \quad (289)$$

We consider each of these final components in turn, writing y for v_y , and x for u_y , for reasons of clarity.

$$\begin{aligned} \langle S'^{1}_{z\text{-triplet}} \rangle &= a \int_a^b dy \int_0^a dx x e^{-xy} \int_0^a dv_x \left[\frac{1}{y - v_x} \right] \\ &= a \int_a^b dy \int_0^a dx x e^{-xy} \ln \left(\frac{y}{y - a} \right) \\ &= a \int_a^b dy \ln \left(\frac{y}{y - a} \right) \left[\frac{1 - e^{-ay} (1 + ay)}{y^2} \right] \\ &= \langle S'^{1a}_{z\text{-triplet}} \rangle + \langle S'^{1b}_{z\text{-triplet}} \rangle + \langle S'^{1c}_{z\text{-triplet}} \rangle \end{aligned} \quad (290)$$

where the final three components equate to the terms in the expansion of the previous line.

Then

$$\begin{aligned}
\langle S'_{z\text{-triplet}}{}^{1a} \rangle &= a \int_a^b dy \frac{1}{y^2} \ln \left(\frac{y}{y-a} \right) \\
&= a \left[\frac{1}{y} \ln \left(\frac{y-a}{y} \right) - \frac{1}{a} \ln \left(\frac{y-a}{a} \right) - \frac{1}{y} \right]_a^b \\
&= a \left[\frac{1}{b} \ln \left(\frac{b-a}{b} \right) - \frac{1}{a} \ln \left(\frac{b-a}{b} \right) - \frac{1}{b} \right. \\
&\quad \left. - \frac{1}{a} \lim_{y \rightarrow a} \ln \left(\frac{y-a}{a} \right) + \frac{1}{a} \lim_{y \rightarrow a} \ln \left(\frac{y-a}{a} \right) + \frac{1}{a} \right] \\
&= 1 + \left(\frac{a}{b} - 1 \right) \ln \left(\frac{b-a}{b} \right) - \frac{a}{b}
\end{aligned} \tag{291}$$

which we will discuss further, below. And

$$\begin{aligned}
|\langle S'_{z\text{-triplet}}{}^{1b} \rangle| &= a \int_a^b dy \frac{e^{-ay}}{y^2} \ln \left(\frac{y}{y-a} \right) \\
&< e^{-a^2} \langle S'_{z\text{-triplet}}{}^{1a} \rangle \\
&\sim 0 \text{ for large } a, \text{ small } a/b.
\end{aligned} \tag{292}$$

And

$$\langle S'_{z\text{-triplet}}{}^{1c} \rangle = -a^2 \int_a^b dy \frac{e^{-ay}}{y} \ln \left(\frac{y}{y-a} \right) \tag{293}$$

which we will show to cancel with another term in $\langle S'_{z\text{-triplet}} \rangle$ below.

Turning to $\langle S'_{z\text{-triplet}}{}^2 \rangle$:

$$\begin{aligned}
\langle S'_{z\text{-triplet}}{}^2 \rangle &= a^2 \int_a^b dy \int_a^b dx e^{-xy} \int_0^a dv_x \left[\frac{1}{y-v_x} \right] \\
&= a^2 \int_a^b dy \int_a^b dx e^{-xy} \ln \left(\frac{y}{y-a} \right) \\
&= a^2 \int_a^b dy \ln \left(\frac{y}{y-a} \right) \frac{1}{y} (e^{-ay} - e^{-by}) \\
&= \langle S'_{z\text{-triplet}}{}^{2a} \rangle + \langle S'_{z\text{-triplet}}{}^{2b} \rangle
\end{aligned} \tag{294}$$

where, again, the final two components equate to the terms in the expansion of the previous line.

Then

$$\begin{aligned}\langle S'_{z\text{-triplet}}{}^{2a} \rangle &= a^2 \int_a^b dy \ln \left(\frac{y}{y-a} \right) \frac{1}{y} e^{-ay} \\ &= -\langle S'_{z\text{-triplet}}{}^{1b} \rangle \quad .\end{aligned}\tag{295}$$

And

$$\begin{aligned}|\langle S'_{z\text{-triplet}}{}^{2b} \rangle| &= a^2 \int_a^b dy \ln \left(\frac{y}{y-a} \right) \frac{1}{y} e^{-by} \\ &< a^2 e^{-ab} \int_a^b dy \frac{1}{y} \ln \left(\frac{y}{y-a} \right) \\ &< a^2 e^{-ab} \langle S_{\text{Dou}} \rangle \\ &< a^2 e^{-ab} \left(\frac{\pi^2}{6} - \ell \left(\frac{a}{b} \right) \right) \\ &\sim 0 \text{ for large } a, \text{ small } a/b \quad .\end{aligned}\tag{296}$$

Substituting back gives us:

$$\begin{aligned}\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle &= -\langle S_{\Lambda\text{-triplet}} \rangle + \langle S'_{z\text{-triplet}} \rangle \\ &= -\langle S_{\Lambda\text{-triplet}} \rangle + \langle S'_{z\text{-triplet}}{}^{1a} \rangle + \langle S'_{z\text{-triplet}}{}^{1c} \rangle - \langle S'_{z\text{-triplet}}{}^{1c} \rangle \\ &= -\langle S_{\Lambda\text{-triplet}} \rangle + \langle S'_{z\text{-triplet}}{}^{1a} \rangle \quad .\end{aligned}\tag{297}$$

In order to take $\lim_{b \rightarrow \infty, b \gg a} \langle S'_{z\text{-triplet}}{}^{1a} \rangle$ we must first expand the logarithm in a :

$$\begin{aligned}1 + \left(\frac{a}{b} - 1 \right) \ln \left(\frac{b-a}{b} \right) - \frac{a}{b} &= 1 + \left(\frac{a}{b} - 1 \right) \left(-\frac{a}{b} - \frac{a^2}{2b^2} - \frac{a^3}{3b^3} - \dots \right) - \frac{a}{b} \\ &= 1 - \left(\frac{a^2}{2 \cdot 1 \cdot b^2} + \frac{a^3}{3 \cdot 2 \cdot b^3} + \frac{a^4}{4 \cdot 3 \cdot b^4} + \dots \right) \\ &= 1 - \sum_{n=2}^{\infty} \frac{1}{n(n-1)} \left(\frac{a}{b} \right)^n \quad .\end{aligned}\tag{298}$$

Thus, our result

$$\langle S_{z\text{-triplet}}^{\text{coll.}} \rangle = 2 - \frac{\pi^2}{6} + \sum_{n=2}^{\infty} \left(\frac{1}{n^2} - \frac{1}{n(n-1)} \right) \left(\frac{a}{b} \right)^n \tag{299}$$

which is in agreement with our calculation of §9.4.

APPENDIX E. CALCULATIONS FOR THE l -TRIPLET.

E.1. **Solving the Integral for $\langle S_{xa}^{\text{static}} \rangle$.** For this integral,

$$\begin{aligned}
 I^{xa}(\alpha) &= \int_{\alpha}^{\infty} dx \frac{x}{x-\alpha} e^{-x^2} \int_{\alpha}^x dy \frac{y^3}{y-\alpha} \int_{\alpha}^y dz e^{z^2} \\
 &= \int_{\alpha}^{\infty} dx x e^{-x^2} \frac{1}{x-\alpha} \int_{\alpha}^x dy \left[\frac{y^3}{1 \cdot 0!} + \frac{y^5 + \alpha y^4 + \alpha^2 y^3}{3 \cdot 1!} + \dots \right] \\
 &= \int_{\alpha}^{\infty} dx x e^{-x^2} \left[\frac{\alpha^3}{1 \cdot 0!} \left(\frac{\left(\frac{x}{\alpha}\right)^3 + \left(\frac{x}{\alpha}\right)^2 + \left(\frac{x}{\alpha}\right) + 1}{4} \right) \right. \\
 &\quad \left. + \frac{\alpha^5}{3 \cdot 1!} \left(\frac{\left(\frac{x}{\alpha}\right)^5 + \dots + 1}{6} + \frac{\left(\frac{x}{\alpha}\right)^5 + \dots + 1}{5} + \frac{\left(\frac{x}{\alpha}\right)^4 + \dots + 1}{4} \right) + \dots \right] \\
 &= \frac{\alpha}{1 \cdot 0!} \left(\frac{J_4 + J_3 + J_2 + J_1}{4} \right) \\
 &\quad + \frac{\alpha^3}{3 \cdot 1!} \left(\frac{J_6 + \dots + J_1}{6} + \frac{J_5 + \dots + J_1}{5} + \frac{J_4 + \dots + J_1}{4} \right) + \dots
 \end{aligned} \tag{300}$$

For our integral,

$$J_n = \frac{\alpha^2}{\alpha^{n-1}} \int_{\alpha}^{\infty} x^n e^{-x^2} dx \tag{301}$$

with the recursion relation

$$J_n = \frac{\alpha^2}{2} e^{-\alpha^2} + \frac{n-1}{2\alpha^2} J_{n-2} \quad . \tag{302}$$

The integrals for J_{2n+1} are exactly solvable:

$$J_{2n+1} = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^n \frac{n!}{(\alpha^2)^m (n-m)!} \right) \quad . \tag{303}$$

The solutions for J_{2n} (expanding $\text{Erfc}(\alpha)$ as eq.281) are of the form

$$\begin{aligned}
 J_{2n} &= \frac{\alpha^2}{2} e^{-\alpha^2} \left(\sum_{m=0}^{n-1} \frac{(2n-1)!!}{2^m (\alpha^2)^m (2n-1-2m)!!} \right. \\
 &\quad \left. + (2n-1)!! \sum_{p=0}^{\infty} \frac{(-1)^p (2p+1)!!}{(2p+1) 2^{p+n} \alpha^{2p+2n}} \right) \quad .
 \end{aligned} \tag{304}$$

Again,

$$I^{xa}(\alpha) = \frac{\alpha^2}{2} e^{-\alpha^2} (I_0 + I_2 + I_4 + I_6 + \dots) \tag{305}$$

where I_{2k} sums terms of order $\frac{1}{\alpha^{2k}}$.

For I_{2k} the components of J_{2n+1} and J_{2n} with which we are concerned are now given by

$$J_{2n+1}|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{2^k n!}{(2\alpha^2)^k (n-k)!} \right) & \forall k \leq n \\ 0 & \text{otherwise} \end{cases} \quad (306)$$

$$\text{and } J_{2n}|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n-1)!!}{(2\alpha^2)^k (2n-1-2k)!!} \right) & \forall k \leq n-1 \\ \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n-1)!! (-1)^{k-n-1} (2k-2n+1)!!}{(2k-2n+1) 2^k (\alpha^2)^k} \right) & k > n-1 \end{cases} \quad (307)$$

Again, these expressions can be simplified. For $k = 0$,

$$J_{2n+1}|_0 = J_{2n}|_0 = \frac{\alpha^2}{2} e^{-\alpha^2} \quad (308)$$

For $k \geq 1$,

$$J_{2n+1}|_k = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{2^k (n) (n-1) \dots (n-(k-1))}{2^k \alpha^{2k}} \right) \quad (309)$$

$$\text{and } J_{2n}|_k = \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(2n-1) (2n-3) \dots (2n-1-(2k-2))}{2^k \alpha^{2k}} \right) \quad (310)$$

And the general result is

$$J_j|_k = \begin{cases} \frac{\alpha^2}{2} e^{-\alpha^2} & k = 0 \\ \frac{\alpha^2}{2} e^{-\alpha^2} \left(\frac{(j-1)(j-3) \dots (j-1-(2k-2))}{2^k \alpha^{2k}} \right) & k \geq 1 \end{cases} \quad (311)$$

from which we derive

$$I_{2k} = \begin{cases} \alpha e^{\alpha^2} & k = 0 \\ \frac{1}{2^k \alpha^{2k}} \sum_{n=0}^{\infty} \frac{\alpha^{2n+1}}{(2n+1)n!} \sum_{u=4}^{2n+4} \frac{1}{u} \\ \quad \times \sum_{j=1}^u (j-1)(j-3) \dots (j-1-(2k-2)) & k \geq 1 \end{cases} \quad (312)$$

Performing this calculation generates

$$\begin{aligned} I_0 &= \alpha e^{\alpha^2} , \\ I_2 &= \alpha e^{\alpha^2} \left(\frac{1}{4} + \frac{3}{4\alpha^2} \right) , \\ I_4 &= \alpha e^{\alpha^2} \left(\frac{1}{9} + \frac{31}{72\alpha^2} - \frac{1}{8\alpha^4} \right) , \\ &\dots \end{aligned}$$

APPENDIX F. NOTATION

F.1. General Symbols.

A	area	$M_{\oplus}$	mass of Earth
A	area of Alexandrov region	$M_{\odot}$	mass of Sun
$A_{\mathcal{H}}$	area of horizon	m	axial Killing vector
$A_{(q)}$	q -form gauge field	$\mathbf{N}^+$	number operator on $\mathfrak{J}^+$
$\mathbf{a}, \mathbf{a}^\dagger$	annihilation, creation operator	$\mathcal{O}()$	terms of order
$\mathbf{b}, \mathbf{b}^\dagger$	annihilation, creation operator	P	pressure
c	speed of light in vacuo	p	probability
D_μ	covariant derivative	$\tilde{p}$	Fourier transform of p
dA	exterior derivative of A	Q	charge
E	energy	Q_ξ	Komar integral w.r.t. ξ
E^μ	electric field	R	Ricci scalar
$F_{\mu\nu}$	electromagnetic field tensor	R_S	Schwarzschild radius
G	Newton constant	$R_{\mu\nu}$	Ricci tensor
$G_{\mu\nu}$	Einstein tensor	$R_{\mu\nu\rho\lambda}$	Riemann tensor
g	determinant of $g_{\mu\nu}$	$R_{\oplus}$	radius of Earth
$g_{\mu\nu}$	spacetime metric	$R_{\odot}$	radius of Sun
$\mathbf{H}$	Hamiltonian	$r_{\mathcal{H}}$	radius of horizon
$h_{\mu\nu}$	deviation of metric from $\eta_{\mu\nu}$	S	action
$\hbar$	Planck constant / 2π	S	entropy
i	$\sqrt{-1}$	S_{BH}	Bekenstein-Hawking entropy
J	angular momentum	T	stress-energy scalar
j^μ	current density	T	temperature
k	Boltzmann's constant	T_H	Hawking temperature
k	timelike Killing vector field	$T_{\mu\nu}$	stress-energy tensor
k	momentum	t_f	fundamental time scale
l_f	fundamental length scale	t_P	Planck time
l_P	Planck length	U	internal energy
M	mass	U	propagator
		V	volume

v_f	fundamental volume scale	r	radial coordinate
α	$2M$ where M = black hole mass	r^*	Regge-Wheeler radial coordinate
α	Bogoliubov matrix	s^2	interval
β	Bogoliubov matrix	t	time coordinate
$\Gamma_{\mu\nu}^\lambda$	connection	U	Kruskal-Szekeres coordinate
$\delta^{(n)}$	Dirac δ -function (n dimensions)	u	Kruskal-Szekeres coordinate
δ_{ij}	Kronecker δ	u	outgoing EF coordinate
$\eta_{\mu\nu}$	Minkowski metric	V	Kruskal-Szekeres coordinate
κ	surface gravity	v	Kruskal-Szekeres coordinate
Λ	cosmological constant	v	ingoing EF coordinate
λ	affine parameter	x^μ	spacetime coordinate
ξ	Killing vector field	Δ	Boyer-Lindquist coordinate
ρ	convergence	θ	angular coordinate
σ	shear	Σ	Boyer-Lindquist coordinate
Φ	scalar field	τ	proper time
Φ_{cp}	Carter-Penrose mapping	ϕ	angular coordinate
$\Phi_{\mathcal{H}}$	electric surface potential		
ϕ	scalar field	F.3. Spacetime Structure.	
Ω	conformal factor	$\mathcal{A}(x, y)$	Alexandrov region of x, y
Ω	solid angle	$\mathcal{B}$	black hole region of spacetime
$\Omega_{\mathcal{H}}$	angular velocity of black hole	$\mathcal{D}^+(\mathcal{S}_C)$	future development of $\mathcal{S}_C$
ω	angular frequency	$\mathcal{H}$	(future) event horizon
ω	twist	$\mathcal{H}^+(\mathcal{S}_C)$	Cauchy horizon of $\mathcal{S}_C$
∂	derivative	I_0	spacelike infinity
$\langle $	Dirac bra	I^+	future timelike infinity
$ \rangle$	Dirac ket	I^-	past timelike infinity
$\langle \mathbf{X} \rangle$	expectation value of $\mathbf{X}$	$\mathcal{I}^+$	future null infinity
∇^2	Laplacian	$\mathcal{I}^-$	past null infinity
$\square$	d'Alembertian	$\mathcal{I}^+(\mathcal{P})$	chronological future of $\mathcal{P}$
		$\mathcal{I}^-(\mathcal{P})$	chronological past of $\mathcal{P}$
		$\dot{\mathcal{I}}^+(\mathcal{P})$	boundary of $\mathcal{I}^+(\mathcal{P})$

F.2. Coordinates and Intervals.

$\dot{\mathcal{I}}^-(\mathcal{P})$	boundary of $\mathcal{I}^-(\mathcal{P})$	$, \mu$	derivative w.r.t. x^μ
$J^+(\mathcal{P})$	causal future of $\mathcal{P}$	$; \mu$	covariant derivative w.r.t. x^μ
$J^-(\mathcal{P})$	causal past of $\mathcal{P}$	$[[$	skew symmetrization
$\dot{J}^+(\mathcal{P})$	boundary of $J^+(\mathcal{P})$	$()$	symmetrization
$\dot{J}^-(\mathcal{P})$	boundary of $J^-(\mathcal{P})$		
F.6. Causal Sets.			
$\mathcal{K}$	null hypersurface	$\mathcal{A}$	Alexandrov set
$\mathcal{M}$	spacetime manifold	$\mathcal{C}$	causal set
$\mathcal{M}_{BH}$	spacetime in region of black hole	N	number of sprinkled points
$\mathcal{M}_\eta$	Minkowski spacetime	$\bar{\mathcal{P}}$	closure of set $\mathcal{P}$
$\mathbb{M}^d$	d -dim. Minkowski spacetime	$\mathbb{R}$	set of real numbers
$\mathcal{N}$	hypersurface	$\mathbb{R}^2$	set of ordered pairs of reals
$\mathcal{P}$	set of spacetime points	$()$	open set
$\mathcal{R}_n$	region of $\mathcal{M}_{BH}$	$[[$	closed set
$\mathcal{S}_C$	Cauchy surface	$x \prec y$	x precedes y
γ	geodesic	$x \succ y$	x is preceded by y
$\gamma_{\mathcal{H}}$	null geodesic generator of horizon	$x \prec\!\!\prec y$	x precedes y , x - y a link
		$x \not\prec y$	x does not precede y
Σ	hypersurface	$x \not\succ y$	x is not preceded by y
$\partial\mathcal{P}$	boundary of region $\mathcal{P}$	$x \in\!\!\!\in \mathcal{P}$	x is maximal in $\mathcal{P}$
F.4. Topological Objects.			
S^1	1-sphere (circle)	$x \in\!\!\!\in \mathcal{P}$	x is minimal in $\mathcal{P}$
F.5. Indices.			
		$\mathcal{X}, \hat{\mathcal{X}}$	spacetime regions in $J^-(\mathcal{H})$
		$\mathcal{Y}, \hat{\mathcal{Y}}$	spacetime regions in $J^+(\mathcal{H})$
		ρ	density of sprinkled points

A Note on Indices. In general, Greek indices represent space and time dimensions, e.g. $\mu \in \{0, 1, 2, 3\}$. Roman indices represent space dimensions only, e.g. $i \in \{1, 2, 3\}$.

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